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Vittorino Pata

# Fixed Point Theorems and Applications

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Vittorino Pata

# Fixed Point Theorems and Applications

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*To Adele, Giulia, Michele and Giovanni*

# Preface

Fixed point theory is a fascinating subject, with a large number of applications in several fields of mathematics. Perhaps because of this crosscutting character, it is usually not so easy to find books that treat the argument in a unitary fashion. In most cases, fixed points pop up when they are most needed. In contrast, I believe that they deserve a relevant place in any general textbook, and particularly in a functional analysis textbook. This is the main consideration that led me to write down these notes. I have been motivated by the idea of producing a sort of user-friendly guide to fixed points and applications, the one that I would have liked to have handy when I was a student, or even now, when I need to recall a certain result.

In this book, I have aimed to collect most of the significant theorems in the field that I have encountered during my mathematical path, and then to present various related applications. For some of the arguments treated here, I have been greatly inspired by the lectures of Professor Hari Bercovici and Professor Ciprian Foias, two extraordinary teachers I had during my Ph.D. at Indiana University.

This work consists of two parts, which, although rather self-contained, require some mathematical background. The reader is indeed supposed to be familiar with measure theory, Banach and Hilbert spaces, locally convex topological vector spaces, and in general, linear functional analysis.

Even if one cannot possibly find here all the information and details on the various aspects of the theory of fixed points, I hope that these notes will provide a quite satisfactory overview of the topic, at least from the point of view of mathematical analysis.

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Milan, Italy
March 2019

Vittorino Pata

# Introduction

Fixed point theorems concern maps $f$ of a set $X$ into itself that, under certain conditions, admit a *fixed point*, that is, a point $x \in X$ such that

$$\boldsymbol{f(x) = x}.$$

The knowledge of the existence of fixed points has relevant applications in many branches of mathematics, and particularly in analysis and topology. Let us show, for instance, the following simple, albeit indicative, example.

**Example** Suppose we are given a system of $n$ equations in $n$ unknowns of the form

$$g_j(x_1, \ldots, x_n) = 0, \qquad j = 1, \ldots, n$$

where the $g_j$ are continuous real-valued functions of the real variables $x_j$. Let

$$f_j(x_1, \ldots, x_n) = g_j(x_1, \ldots, x_n) + x_j,$$

and, for any point $x = (x_1, \ldots, x_n)$, define

$$f(x) = (f_1(x), \ldots, f_n(x)).$$

Assume now that $f$ has a fixed point $\bar{x} \in \mathbb{R}^n$. Then it is apparent that $\bar{x}$ is a solution to the system of equations.

In Part I of this book, of more theoretical flavor, we will discuss several abstract results concerning fixed points. Various applications of the theorems will be given in Part II. The final Part III contains 50 problems, many of which include helpful hints.

## Notation

Throughout this book, the basic properties of metric spaces (such as the notion of compactness), as well as those of Banach, Hilbert, and topological vector spaces, are taken for granted. Whenever we speak of a generic space $X$, it is understood that $X$ is nonempty.

We define *diameter* of a metric space $X$ endowed with a distance $d$ the quantity

$$\operatorname{diam}(X) = \sup_{x,y\in X} d(x,y).$$

The space $X$ is called bounded when it has finite diameter. In particular, compact metric spaces are bounded.

We will use the symbol $\mathbb{N}$ to denote the set $\{0,1,2,\ldots\}$. Banach spaces (or, more generally, topological vector spaces) are meant to be vector spaces indifferently on the real field $\mathbb{R}$ or on the complex field $\mathbb{C}$, unless otherwise specified. Given a Banach space $X$, $x \in X$ and $r > 0$ we set

$$B_X(x,r) = \{y \in X : \|y-x\| < r\},$$

$$\overline{B}_X(x,r) = \{y \in X : \|y-x\| \le r\},$$

$$\partial B_X(x,r) = \{y \in X : \|y-x\| = r\}.$$

Whenever misunderstandings might occur, we write $\|x\|_X$ to stress that the norm is taken in $X$. Similarly, if $X$ is a Hilbert space, we write the scalar product as $\langle\cdot,\cdot\rangle$, or as $\langle\cdot,\cdot\rangle_X$ to avoid misunderstandings. For a subset $Z \subset X$, we denote:

- By $\overline{Z}$ the closure of $Z$.
- By $Z^C$ the complement of $Z$ in $X$.
- By $\operatorname{span}(Z)$ the vector space generated by $Z$.
- By $\operatorname{co}(Z)$ the *convex hull* of $Z$, that is, the set of all finite convex combinations of elements of $Z$.
- By $\overline{\operatorname{co}}(Z)$ the closure of the convex hull of $Z$ in $X$.

Given two Banach spaces $X$, $Y$, the symbol $L(X, Y)$ stands for the space of bounded linear maps $T : X \to Y$. This is a Banach space with the norm

$$\|T\|_{L(X,Y)} = \sup_{\|x\|_X \le 1} \|Tx\|_Y.$$

As usual, if $X = Y$ we simply write $L(X)$. The dual space of a Banach space $X$, that is, the space $L(X,\mathbb{R})$ or $L(X,\mathbb{C})$ (depending whether $X$ is a real or a complex vector space) is denoted by $X^*$. The elements of $X^*$ are called continuous linear functionals.

Given a locally compact Hausdorff space $K$, we denote by $\mathcal{C}(K)$ the space of continuous functions $f : K \to \mathbb{R}$ or $f : K \to \mathbb{C}$. Locally compact means that every point has a compact neighborhood. If $I$ is a closed bounded interval of the real line and $X$ is a Banach space, then $\mathcal{C}(I, X)$ is the Banach space of continuous functions $f : I \to X$ with the norm

$$\|f\|_{\mathcal{C}(I,X)} = \max_{t \in I} \|f(t)\|_X.$$

We will often use the notion of *uniformly convex* Banach space. Recall that a Banach space $X$ is uniformly convex if, given any two sequences $x_n, y_n \in X$ with

$$\|x_n\| \leq 1, \qquad \|y_n\| \leq 1, \qquad \lim_{n\to\infty} \|x_n + y_n\| = 2,$$

it follows that

$$\lim_{n\to\infty} \|x_n - y_n\| = 0.$$

In particular, we will exploit the property (coming directly from the definition of uniform convexity) that minimizing sequences in closed convex subsets are convergent. Namely, if $C \subset X$ is (nonempty) closed and convex and $x_n \in C$ is such that

$$\lim_{n\to\infty} \|x_n\| = \inf_{y \in C} \|y\|,$$

then there exists a unique $x \in C$ such that

$$\|x\| = \inf_{y \in C} \|y\| = \min_{y \in C} \|y\|,$$

and

$$\lim_{n\to\infty} x_n = x.$$

A remarkable fact is that every uniformly convex Banach space is reflexive (but not the other way around). Typical examples of uniformly convex spaces are the Lebesgue $L^p$ spaces, provided that $p \in (1, \infty)$. Clearly, being Hilbert spaces uniformly convex, all the results involving uniformly convex Banach spaces can be read in terms of Hilbert spaces.

A weaker notion is strict convexity: A Banach space $X$ is *strictly convex* if, for all $x, y \in X$ with $x \neq y$,

$$\|x\| = \|y\| \leq 1 \qquad \Rightarrow \qquad \|x + y\| < 2.$$

It is immediate to check from the definitions that uniform convexity implies strict convexity.

We now report a first version of the Hahn–Banach theorem that applies to real vector spaces (no topology inside):

Let $Y$ be a subspace of a *real* vector space $X$, and let $p : X \to \mathbb{R}$ be a function satisfying

$$p(x+y) \le p(x) + p(y) \qquad \text{and} \qquad p(\lambda x) = \lambda p(x),$$

for every $x, y \in X$ and every $\lambda \ge 0$. If $\Lambda_0 : Y \to \mathbb{R}$ is linear functional and

$$\Lambda_0 x \le p(x), \qquad \forall x \in Y,$$

then $\Lambda_0$ extends to a linear functional $\Lambda$ on $X$ such that

$$-p(-x) \le \Lambda x \le p(x), \qquad \forall x \in X.$$

A *seminorm* on a (real or complex) vector space $X$ is a function $p : X \to \mathbb{R}$ such that

1. $p(x+y) \le p(x) + p(y)$,
2. $p(\lambda x) = |\lambda| p(x)$,

for all $x, y \in X$ and all scalar $\lambda$. A seminorm $p$ is easily seen to fulfill the following properties:

- $p(0) = 0$.
- $p(x) \ge 0$.
- $|p(x) - p(y)| \le p(x - y)$.
- The set $\{x : p(x) = 0\}$ is a subspace of $X$.
- The sets of the form $\{x \in X : p(x) < \varepsilon\}$ for some $\varepsilon > 0$ are convex.

As a consequence of the first two properties above, we deduce that if $p(x) \ne 0$ whenever $x \ne 0$, then $p$ is actually a norm.

In connection with seminorms, the Hahn–Banach theorem takes the more familiar form:

Give a (real or complex) vector space $X$, let $Y$ be a subspace of $X$, let $p$ be a seminorm on $X$, and let $\Lambda_0$ be a linear functional such that

$$|\Lambda_0 x| \leq p(x), \qquad \forall x \in Y.$$

Then $\Lambda_0$ extends to a linear functional $\Lambda$ on $X$ such that

$$|\Lambda x| \leq p(x), \qquad \forall x \in X.$$

In particular, if $X$ is a Banach space and $p(\cdot) = c\|\cdot\|$, the theorem tells that the extension $\Lambda$ belongs to $X^*$ and its norm does not exceed $c$. Let us recall two important corollaries of the Hahn–Banach theorem in the Banach space setting.

Let $X$ be a Banach space. Then the following hold:

- For every $x \in X$ there exists $\Lambda \in X^*$ of unit norm such that $\Lambda x = \|x\|$.
- Let $Y$ be a closed subspace of $X$ with $Y \neq X$. Then exists $\Lambda \in X^*$ of unit norm such that

$$\Lambda x = 0, \qquad \forall x \in Y.$$

Other spaces widely used here are *locally convex spaces*. A locally convex space $X$ is a (real or complex) vector space endowed with a separating family $\mathcal{P}$ of seminorms. Separating means that for every element $x \in X$ there is $p \in \mathcal{P}$ such that $p(x) \neq 0$. In which case, $\mathcal{P}$ gives $X$ the structure of (Hausdorff) topological vector space in which there is a local base whose members are convex (a local base is a family $\mathcal{B}$ of open neighborhoods of 0 such that every neighborhood of 0 contains a member of $\mathcal{B}$). Besides, every $p \in \mathcal{P}$ is continuous. A local base $\mathcal{B}$ for the topology is given by finite intersections of sets of the form

$$\{x \in X : p(x) < \varepsilon\},$$

for some $p \in \mathcal{P}$ and some $\varepsilon > 0$. Note that, given $U \in \mathcal{B}$, we have

$$U + U \doteq \{x + y : x \in U, y \in U\} = 2U \doteq \{2x : x \in U\}.$$

The dual space of a locally convex space $X$, that is, the vector space of continuous linear functionals on $X$, is still denoted by $X^*$. An important result, consequence of the Hahn–Banach theorem, is the following separation theorem:

Let $C$ be a nonempty closed convex set in a locally convex space $X$, and let $x \notin C$. Then there exists $\Lambda \in X^*$ and $\alpha \in \mathbb{R}$ such that

$$\sup_{y \in C} \mathrm{Re}\, \Lambda y < \alpha < \mathrm{Re}\, \Lambda x.$$

If the scalar field is $\mathbb{R}$, then $\mathrm{Re}\, \Lambda = \Lambda$.

A striking consequence is that, in a locally convex space, the dual $X^*$ separates the points of $X$, namely for every $x, y \in X$ with $x \neq y$ there is $\Lambda \in X^*$ such that $\Lambda x \neq \Lambda y$.

Good references for these arguments are, e.g., the books [9, 23, 66]. Concerning measure theory, we address the reader to [34, 65, 68].

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# About the Author

**Vittorino Pata** completed his Ph.D. in pure mathematics at Indiana University, USA, in December 1994. Since 2000, he has been a professor of mathematical analysis at the Politecnico di Milano, Italy. His research interests touch several areas of pure and applied mathematics, including ordinary and partial differential equations (with particular emphasis on the asymptotic behavior of solutions), infinite-dimensional dynamical systems, real and functional analysis, operator theory, and noncommutative probability.

# Part I
# Fixed Point Theorems

# Chapter 1
# The Banach Contraction Principle




**Definition 1.1** Let $X$ be a metric space equipped with a distance $d$. A map

$$f : X \to X$$

is said to be *Lipschitz continuous* if there is $\lambda \geq 0$ such that

$$d(f(x), f(y)) \leq \lambda d(x, y), \qquad \forall x, y \in X.$$

In this case, it is readily seen that there exists the smallest value $\lambda$ for which the inequality holds, called the *Lipschitz constant* of $f$.

- If $\lambda = 1$, then $f$ is said to be *non-expansive*.
- If $\lambda < 1$, then $f$ is said to be a *contraction*.

Perhaps the most important result in the theory of fixed points is the celebrated Banach contraction principle (BCP), stated and proved by Banach [4] in 1922 and subsequently -and independently- rediscovered by Caccioppoli [13].

**Theorem 1.1** (BCP) *Let $f$ be a contraction on a complete metric space $X$. Then $f$ has a unique fixed point $\bar{x} \in X$.*

*Proof* Note first that if $x, y \in X$ are fixed points of $f$, then

$$d(x, y) = d(f(x), f(y)) \leq \lambda d(x, y),$$

which implies $x = y$. Choose now any $x_0 \in X$, and define the iterated sequence

$$x_{n+1} = f(x_n).$$

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By induction on $n$,

$$d(x_{n+1}, x_n) \leq \lambda^n d(f(x_0), x_0), \qquad \forall n \geq 1.$$

Thus, if $n \in \mathbb{N}$ and $m \geq 1$, recalling that $\lambda < 1$ we have

$$\begin{aligned} d(x_{n+m}, x_n) &\leq d(x_{n+m}, x_{n+m-1}) + \cdots + d(x_{n+1}, x_n) \\ &\leq (\lambda^{n+m-1} + \cdots + \lambda^n) d(f(x_0), x_0) \\ &\leq \lambda^n \left( \sum_{j=0}^{\infty} \lambda^j \right) d(f(x_0), x_0) \\ &= \frac{\lambda^n}{1-\lambda} d(f(x_0), x_0). \end{aligned}$$

Hence $x_n$ is a Cauchy sequence, and admits a limit $\bar{x} \in X$, for $X$ is complete. Since $f$ is continuous, we have

$$f(\bar{x}) = \lim_{n \to \infty} f(x_n) = \lim_{n \to \infty} x_{n+1} = \bar{x},$$

as claimed. □

The construction of the approximating sequence $x_n$ above is known as *Picard iteration method* [62].

**Corollary 1.1** *The approximating sequence $x_n$ fulfills the estimate*

$$d(\bar{x}, x_n) \leq \frac{\lambda^n}{1-\lambda} d(f(x_0), x_0).$$

*Proof* From the previous proof,

$$d(x_{n+m}, x_n) \leq \frac{\lambda^n}{1-\lambda} d(f(x_0), x_0).$$

Knowing now that $x_n \to \bar{x}$, the claim follows by letting $m \to \infty$. □

*Remark* As noted in [56], the inequality established in Corollary 1.1 is important also for the following reason: assume we want to find the fixed point up to an "error" $\varepsilon > 0$, namely, we want to find a point $\hat{x}$ such that

$$d(\bar{x}, \hat{x}) < \varepsilon,$$

where $\bar{x}$ is the actual fixed point. Then, the inequality above allows to find an explicit value of $n \in \mathbb{N}$ for which $\hat{x} = x_n$ will do. Indeed, such an $n$ has to comply with the relation $d(x_n, \bar{x}) < \varepsilon$. Accordingly, we have to take $n$ large enough that

$$\frac{\lambda^n}{1-\lambda}\, d(f(x_0), x_0) < \varepsilon.$$

The quantity $\varrho = d(f(x_0), x_0)$ is something known right after the first iteration. Hence, recalling that $\log \lambda < 0$, the desired $n$ fulfills

$$n > \frac{\log \varepsilon + \log(1-\lambda) - \log \varrho}{\log \lambda}.$$

The completeness of $X$ plays a crucial role in the BCP. Indeed, contractions on incomplete metric spaces may fail to have fixed points.

**Example** Let $X = (0, 1]$ with the usual distance. Define $f : X \to X$ as $f(x) = x/2$.

**Corollary 1.2** *Let $X$ be a complete metric space and $W$ be a topological space. Let $f : X \times W \to X$ be a continuous function. Assume that $f$ is a contraction on $X$ uniformly in $W$, that is,*

$$d(f(x, w), f(y, w)) \le \lambda d(x, y), \qquad \forall x, y \in X, \ \forall w \in W,$$

*for some $\lambda < 1$. Then, for every fixed $w \in W$, the map $x \mapsto f(x, w)$ has a unique fixed point $\varphi(w)$. Moreover, the function $w \mapsto \varphi(w)$ is continuous from $W$ to $X$.*

Note that if $f : X \times W \to X$ is continuous on $W$ for every fixed $x \in X$ and is a contraction on $X$ uniformly in $W$, then $f$ is automatically continuous on $X \times W$.

*Proof* In light of Theorem 1.1, we only have to prove the continuity of $\varphi$. For $w, w_0 \in W$, we have

$$\begin{aligned} d(\varphi(w), \varphi(w_0)) &= d(f(\varphi(w), w), f(\varphi(w_0), w_0)) \\ &\le d(f(\varphi(w), w), f(\varphi(w_0), w)) + d(f(\varphi(w_0), w), f(\varphi(w_0), w_0)) \\ &\le \lambda d(\varphi(w), \varphi(w_0)) + d(f(\varphi(w_0), w), f(\varphi(w_0), w_0)), \end{aligned}$$

which implies

$$d(\varphi(w), \varphi(w_0)) \le \frac{1}{1-\lambda}\, d(f(\varphi(w_0), w), f(\varphi(w_0), w_0)).$$

Since the above right-hand side goes to zero as $w \to w_0$, we have the desired continuity. □

*Remark* If in addition $W$ is a metric space and $f$ is Lipschitz continuous in $W$, uniformly with respect to $X$, with Lipschitz constant $L \ge 0$, then the function $w \mapsto \varphi(w)$ is Lipschitz continuous with Lipschitz constant less than or equal to $L/(1-\lambda)$.

Theorem 1.1 establishes a sufficient condition in order for $f$ to have a unique fixed point.

**Example** Consider the map

$$f(x) = \begin{cases} 1/2 + 2x & x \in [0, 1/4], \\ 1/2 & x \in (1/4, 1], \end{cases}$$

mapping $[0, 1]$ onto itself. Then $f$ is not even continuous, but it has the unique fixed point $\bar{x} = 1/2$.

The next corollary takes into account this situation, providing existence and uniqueness of a fixed point under more general conditions. We first need a definition.

**Definition 1.2** For $f : X \to X$ and $n \in \mathbb{N}$, we denote by $f^n$ the *nth-iteration* of $f$, namely,

$$f^n = \underbrace{f \circ \cdots \circ f}_{n \text{ times}},$$

where $f^0$ is understood to be the identity map.

**Corollary 1.3** *Let $X$ be a complete metric space, and let $f : X \to X$. If $f^m$ is a contraction for some $m \geq 1$, then $f$ has a unique fixed point $\bar{x} \in X$. Moreover, for every $x_0 \in X$, the sequence $f^n(x_0)$ converges to $\bar{x}$.*

Note that in the previous example $f^2(x) \equiv 1/2$.

*Proof* Let $\bar{x}$ be the unique fixed point of $f^m$, given by Theorem 1.1. Then

$$f^m(f(\bar{x})) = f(f^m(\bar{x})) = f(\bar{x}),$$

which implies $f(\bar{x}) = \bar{x}$. Since a fixed point of $f$ is clearly a fixed point of $f^m$, we have uniqueness as well. The proof of the convergence $f^n(x_0) \to \bar{x}$ is left as an exercise. □

We conclude the chapter discussing the converse to the BCP. Assume we are given a set $X$ and a map $f : X \to X$. We are interested to find a metric $d$ on $X$ such that $(X, d)$ is a complete metric space and $f$ is a contraction on $X$. Clearly, in light of Theorem 1.1, a necessary condition is that each iteration $f^n$ has a unique fixed point. Surprisingly enough, the condition turns out to be sufficient as well.

**Theorem 1.2** *Let $X$ be an arbitrary set, and let $f : X \to X$ be a map such that $f^n$ has a unique fixed point $\bar{x} \in X$ for every $n \geq 1$. Then for every $\varepsilon \in (0, 1)$, there is a metric $d = d_\varepsilon$ on $X$ that makes $X$ a complete metric space, and $f$ is a contraction on $X$ with Lipschitz constant less than or equal to $\varepsilon$.*

*Proof* Choose $\varepsilon \in (0, 1)$. Let $Z$ be the subset of $X$ consisting of all elements $x$ such that $f^k(x) = \bar{x}$ for some $k \in \mathbb{N}$. We define the following equivalence relation on the (possibly empty) set $X \setminus Z$: we say that $x \sim y$ if and only if $f^n(x) = f^m(y)$ for some $n, m \in \mathbb{N}$. Note that if

$$f^n(x) = f^m(y) \qquad \text{and} \qquad f^{n'}(x) = f^{m'}(y),$$

then

$$f^{n+m'}(x) = f^{m+n'}(x).$$

But since $x \notin Z$, this yields $n + m' = m + n'$, that is,

$$n - m = n' - m'.$$

At this point, by means of the axiom of choice, we select an element from each equivalence class. We now define the distance of $\bar{x}$ from a generic $x \in X$ by setting

$$d(x, \bar{x}) = \begin{cases} 0 & \text{if } x = \bar{x}, \\ \varepsilon^{-k} & \text{if } \bar{x} \neq x \in Z, \\ \varepsilon^{n-m} & \text{if } x \notin Z, \end{cases}$$

where $k = \min\left\{p \geq 1 : f^p(x) = \bar{x}\right\}$, while $n, m \in \mathbb{N}$ are such that $f^n(\hat{x}) = f^m(x)$, where $\hat{x}$ is the selected representative of the equivalence class $[x]$. The definition is unambiguous, due to the discussion above. Finally, for any $x, y \in X$, we set

$$d(x, y) = \begin{cases} d(x, \bar{x}) + d(y, \bar{x}) & \text{if } x \neq y, \\ 0 & \text{if } x = y. \end{cases}$$

It is straightforward to verify that $d$ is a metric. To see that $d$ is complete, observe that the only Cauchy sequences which do not converge to $\bar{x}$ are ultimately constant. We are left to show that $f$ is a contraction with Lipschitz constant equal to $\varepsilon$. Let then $x \in X$, with $x \neq \bar{x}$. We shall distinguish three cases:

- If $x \in Z$ and $f(x) = \bar{x}$, then

$$0 = d(f(x), \bar{x}) < \varepsilon d(x, \bar{x}).$$

- If $x \in Z$ and $f(x) \neq \bar{x}$, then there is the smallest $k \geq 2$ such that

$$\bar{x} = f^k(x) = f^{k-1}(f(x)).$$

Hence,

$$d(f(x), \bar{x}) = \varepsilon^{-k+1} = \varepsilon d(x, \bar{x}).$$

- If $x \notin Z$, then there are $n, m \in \mathbb{N}$, and we can take $m \geq 1$, such that

$$f^n(\hat{x}) = f^m(x) = f^{m-1}(f(x)),$$

where $\hat{x}$ is the representative of $[x]$. Hence,

$$d(f(x), \bar{x}) = \varepsilon^{n-m+1} = \varepsilon d(x, \bar{x}).$$

In summary, for every $x \in X$,

$$d(f(x), \bar{x}) \leq \varepsilon d(x, \bar{x}).$$

From the definition of the distance, given any $x \neq y \in X$ we conclude that

$$\begin{aligned} d(f(x), f(y)) &\leq d(f(x), \bar{x}) + d(f(y), \bar{x}) \\ &\leq \varepsilon\big[d(x, \bar{x}) + d(y, \bar{x})\big] \\ &= \varepsilon d(x, y), \end{aligned}$$

as desired. □

Theorem 1.2 is due to Bessaga [6]. An alternative proof can be found in the book [18] (pp. 191–192). The argument presented here, indeed much simpler, is due to Peirone [61]. There is an interesting related result for compact metric spaces due to Janoš [37], where an equivalent metric that makes $f$ a contraction is constructed.

# Chapter 2
# The Boyd-Wong Theorem




The following result is proved in [8].

**Theorem 2.1** (Boyd-Wong) *Let $(X, d)$ be a complete metric space, and let $f : X \to X$. Assume there exists a continuous function $\varphi : [0, \infty) \to [0, \infty)$ such that $\varphi(r) < r$ if $r > 0$, and*

$$d(f(x), f(y)) \leq \varphi(d(x, y)), \qquad \forall x, y \in X.$$

*Then $f$ has a unique fixed point $\bar{x} \in X$. Moreover, for any $x_0 \in X$, the sequence $x_n = f^n(x_0)$ converges to $\bar{x}$.*

*Remark* The BCP is a particular case of this result, obtained by taking

$$\varphi(r) = \lambda r.$$

On the other hand, the function $f(x) = x - x^2$ mapping $X = [0, 1]$ into itself, and possessing the unique fixed point $\bar{x} = 0$, does not satisfy the assumptions of the BCP (the smallest possible $\lambda$ equals 1), whereas it satisfies those of Theorem 2.1 with

$$\varphi(r) = r - r^2.$$

*Proof* If $x, y \in X$ are fixed points of $f$, then

$$d(x, y) = d(f(x), f(y)) \leq \varphi(d(x, y)),$$

so $x = y$. To prove existence, fix any $x_0 \in X$ and define the sequence $x_n = f^n(x_0)$. We will show that $x_n$ is a Cauchy sequence, and the desired conclusion follows arguing as in the proof of Theorem 1.1. Indeed, it is clear from the hypotheses that $f$ is continuous on $X$. For $n \in \mathbb{N}$, define the positive sequence

$$a_n = d(x_{n+1}, x_n).$$

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It is apparent that

$$a_{n+1} = d(f(x_{n+1}), f(x_n)) \leq \varphi(a_n) \leq a_n.$$

Therefore, $a_n$ converges monotonically to some $a \geq 0$. By comparison, $\varphi(a_n) \to a$ as well, and from the continuity of $\varphi$ we draw the equality $a = \varphi(a)$, implying in turn $a = 0$. At this point, by contradiction, assume that $x_n$ is not a Cauchy sequence. Then, there are $\varepsilon > 0$ and integers $m_k > n_k \geq k$ such that

$$d_k \doteq d(x_{m_k}, x_{n_k}) \geq \varepsilon, \qquad \forall k \geq 1.$$

Upon choosing the smallest possible $m_k$, we may (and do) assume that

$$d(x_{m_k-1}, x_{n_k}) < \varepsilon,$$

for $k$ large enough (here we are using the fact that $a_n \to 0$). Therefore, for $k$ large enough,

$$\varepsilon \leq d_k \leq d(x_{m_k}, x_{m_k-1}) + d(x_{m_k-1}, x_{n_k}) < a_{m_k-1} + \varepsilon,$$

implying that $d_k \to \varepsilon$ from above as $k \to \infty$. Moreover,

$$\begin{aligned} d_k &\leq d(x_{m_k+1}, x_{n_k+1}) + d(x_{m_k+1}, x_{m_k}) + d(x_{n_k+1}, x_{n_k}) \\ &= d(f(x_{m_k}), f(x_{n_k})) + a_{m_k} + a_{n_k} \\ &\leq \varphi(d_k) + a_{m_k} + a_{n_k}, \end{aligned}$$

and taking the limit $k \to \infty$ we obtain the relation $\varepsilon \leq \varphi(\varepsilon)$, which has to be false since $\varepsilon > 0$. □

*Remark* Actually, a closer look at the proof above shows that the continuity of $\varphi$ can be replaced by the right continuity, or even by the right upper semicontinuity. However, for practical purposes such a distinction is really inessential. Under quite general assumptions on the space $X$, if one has a right continuous (or upper semicontinuous) $\varphi$ matching the hypotheses of Theorem 2.1, then such a $\varphi$ can be modified to be continuous and still complying with those hypotheses (see [57]). This is always true, for instance, if $X$ is a closed convex subset of a Banach space.

In some cases, it is possible to determine the explicit convergence rate of the sequence $x_n = f^n(x_0)$ to the fixed point $\bar{x}$.

**Theorem 2.2** *Let the hypotheses of Theorem 2.1 hold. Assume in addition that, for some $a > 0$ and $p > 0$,*

$$\varphi(r) = r - ar^{1+p} + o(r^{1+p}),$$

*where "o" stands for the little-o of Landau (in a neighborhood of zero). Then the sequence*

$$\delta_n = d(\bar{x}, x_n)$$

*fulfills the polynomial decay rate*

$$\delta_n \leq \frac{c}{n^{1/p}},$$

*for some* $c > 0$.

*Proof* Applying the inequality $d(f(x), f(y)) \leq \varphi(d(x, y))$ with $x = \bar{x} = f(\bar{x})$ and $y = x_n$, we readily get

$$\delta_{n+1} \leq \varphi(\delta_n) = \delta_n - a\delta_n^{1+p} + o(\delta_n^{1+p}).$$

If $\delta_n$ eventually vanishes there is nothing to prove. Otherwise, setting

$$z_n = \frac{1}{\delta_n^p},$$

and knowing that $z_n \to \infty$, we obtain the inequality

$$z_{n+1} \geq z_n \left[1 - \frac{a + o(1)}{z_n}\right]^{-p} = z_n + ap + \varepsilon_n,$$

for some $\varepsilon_n \to 0$. This yields

$$z_n \geq z_0 + nap + \sum_{j=0}^{n-1} \varepsilon_j = nap\big(1 + o(1)\big).$$

Accordingly,

$$\delta_n \leq \frac{1}{n^{1/p}} \left[\frac{1 + o(1)}{ap}\right]^{1/p},$$

which clearly implies the claim. □

# Chapter 3
# Further Extensions of the Contraction Principle




Further generalizations of the BCP Theorem 1.1 can be found in the literature (see, e.g., [30, 36]). We report here a couple of results in that direction.

**Theorem 3.1** (Caristi) *Let $(X, d)$ be a complete metric space, and let $f : X \to X$. Assume there exists a lower semicontinuous function $\psi : X \to [0, \infty)$ such that*

$$d(x, f(x)) \leq \psi(x) - \psi(f(x)), \qquad \forall x \in X.$$

*Then $f$ has (at least) a fixed point in $X$.*

Recall that $\psi$ lower semicontinuous means that the set $\{x : \psi(x) > \alpha\}$ is open in $X$ for every $\alpha$. A direct consequence of the definition is that, for every sequence $x_n \in X$ and every $x_* \in X$, we have the implication

$$x_n \to x_* \qquad \Rightarrow \qquad \psi(x_*) \leq \liminf_{n\to\infty} \psi(x_n).$$

*Remark* The hypotheses of the BCP imply those of Theorem 3.1 with

$$\psi(x) = \frac{1}{1-\lambda}\, d(x, f(x)).$$

Indeed, within the hypotheses of the BCP, we have

$$d(f(x), f^2(x)) \leq \lambda d(x, f(x)) = d(x, f(x)) - (1-\lambda) d(x, f(x)).$$

Hence,

$$(1-\lambda) d(x, f(x)) \leq d(x, f(x)) - d(f(x), f^2(x)).$$

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*Proof* We introduce a partial ordering on $X$ by setting

$$x \preceq y \qquad \Leftrightarrow \qquad d(x, y) \leq \psi(x) - \psi(y).$$

Let $\emptyset \neq X_0 \subset X$ be totally ordered, and consider a sequence $x_n \in X_0$ such that

$$\psi(x_{n+1}) \leq \psi(x_n) \qquad \text{and} \qquad \psi(x_n) \to \mu,$$

where

$$\mu = \inf \big\{\psi(x) : x \in X_0\big\}.$$

Due to the fact that $X_0$ is totally ordered, either $x_{n+1} \preceq x_n$ or $x_n \preceq x_{n+1}$. But knowing that $\psi(x_{n+1}) - \psi(x_n) \leq 0$, we necessarily have $x_n \preceq x_{n+1}$, that is,

$$d(x_n, x_{n+1}) \leq \psi(x_n) - \psi(x_{n+1}).$$

Hence, if $n \in \mathbb{N}$ and $m \geq 1$,

$$\begin{aligned} d(x_n, x_{n+m}) &\leq \sum_{i=0}^{m-1} d(x_{n+i}, x_{n+i+1}) \\ &= \sum_{i=0}^{m-1} d(x_{n+i}, f(x_{n+i})) \\ &\leq \sum_{i=0}^{m-1} \psi(x_{n+i}) - \psi(x_{n+i+1}) \\ &= \psi(x_n) - \psi(x_{n+m}). \end{aligned}$$

This shows that $x_n$ is a Cauchy sequence, and so it converges to some limit $x_* \in X$, for $X$ is complete. Let now $x \in X_0$ be such that $d(x, x_*) > 0$. We claim that $x \preceq x_n$ for $n$ large enough. Indeed, observing that $\psi(x) \geq \mu$, there exists $\varepsilon > 0$ such that, for $n$ large enough,

$$d(x, x_n) \geq \varepsilon \qquad \text{and} \qquad \psi(x_n) - \psi(x) < \varepsilon.$$

Accordingly, $x_n \preceq x$ would lead to the contradiction

$$\varepsilon \leq d(x, x_n) \leq \psi(x_n) - \psi(x) < \varepsilon.$$

Therefore, for $n$ large enough,

$$d(x, x_n) \leq \psi(x) - \psi(x_n),$$

and letting $n \to \infty$ we end up with

$$d(x, x_*) \leq \psi(x) - \mu.$$

At this point, we invoke the lower semicontinuity of $\psi$, yielding

$$\psi(x_*) \le \lim_{n\to\infty} \psi(x_n) = \mu,$$

which allows us to conclude that

$$d(x, x_*) \le \psi(x) - \psi(x_*) \qquad \Rightarrow \qquad x \preceq x_*.$$

This means that $x_*$ is an upper bound for $X_0$, relative to the partial ordering $\preceq$ under consideration. Then, by the Zorn lemma, there exists a maximal element $\bar{x}$. On the other hand,

$$d(\bar{x}, f(\bar{x})) \le \psi(\bar{x}) - \psi(f(\bar{x})) \qquad \Rightarrow \qquad \bar{x} \preceq f(\bar{x}),$$

and the maximality of $\bar{x}$ forces the equality $\bar{x} = f(\bar{x})$. □

Theorem 3.1 has been proved more or less explicitly by many authors. We refer to [14] where some applications of the theorem are also provided. Caristi's original proof involves an intricate transfinite induction argument, whereas the more direct proof reported here follows the lines of [18].

Observe that in Theorem 3.1 the continuity of the function $f$ is not required.

**Example** Consider the complete metric space $X = [0, 2]$ with the usual distance, and let $f : X \to X$ be defined as

$$f(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1 + x & \text{if } 0 < x < 1, \\ 0 & \text{if } x \ge 1. \end{cases}$$

The function $f$ is not continuous and admits the (unique) fixed point $\bar{x} = 0$. The reader will have no difficulties to verify that the hypotheses of Theorem 3.1 hold for the lower semicontinuous function $\psi : X \to [0, \infty)$ defined as

$$\psi(x) = \begin{cases} 0 & \text{if } x = 0, \\ 4/x & \text{if } x > 0. \end{cases}$$

If instead we assume the continuity of $f$, we obtain a slightly stronger result, even relaxing the continuity hypothesis on $\psi$.

**Theorem 3.2** *Let $(X, d)$ be a complete metric space, and let $f : X \to X$ be a continuous map. Assume there exists a function $\psi : X \to [0, \infty)$ such that*

$$d(x, f(x)) \le \psi(x) - \psi(f(x)), \qquad \forall x \in X.$$

*Then $f$ has a fixed point in $X$. Moreover, for any $x_0 \in X$, the sequence $f^n(x_0)$ converges to a fixed point of $f$.*

*Proof* Choose $x_0 \in X$, and call $x_n = f^n(x_0)$. Since

$$0 \le d(x_n, f(x_n)) \le \psi(x_n) - \psi(f(x_n)) = \psi(x_n) - \psi(x_{n+1}),$$

the sequence $\psi(x_n)$ is decreasing, and thus convergent. For $n \in \mathbb{N}$ and $m \ge 1$, we have

$$\begin{aligned} d(x_n, x_{n+m}) &\le \sum_{i=0}^{m-1} d(x_{n+i}, x_{n+i+1}) \\ &\le \sum_{i=0}^{m-1} \psi(x_{n+i}) - \psi(x_{n+i+1}) \\ &= \psi(x_n) - \psi(x_{n+m}). \end{aligned}$$

Hence $x_n$ is Cauchy and admits a limit $\bar{x} \in X$, for $X$ is complete. The continuity of $f$ then entails

$$f(\bar{x}) = \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_{n+1} = \bar{x},$$

as claimed. □

We conclude with a generalization of the BCP to the case of quasi-contractions devised by Ćirić [16].

**Definition 3.1** A map $f : X \to X$ on a metric space $X$ is called a *quasi-contraction* if there is $\lambda < 1$ such that

$$\begin{aligned} &d(f(x), f(y)) \\ &\quad \le \lambda \max\left\{d(x, y),\, d(x, f(x)),\, d(y, f(y)),\, d(x, f(y)),\, d(y, f(x))\right\}, \end{aligned}$$

for every $x, y \in X$.

**Theorem 3.3** (Ćirić) *Let $(X, d)$ be a complete metric space, and let $f : X \to X$ be a quasi-contraction. Then $f$ has a unique fixed point $\bar{x} \in X$. Moreover, for any $x_0 \in X$,*

$$d(\bar{x}, f^n(x_0)) \le \frac{\lambda^n}{1-\lambda}\, d(f(x_0), x_0).$$

**Example** The function $f : [0, 1] \to [0, 1]$, given by

$$f(x) = \begin{cases} 1/2 + 2x & x \in [0, 1/4], \\ 1/2 & x \in (1/4, 1], \end{cases}$$

fulfills the hypotheses of Theorem 3.3 with $\lambda = 2/3$, being $\bar{x} = 1/2$ the unique fixed point.

*Proof* The uniqueness of the fixed point follows directly from the fact that $f$ is quasi-contractive (by the same argument of Theorem 1.1). Let then $x_0 \in X$ be arbitrarily fixed, and consider the sequence $x_n = f^n(x_0)$. For every $n, k \in \mathbb{N}$, define

$$D_n^k = \max \left\{ d(x_{i+k}, x_{j+k}) \, : \, 0 \leq i, \, j \leq n \right\}.$$

We proceed with some lemmas.

**Lemma 3.1** *For all integers $1 \leq i, \, j \leq n$ and every $k \in \mathbb{N}$ we have*

$$d(x_{k+i}, x_{k+j}) \leq \lambda D_n^k.$$

*Proof* Consider the inequality

$$\begin{aligned} d(x_{k+i}, x_{k+j}) &= d(f(x_{k+i-1}), \, f(x_{k+j-1})) \\ &\leq \lambda \max \left\{ d(x_{k+i-1}, x_{k+j-1}), \, d(x_{k+i-1}, x_{k+i}), \, d(x_{k+j-1}, x_{k+j}), \right. \\ &\qquad \left. d(x_{k+i-1}, x_{k+j}), \, d(x_{k+j-1}, x_{k+i}) \right\}. \end{aligned}$$

The claim follows from the very definition of $D_n^k$. □

Lemma 3.1 produces an immediate corollary.

**Corollary 3.1** *For every $n \geq 1$ and $k \in \mathbb{N}$ there is an integer $1 \leq q \leq n$ such that*

$$D_n^k = d(x_{k+q}, x_k).$$

**Lemma 3.2** *The estimate*

$$D_n^0 \leq \frac{1}{1 - \lambda} \, d(x_1, x_0)$$

*holds for every $n \geq 1$.*

*Proof* Let $n \geq 1$ be fixed. From Corollary 3.1 there is $1 \leq q \leq n$ such that

$$D_n^0 = d(x_q, x_0).$$

Applying the triangle inequality and Lemma 3.1 we get

$$d(x_q, x_0) \leq d(x_q, x_1) + d(x_1, x_0) \leq \lambda D_n^0 + d(x_1, x_0).$$

Therefore,

$$(1 - \lambda) D_n^0 \leq d(x_1, x_0),$$

which is what we wanted to prove. □

**Lemma 3.3** *For every $n \in \mathbb{N}$ and $m \geq 1$ we have*

$$d(x_{n+m}, x_n) \leq \lambda^n D_{n+m}^0.$$

*Proof* The case $n = 0$ is obviously true. Assume then $n > 0$. From Lemma 3.1 with $k = n - 1$ we get

$$d(x_{n+m}, x_n) = d(x_{n-1+1+m}, x_{n-1+1}) \leq \lambda D_{1+m}^{n-1}.$$

If $n = 1$ we are finished. Otherwise, by Corollary 3.1, there is $1 \leq q \leq 1 + m$ such that

$$D_{1+m}^{n-1} = d(x_{n-1+q}, x_{n-1}).$$

By a further use of Lemma 3.1, this time with $k = n - 2$,

$$d(x_{n-1+q}, x_{n-1}) = d(x_{n-2+1+q}, x_{n-2+1}) \leq \lambda D_{1+q}^{n-2}.$$

On the other hand, it is apparent that

$$D_{1+q}^{n-2} \leq D_{2+m}^{n-2}.$$

Putting all the pieces together, we learn that

$$d(x_{n+m}, x_n) \leq \lambda D_{1+m}^{n-1} \leq \lambda^2 D_{2+m}^{n-2}.$$

Again, if $n = 2$ we are finished. Otherwise we iterate the procedure, and we end up with the chain of inequalities

$$d(x_{n+m}, x_n) \leq \lambda D_{1+m}^{n-1} \leq \lambda^2 D_{2+m}^{n-2} \leq \ldots \leq \lambda^{n-1} D_{n-1+m}^1 \leq \lambda^n D_{n+m}^0,$$

as claimed. □

Collecting now Lemmas 3.2 and 3.3, we produce for every $n \in \mathbb{N}$ and $m \geq 1$ the estimate

$$d(x_{n+m}, x_n) \leq \frac{\lambda^n}{1 - \lambda} d(x_1, x_0).$$

This tells that $x_n$ is a Cauchy sequence, hence $x_n \to \bar{x}$, for some $\bar{x} \in X$. We are left to show that $\bar{x}$ is a fixed point. In that case, the desired convergence estimate follows by letting $m \to \infty$ in the inequality above. The argument here is a little bit more involved than in Theorem 1.1, since $f$ may fail to be continuous (as in the previous example). To this end, let us write

$$\begin{aligned} &d(\bar{x}, f(\bar{x})) \\ &\quad \le d(\bar{x}, x_{n+1}) + d(x_{n+1}, f(\bar{x})) \\ &\quad = d(\bar{x}, x_{n+1}) + d(f(x_n), f(\bar{x})) \\ &\quad \le d(\bar{x}, x_{n+1}) + \lambda \max\left\{d(x_n, \bar{x}),\ d(x_n, x_{n+1}),\ d(\bar{x}, f(\bar{x})),\ d(x_n, f(\bar{x})),\ d(\bar{x}, x_{n+1})\right\}. \end{aligned}$$

Letting $n \to \infty$, we readily obtain

$$d(\bar{x}, f(\bar{x})) \le \lambda d(\bar{x}, f(\bar{x})),$$

forcing the equality $\bar{x} = f(\bar{x})$. This finishes the proof of Theorem 3.3. □

*Remark* A quasi-contraction $f$, although possibly not continuous on the whole space $X$, is however continuous at the fixed point $\bar{x}$. Indeed, if $\bar{x} = f(\bar{x})$, for any $x \in X$ we infer from the definition of quasi-contraction that

$$d(f(x), f(\bar{x})) \le \lambda \max\left\{d(x, \bar{x}),\ d(x, f(x)),\ d(\bar{x}, f(x))\right\}.$$

Since $\lambda < 1$, we necessarily have

$$\begin{aligned} \max\left\{d(x, \bar{x}),\ d(x, f(x)),\ d(\bar{x}, f(x))\right\} &= \max\left\{d(x, \bar{x}),\ d(x, f(x))\right\} \\ &\le d(x, \bar{x}) + d(x, f(x)) \\ &\le d(x, \bar{x}) + d(x, \bar{x}) + d(f(x), \bar{x}) \\ &\le 2d(x, \bar{x}) + d(f(x), f(\bar{x})). \end{aligned}$$

Hence,

$$d(f(x), f(\bar{x})) \le \frac{2\lambda}{1-\lambda} d(x, \bar{x}),$$

implying the continuity of $f$ at $\bar{x}$.

We also have the analogue of Corollary 1.3 for quasi-contractions (the proof is identical).

**Corollary 3.2** *Let $(X, d)$ be a complete metric space, and let $f : X \to X$. If $f^m$ is a quasi-contraction for some $m \ge 1$, then $f$ has a unique fixed point $\bar{x} \in X$. Moreover, for every $x_0 \in X$, the sequence $f^n(x_0)$ converges to $\bar{x}$.*

# Chapter 4
# Weak Contractions




We now dwell on the case of maps on a metric space which are contractive without being contractions.

**Definition 4.1** Let $(X, d)$ be a metric space. A map $f : X \to X$ is a *weak contraction* if

$$d(f(x), f(y)) < d(x, y), \qquad \forall x \neq y \in X.$$

*Remark* If $f$ is a weak contraction, then it has *at most* one fixed point. This follows immediately by contradiction, exactly as in Theorem 1.1.

Nonetheless, being a weak contraction is not in general a sufficient condition for $f$ in order to have a fixed point, as shown in the following example.

**Example** Consider the complete metric space $X = [1, +\infty)$, and let $f : X \to X$ be defined as

$$f(x) = x + \frac{1}{x}.$$

It is immediate to check that $f$ is a weak contraction with no fixed points.

The condition turns out to be sufficient when $X$ is compact.

**Theorem 4.1** *Let $f$ be a weak contraction on a compact metric space $X$. Then $f$ has a unique fixed point $\bar{x} \in X$. Moreover, for any $x_0 \in X$, the sequence $f^n(x_0)$ converges to $\bar{x}$.*

*Proof* From the compactness of $X$, the continuous function $x \mapsto d(x, f(x))$ attains its minimum at some $\bar{x} \in X$. If $\bar{x} \neq f(\bar{x})$, we get

$$d(\bar{x}, f(\bar{x})) = \min_{x \in X} d(x, f(x)) \leq d(f(\bar{x}), f(f(\bar{x}))) < d(\bar{x}, f(\bar{x})),$$

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which is impossible. Thus $\bar{x}$ is the unique fixed point of $f$ (and so of $f^n$ for all $n \geq 2$). Let now $x_0 \in X$ with $x_0 \neq \bar{x}$ be given, and define

$$\delta_n = d(\bar{x}, f^n(x_0)).$$

Observe that

$$\delta_{n+1} = d(f(\bar{x}), f^{n+1}(x_0)) < d(\bar{x}, f^n(x_0)) = \delta_n.$$

Hence $\delta_n$ is strictly decreasing, and admits a limit $r \geq 0$. Let now $f^{n_k}(x_0)$ be a subsequence of $f^n(x_0)$ converging to some $z \in X$. Then

$$r = d(\bar{x}, z) = \lim_{k\to\infty} \delta_{n_k} = \lim_{k\to\infty} \delta_{n_k+1} = \lim_{k\to\infty} d(\bar{x}, f(f^{n_k}(x_0))) = d(\bar{x}, f(z)).$$

But if $z \neq \bar{x}$, then

$$d(\bar{x}, f(z)) = d(f(\bar{x}), f(z)) < d(\bar{x}, z),$$

a contradiction. Therefore any convergent subsequence of $f^n(x_0)$ has limit $\bar{x}$, which, along with the compactness of $X$, implies that $f^n(x_0)$ converges to $\bar{x}$. □

*Remark* The compactness of $X$ can be relaxed by requiring the compactness of the closure of $f(X)$. In this case, we just apply the theorem on the restriction of $f$ on $\overline{f(X)}$. Here we use the fact that, since $f$ is continuous on $X$, for every set $A \subset X$

$$f(\overline{A}) \subset \overline{f(A)}.$$

This condition is actually necessary and sufficient in order for $f$ to be continuous on $X$. Accordingly,

$$f(\overline{f(X)}) \subset \overline{f(f(X))} \subset \overline{f(X)},$$

ensuring that $f$ is a selfmap on $\overline{f(X)}$.

Arguing as in Corollary 1.3, it is also immediate to prove the following

**Corollary 4.1** *Let $X$ be a compact metric space and let $f : X \to X$. If $f^m$ is a weak contraction, for some $m \geq 1$, then $f$ has a unique fixed point $\bar{x} \in X$. Moreover, for any $x_0 \in X$, the sequence $f^n(x_0)$ converges to $\bar{x}$.*

Actually, when $X$ is a compact space, Theorem 4.1 turns out to be fully equivalent to the Boyd-Wong Theorem 2.1. Indeed, it is apparent that any function $f$ complying with the hypotheses of Theorem 2.1 is a weak contraction, for

$$d(f(x), f(y)) \leq \varphi(d(x, y)) < d(x, y) \qquad \forall x \neq y \in X.$$

If $X$ is compact, the converse holds as well [27]. Incidentally, this also provides an alternative proof of Theorem 4.1.

**Theorem 4.2** *Let $f$ be a weak contraction on a compact metric space $X$. Then there exists a strictly increasing continuous function $\varphi : [0, \infty) \to [0, \infty)$ such that*

$$\varphi(r) < r, \qquad \forall r > 0,$$

*and*

$$d(f(x), f(y)) \le \varphi(d(x, y)), \qquad \forall x, y \in X.$$

The proof requires a technical lemma.

**Lemma 4.1** *Let $\varrho : [0, 1] \to [0, 1]$ be a right continuous increasing function such that $\varrho(r) < r$ if $r > 0$. Then there exists a strictly increasing continuous function $\varphi : [0, 1] \to [0, 1]$ such that*

$$\varrho(r) \le \varphi(r) < r, \qquad \forall r > 0.$$

*Proof* We claim that, for every interval $[a, b]$ with $0 < a < b \le 1$,

$$\inf_{r \in [a,b]} \big[r - \varrho(r)\big] > 0.$$

Assume it is not so. Then there exists a sequence $r_n \in [a, b]$ such that

$$r_n - \varrho(r_n) \to 0.$$

By compactness, there exists $r \in [a, b]$ such that, up to a subsequence, $r_n \to r$. We first show that $r_n \ge r$ cannot occur infinitely many time. Otherwise, there would be a subsequence $r_{n_k} \to r^+$. But then, from the right continuity of $\varrho$, we would get the contradiction

$$0 = \lim_{k \to \infty} r_{n_k} - \varrho(r_{n_k}) = r - \varrho(r).$$

Accordingly, we may suppose without loss of generality that $r_n < r$ for all $n$. In this case, setting

$$\varepsilon = r - \varrho(r) > 0,$$

we can choose $m$ large enough that

$$r - r_m < \frac{\varepsilon}{2} \qquad \text{and} \qquad r_m - \varrho(r_m) < \frac{\varepsilon}{2}.$$

Accordingly

$$\varrho(r) = r - \varepsilon = r - r_m + r_m - \varrho(r_m) + \varrho(r_m) - \varepsilon < \varrho(r_m),$$

against the fact that $\varrho$ is increasing. The claim is proven.

Next, for $n = 0, 1, 2, \ldots$, we define $r_n = 2^{-n}$. In light of the previous step, we call

$$\varepsilon_n = \inf_{r \in [r_{n+1}, r_n]} \big[ r - \varrho(r) \big] > 0.$$

We now construct inductively $\hat{\varphi} : [0, 1] \to [0, 1]$ such that

$$\varrho(r) \le \hat{\varphi}(r) < r, \qquad \forall r > 0.$$

To this end, we consider the half-open intervals $I_n = (r_{n+1}, r_n]$. Setting

$$\omega_0 = \varepsilon_0 \le r_1,$$

we define

$$\hat{\varphi}(r) = r - \omega_0, \qquad \forall r \in I_0.$$

Assuming then $\hat{\varphi}$ given on $I_0, I_1, \ldots, I_n$, let

$$\hat{\varphi}(r) = r - \omega_{n+1}, \qquad \forall r \in I_{n+1},$$

where

$$\omega_{n+1} = \min\{\omega_n, \varepsilon_{n+1}\} \le r_{n+2}.$$

Finally, we take $\hat{\varphi}(0) = 0$. Such a $\hat{\varphi}$ is *almost* what we are looking for. Indeed, besides satisfying $\hat{\varphi}(r) < r$ for $r > 0$, it is strictly increasing on each $I_n$, but it has downward jumps at (possibly not all) the points $r_n$, for $n \ge 1$. Accordingly, we modify it in the following manner: we define

$$\varphi(r) = \begin{cases} q_n(r) & \text{if } r \in I_{n-1} \text{ and } r_n \text{ is a jump point of } \hat{\varphi}, \\ \hat{\varphi}(r) & \text{otherwise,} \end{cases}$$

where $q_n(r)$ is the segment of line connecting the points

$$(r_n, \hat{\varphi}(r_n)) \qquad \text{and} \qquad (r_{n-1}, \hat{\varphi}(r_{n-1})).$$

By construction, $\varphi$ lies above $\hat{\varphi}$ (hence above $\varrho$), and $\varphi(r) < r$ for $r > 0$. Moreover, it is clearly continuous and strictly increasing. Indeed, if $r_n$ is a jump point of $\hat{\varphi}$, we have that

$$q_n(r_n) = \hat{\varphi}(r_n) < r_n,$$

whereas

$$q_n(r_{n-1}) = \hat{\varphi}(r_{n-1}) \ge r_{n-1} - \varepsilon_{n-1} \ge r_n.$$

The proof is completed. □

We are now ready to prove Theorem 4.2.

*Proof* Since $X$ is compact, it is bounded, and without loss of generality we may assume $\operatorname{diam}(X) \leq 1$. Define the increasing function $\varrho : [0, 1] \to [0, 1]$ as

$$\varrho(r) = \sup_{d(x,y)\leq r} d(f(x), f(y)).$$

We preliminarily observe that, for every fixed $r > 0$, there exist $x, y \in X$ with $x \neq y$ such that

$$d(x, y) \leq r \qquad \text{and} \qquad \varrho(r) = d(f(x), f(y)).$$

Indeed, by the definition of $\varrho$, there are sequences $x_n, y_n \in X$ with $d(x_n, y_n) \leq r$ such that

$$d(f(x_n), f(y_n)) \to \varrho(r).$$

But then the compactness of $X$ implies the convergences (up to a subsequence) $x_n \to x$ and $y_n \to y$, for some $x, y \in X$. The next step is showing that

$$\varrho(r) < r, \qquad \forall r > 0.$$

To this end, for a fixed $r > 0$, we write

$$\varrho(r) = d(f(x), f(y)),$$

for some $x, y \in X$ with $d(x, y) \leq r$. Exploiting the fact that $f$ is weak contraction, we readily get

$$\varrho(r) = d(f(x), f(y)) < d(x, y) \leq r.$$

Finally, we show that $\varrho$ is right continuous. Let $r \geq 0$ be fixed, and let $\varepsilon_n \to 0^+$ be an arbitrary sequence. Then there exist sequences $x_n, y_n \in X$, with

$$d(x_n, y_n) \leq r + \varepsilon_n,$$

such that

$$\varrho(r + \varepsilon_n) = d(f(x_n), f(y_n)).$$

Invoking again the compactness of $X$, we have that (up to a subsequence) $x_n \to x$ and $y_n \to y$, for some $x, y \in X$. Hence, by the monotonicity of $\varrho$,

$$\varrho(r) \leq \varrho(r + \varepsilon_n) \to d(f(x), f(y)).$$

On the other hand,

$$r + \varepsilon_n \geq d(x_n, y_n) \to d(x, y),$$

implying the inequality

$$d(x, y) \leq r.$$

In turn, this means that

$$d(f(x), f(y)) \leq \varrho(r).$$

This forces the convergence

$$\varrho(r + \varepsilon_n) \to \varrho(r),$$

establishing the claimed right continuity of $\varrho$. Summarizing, the hypotheses of Lemma 4.1 are satisfied, so there exists a strictly increasing continuous function $\varphi : [0, 1] \to [0, 1]$ such that

$$\varrho(r) \leq \varphi(r) < r, \qquad \forall r > 0.$$

From the very definition of $\varrho$, it is clear that

$$d(f(x), f(y)) \leq \varphi(d(x, y)), \qquad \forall x, y \in X.$$

This $\varphi$ can be obviously extended on the whole $[0, \infty)$ by setting, say,

$$\varphi(r) = r - 1 + \varphi(1),$$

for $r > 1$. □

# Chapter 5
# Contractions of $\varepsilon$-Type




Let $(X, d)$ be a complete metric space. Selecting an arbitrary $x_0 \in X$, that we may call the zero of $X$, we denote

$$\|x\| = d(x, x_0), \qquad \forall x \in X.$$

Moreover, let $\Lambda \geq 0$, $\alpha \geq 1$ and $\beta \in [0, \alpha]$ be fixed constants, and let $\psi : [0, \alpha] \to [0, \infty)$ be an assigned increasing function vanishing with continuity at zero.

**Definition 5.1** A function $f : X \to X$ is said to be an *$\varepsilon$-contraction* if the inequality

$$d(f(x), f(y)) \leq (1 - \varepsilon)d(x, y) + \Lambda\varepsilon^{\alpha}\psi(\varepsilon)\big[1 + \|x\| + \|y\|\big]^{\beta} \tag{5.1}$$

is satisfied for every $\varepsilon \in [0, 1]$ and every $x, y \in X$.

Observe that taking $x_0$ rather than a generic $x \in X$ is not restrictive, since the validity of (5.1) is clearly independent of the particular choice of the zero of $X$, up to redefining $\Lambda$.

Defining the (vanishing) sequence depending on $\alpha \geq 1$ as

$$\omega_n(\alpha) = (\alpha/n)^{\alpha} \sum_{k=1}^{n} \psi(\alpha/k),$$

the following theorem holds [57].

**Theorem 5.1** *Let $f$ be an $\varepsilon$-contraction. Then $f$ possesses a unique fixed point $\bar{x}$. Furthermore, the iteration $f^n(x_0)$ fulfills the estimate*

$$d(\bar{x}, f^n(x_0)) \leq c\omega_n(\alpha) \tag{5.2}$$

*for some positive constant $c \leq \Lambda(1 + 4\|\bar{x}\|)^{\beta}$.*

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*Proof* (*Uniqueness*) This is a byproduct of $f$ being a weak contraction, namely,

$$d(f(x), f(y)) < d(x, y), \quad \forall x \neq y.$$

Indeed, for any two fixed $x, y \in X$, we can write (5.1) in the form

$$d(f(x), f(y)) \leq (1 - \varepsilon)d(x, y) + K\varepsilon\psi(\varepsilon), \qquad K > 0.$$

Setting $\varepsilon = 0$ we find

$$d(f(x), f(y)) \leq d(x, y).$$

If equality occurs, the relation

$$d(x, y) \leq K\psi(\varepsilon)$$

is valid for every $\varepsilon \in (0, 1]$, and letting $\varepsilon \to 0$ we get $d(x, y) = 0$. □

*Proof* (*Existence*) Starting from $x_0$, we introduce the sequences

$$x_n = f(x_{n-1}) = f^n(x_0) \qquad \text{and} \qquad c_n = \|x_n\|.$$

We need two lemmas.

**Lemma 5.1** *The sequence $c_n$ is bounded.*

*Proof* Exploiting the inequalities

$$d(x_{n+1}, x_n) \leq d(x_n, x_{n-1}) \leq \cdots \leq d(x_1, x_0) = c_1,$$

we deduce the bound

$$c_n \leq d(x_{n+1}, x_1) + 2c_1 = d(f(x_n), f(x_0)) + 2c_1.$$

Therefore, as $\beta \leq \alpha$, we infer from (5.1) that

$$c_n \leq (1 - \varepsilon)c_n + \Lambda\varepsilon^\alpha\psi(\varepsilon)[1 + c_n]^\beta + 2c_1 \leq (1 - \varepsilon)c_n + a\varepsilon^\alpha\psi(\varepsilon)c_n^\alpha + b$$

for some $a, b > 0$. Accordingly,

$$\varepsilon c_n \leq a\varepsilon^\alpha\psi(\varepsilon)c_n^\alpha + b.$$

If there is a subsequence $c_{n_\iota} \to \infty$, the choice $\varepsilon = \varepsilon_\iota = (1 + b)/c_{n_\iota}$ leads to

$$1 \leq a(1 + b)^\alpha\psi(\varepsilon_\iota) \to 0,$$

which is a contradiction. □

**Lemma 5.2** *For every $n, m \in \mathbb{N}$, we have the inequality*

$$d(x_{n+m}, x_n) \leq c\omega_n(\alpha),$$

*where $c = \sup_n \Lambda(1 + 2c_n)^\beta$, which is finite by virtue of Lemma 5.1.*

*Proof* Let $m$ be fixed. In light of (5.1), the sequence $p_n = n^\alpha d(x_{n+m}, x_n)$ fulfills

$$\begin{aligned} p_{n+1} &= (n+1)^\alpha d(f(x_{n+m}), f(x_n)) \\ &\leq (n+1)^\alpha (1-\varepsilon) d(x_{n+m}, x_n) + c(n+1)^\alpha \varepsilon^\alpha \psi(\varepsilon). \end{aligned}$$

Choosing at each $n$

$$\varepsilon = 1 - \Big(\frac{n}{n+1}\Big)^\alpha \leq \frac{\alpha}{n+1},$$

we end up with

$$p_{n+1} \leq p_n + c\alpha^\alpha \psi\big(\tfrac{\alpha}{n+1}\big).$$

Since $p_0 = 0$, this gives

$$p_n \leq c\alpha^\alpha \sum_{k=1}^{n} \psi(\alpha/k),$$

and a final division by $n^\alpha$ will do. □

Lemma 5.2 tells in particular that $x_n$ is a Cauchy sequence, so having limit $\bar{x} \in X$. Then, by a further application of Lemma 5.2,

$$d(\bar{x}, f^n(x_0)) = d(\bar{x}, f(x_{n-1})) = d(\bar{x}, x_n) = \lim_{m\to\infty} d(x_{n+m}, x_n) \leq c\omega_n(\alpha).$$

Due to the continuity of $f$, letting $n \to \infty$ we establish both the equality $\bar{x} = f(\bar{x})$ and the convergence rate estimate (5.2). We are left to show that $c$ fulfills the required bound. To this end, knowing now that $\bar{x}$ is a fixed point, we write

$$d(\bar{x}, x_n) = d(f(\bar{x}), f(x_{n-1})) \leq d(\bar{x}, x_{n-1}) \leq \cdots \leq d(\bar{x}, x_0) = \|\bar{x}\|,$$

which entails

$$c_n \leq \|\bar{x}\| + d(\bar{x}, x_n) \leq 2\|\bar{x}\|.$$

This finishes the proof of Theorem 5.1. □

If the function $\psi$ is a power, then the convergence of the iterated sequence to the fixed point occurs polynomially fast.

**Corollary 5.1** *Let $f$ be an $\varepsilon$-contraction. If*

$$\psi(\varepsilon) = \varepsilon^\gamma,$$

*for some $\gamma > 0$, setting $\vartheta = \alpha + \gamma$ the convergence rate estimate (5.2) becomes*

$$d(\bar{x},\, f^n(x_0)) \le \frac{c\vartheta^\vartheta}{n^{\vartheta-1}}. \tag{5.3}$$

*Proof* For $\vartheta = \alpha + \gamma$, it is readily seen that (5.1) remains valid with $\vartheta - \nu$ and $\nu$ in place of $\alpha$ and $\gamma$, respectively, with $\nu > 0$ arbitrarily small. In which case,

$$\omega_n(\vartheta - \nu) = \frac{(\vartheta-\nu)^\vartheta}{n^{\vartheta-\nu}} \sum_{k=1}^{n} \frac{1}{k^\nu} \le \frac{(\vartheta-\nu)^\vartheta}{n^{\vartheta-\nu-1}},$$

and Theorem 5.1 together with a final limit $\nu \to 0$ entail the desired conclusion. □

The hypotheses of Theorem 5.1 are weaker than those of the BCP. Indeed, the following proposition holds.

**Proposition 5.1** *Assume that $d(f(x),\, f(y)) \le \lambda d(x, y)$ for some $\lambda < 1$ and every $x, y \in X$. Then, for every $\gamma > 0$,*

$$d(f(x),\, f(y)) \le (1-\varepsilon)d(x, y) + \Lambda\varepsilon^{1+\gamma}\big[1 + \|x\| + \|y\|\big], \tag{5.4}$$

*where*

$$\Lambda = \Lambda(\gamma, \lambda) = \frac{\gamma^\gamma}{(1+\gamma)^{1+\gamma}} \frac{1}{(1-\lambda)^\gamma}.$$

*Proof* For any fixed $\gamma$, all we need is finding $\Lambda$ for which the inequality

$$\lambda \le (1-\varepsilon) + \Lambda\varepsilon^{1+\gamma}$$

holds for all $\varepsilon \in [0, 1]$. To this end, consider the function

$$F(\varepsilon) = (1-\varepsilon) + \Lambda\varepsilon^{1+\gamma} - \lambda, \qquad \varepsilon \ge 0,$$

which attains the minimum at $\varepsilon = \varepsilon_0$, with

$$\varepsilon_0 = \left[\frac{1}{(1+\gamma)\Lambda}\right]^{1/\gamma}.$$

Imposing the condition $F(\varepsilon_0) = 0$ one finds the value $\Lambda = \Lambda(\gamma, \lambda)$ above. □

*Remark* It is interesting to note that (5.4) provides the exponential decay of the sequence $d(\bar{x},\, f^n(x_0))$. Indeed, for every $\gamma > 0$ we deduce from Corollary 5.1 with $\vartheta = 1 + \gamma$ the estimate

$$d(\bar{x},\, f^n(x_0)) \le M\left[\frac{\gamma}{n(1-\lambda)}\right]^\gamma,$$

having set $M = 1 + 4\|\bar{x}\|$. Choosing then $\gamma = n\tau$, for $\tau > 0$ to be fixed, we get

$$d(\bar{x}, f^n(x_0)) \le M q_\tau^n \qquad \text{with} \qquad q_\tau = \left[\frac{\tau}{1-\lambda}\right]^\tau.$$

The function $\tau \mapsto q_\tau$ attains its minimum at the point $\tau_{\min} = (1-\lambda)e^{-1}$, where

$$q_{\tau_{\min}} = e^{-(1-\lambda)e^{-1}} < 1.$$

We conclude that

$$d(\bar{x}, f^n(x_0)) \le M q_{\tau_{\min}}^n.$$

Let aside the explicit convergence rate of $d(f^n(x_0), \bar{x})$, but limited to the existence and uniqueness of the fixed point, we have the equivalence of Theorem 5.1 and the Boyd-Wong Theorem 2.1 when $X$ is bounded.

**Proposition 5.2** *If $X$ is a bounded space, Theorems 5.1 and 2.1 imply each other.*

*Proof* Without loss of generality, we may assume $\operatorname{diam}(X) \le 1$. Up to redefining $\Lambda$, when $X$ is bounded the inequality (5.1) becomes

$$d(f(x), f(y)) \le (1-\varepsilon)d(x, y) + \Lambda \varepsilon^\alpha \psi(\varepsilon), \tag{5.5}$$

whereas we recall that the condition of Theorem 2.1 reads

$$d(f(x), f(y)) \le \varphi(d(x, y)). \tag{5.6}$$

We first prove the implication (5.5) $\Rightarrow$ (5.6). Without loss of generality, we can take $\psi$ strictly increasing and continuous. Defining then $\zeta(r) = \psi^{-1}(\nu r)$ with $\nu > 0$ suitably small, and choosing $\varepsilon = \zeta(d(x, y))$ in (5.5) we recover (5.6) with

$$\varphi(r) = r - r\zeta(r)\Big[1 - \nu\Lambda[\zeta(r)]^{\alpha-1}\Big].$$

Conversely, suppose that (5.6) holds. Since $\varphi$ is continuous and $\varphi(r) < r$ on $(0, 1]$, we can easily construct a strictly increasing continuous function $\mu$ such that $\mu(0) = 0$ and

$$\mu(r) \le 1 - \frac{\varphi(r)}{r}, \qquad \forall r \in (0, 1].$$

Let $x, y \in X$ with $d(x, y) = r \in (0, 1]$ be given. By virtue of (5.6),

$$d(f(x), f(y)) \le \varphi(r) \le r - r\mu(r) < r.$$

Fix now any $\varepsilon \in [0, 1]$. If $\mu(r) \geq \varepsilon$, we readily obtain

$$d(f(x), f(y)) \leq (1 - \varepsilon)r.$$

If $\mu(r) < \varepsilon \leq \mu(1)$, we get

$$d(f(x), f(y)) < (1 - \varepsilon)r + \varepsilon r < (1 - \varepsilon)r + \varepsilon\mu^{-1}(\varepsilon),$$

whereas if $\mu(1) < \varepsilon$, we simply write

$$d(f(x), f(y)) < (1 - \varepsilon)r + \varepsilon.$$

Collecting the three inequalities, (5.5) follows for $\Lambda = 1$, $\alpha = 1$, and $\psi(\varepsilon) = \mu^{-1}(\varepsilon)$ if $\varepsilon \leq \mu(1)$, and $\psi(\varepsilon) = 1$ otherwise. □

*Remark* In fact, assuming $X$ bounded in the statement of Theorem 2.1 is really no loss of generality. To see that, let us fix an arbitrary $z \in X$. Arguing exactly as in the proof of Theorem 2.1, we have the convergence

$$d(f^n(z), f^{n-1}(z)) \to 0.$$

Then we can select $x = f^n(z)$ with $n$ large enough that

$$d(f(x), x) \leq 1 - \varkappa,$$

having set

$$\varkappa = \sup_{r \leq 1} \varphi(r) < 1.$$

The fact that $\varkappa < 1$ is an immediate consequence of the strict inequality $\varphi(r) < r$ for $r > 0$. Indeed, if $\varkappa = 1$ there is a positive sequence $r_n \leq 1$ such that $\varphi(r_n) \to 1$. Then, up to a subsequence, $r_n \to r \leq 1$, in turn implying $\varphi(r) = 1$, which is impossible. At this point, denoting

$$B = \big\{y \in X : d(y, x) \leq 1\big\},$$

we observe that $f(B) \subset B$, meaning that the whole space $X$ can be replaced by the unit ball $B$. Indeed, for any fixed $y \in B$, we have

$$d(f(y), x) \leq d(f(y), f(x)) + d(f(x), x) \leq \varphi(d(y, x)) + 1 - \varkappa \leq 1.$$

On the contrary, when $X$ is unbounded, there are situations where the simultaneous occurrences of $d(x, y) \ll 1$ and $\|x\| + \|y\| \gg 1$ can be handled by Theorem 5.1 but not by Theorem 2.1.

**Example** The function $f : [1, \infty) \to [1, \infty)$ defined as

$$f(x) = -2 + x - 2\sqrt{x} + 4\sqrt[4]{x}$$

has a unique fixed point $\bar{x} = 1$ but fails to be a strict contraction on any neighborhood both of $\bar{x}$ and of $\infty$. For any given $r > 0$ and $x \geq 1$, setting

$$F(x, r) = 2\big[\sqrt{x + r} - \sqrt{x}\,\big] - 4\big[\sqrt[4]{x + r} - \sqrt[4]{x}\,\big],$$

we have the identity

$$|f(x + r) - f(x)| = r - F(x, r).$$

• Assume there is $\varphi$ as in Theorem 2.1. Choosing any $r > 0$, we find the contradiction

$$r > \varphi(r) \geq \lim_{x \to \infty} \big[r - F(x, r)\big] = r.$$

• For every $\varepsilon \in [0, 1]$,

$$\varepsilon r - \varepsilon^2 (r + 2x)^{3/2} \leq \frac{r^2}{4(r + 2x)^{3/2}}.$$

Besides, we claim that

$$\frac{r^2}{4(r + 2x)^{3/2}} \leq F(x, r).$$

Indeed, setting $y = r/x$, the inequality above can be written as

$$4\Big(\sqrt[4]{1 + y} - 1\Big) \leq \sqrt[4]{x}\left[2\Big(\sqrt{1 + y} - 1\Big) - \frac{y^2}{4(y + 2)^{3/2}}\right].$$

Since $x \geq 1$, this is implied by

$$4\Big(\sqrt[4]{1 + y} - 1\Big) \leq \left[2\Big(\sqrt{1 + y} - 1\Big) - \frac{y^2}{4(y + 2)^{3/2}}\right],$$

which is equivalent to

$$\frac{y}{\sqrt{8}(y + 2)^{3/4}} \leq \sqrt[4]{1 + y} - 1.$$

By a standard (although quite cumbersome) calculations, one can show that the latter inequality holds for all $y \geq 0$. Therefore, we conclude that

$$|f(x+r) - f(x)| = r - F(x,r) \leq (1-\varepsilon)r + \varepsilon^2(r+2x)^{3/2},$$

and Theorem 5.1 applies. By Corollary 5.1, we also deduce (5.3) with $\vartheta = 2$. Incidentally, since

$$f(1+r) = 1 + r - \frac{1}{8}r^2 + o(r^2),$$

the predicted convergence rate $1/n$ turns out to be optimal (to see that, argue exactly as in the proof of Theorem 2.2).

# Chapter 6
# Sequences of Maps and Fixed Points




Let $(X, d)$ be a complete metric space. We consider the problem of convergence of fixed points for a sequence of maps $f_n : X \to X$. Corollary 1.3 will be implicitly used in the statements of the next two theorems.

**Theorem 6.1** *Assume that each $f_n$ has at least a fixed point $\bar{x}_n = f_n(\bar{x}_n)$. Let $f : X \to X$ be a uniformly continuous map such that $f^m$ is a contraction for some $m \geq 1$. If $f_n$ converges uniformly to $f$, then $\bar{x}_n$ converges to $\bar{x} = f(\bar{x})$.*

*Proof* We first assume that $f$ is a contraction (i.e., $m = 1$). Let $\lambda < 1$ be the Lipschitz constant of $f$. Given $\varepsilon > 0$, choose $n_0 = n_0(\varepsilon)$ such that

$$d(f_n(x), f(x)) \leq \varepsilon(1 - \lambda), \qquad \forall n \geq n_0, \forall x \in X.$$

Then, for $n \geq n_0$,

$$\begin{aligned} d(\bar{x}_n, \bar{x}) &= d(f_n(\bar{x}_n), f(\bar{x})) \\ &\leq d(f_n(\bar{x}_n), f(\bar{x}_n)) + d(f(\bar{x}_n), f(\bar{x})) \\ &\leq \varepsilon(1 - \lambda) + \lambda d(\bar{x}_n, \bar{x}). \end{aligned}$$

Therefore $d(\bar{x}_n, \bar{x}) \leq \varepsilon$, which proves the convergence.

To prove the general case it is enough to observe that if

$$d(f^m(x), f^m(y)) \leq \lambda^m d(x, y),$$

for some $\lambda < 1$, then we can define a new metric $\hat{d}$ on $X$ equivalent to $d$ (i.e., inducing the same topology on $X$) by setting

$$\hat{d}(x, y) = \sum_{k=0}^{m-1} \frac{1}{\lambda^k} d(f^k(x), f^k(y)).$$

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Moreover, since $f$ is uniformly continuous, $f_n$ converges uniformly to $f$ also with respect to $\hat{d}$. Finally, $f$ is a contraction with respect to $\hat{d}$. Indeed,

$$\begin{aligned}\hat{d}(f(x), f(y)) &= \sum_{k=0}^{m-1} \frac{1}{\lambda^k} d(f^{k+1}(x), f^{k+1}(y)) \\ &= \lambda \sum_{k=1}^{m-1} \frac{1}{\lambda^k} d(f^k(x), f^k(y)) + \frac{1}{\lambda^{m-1}} d(f^m(x), f^m(y)) \\ &\le \lambda \sum_{k=0}^{m-1} \frac{1}{\lambda^k} d(f^k(x), f^k(y)) = \lambda \hat{d}(x, y).\end{aligned}$$

So the problem is reduced to the previous case $m = 1$. □

The next result refers to a special class of complete metric spaces.

**Theorem 6.2** *Let $X$ be locally compact. Assume that for each $n \in \mathbb{N}$ there is $m_n \ge 1$ such that $f_n^{m_n}$ is a contraction. Let $f : X \to X$ be a map such that $f^m$ is a contraction for some $m \ge 1$. If $f_n$ converges pointwise to $f$, and $f_n$ is an equicontinuous family, then $\bar{x}_n = f_n(\bar{x}_n)$ converges to $\bar{x} = f(\bar{x})$.*

*Proof* Let $\varepsilon > 0$ be sufficiently small such that

$$K(\bar{x}, \varepsilon) \doteq \left\{x \in X : d(x, \bar{x}) \le \varepsilon\right\} \subset X$$

is compact. As a byproduct of the Ascoli theorem, $f_n$ converges to $f$ uniformly on $K(\bar{x}, \varepsilon)$, since it is equicontinuous and pointwise convergent. Choose $n_0 = n_0(\varepsilon)$ such that

$$d(f_n^m(x), f^m(x)) \le \varepsilon(1 - \lambda), \qquad \forall n \ge n_0,\ \forall x \in K(\bar{x}, \varepsilon),$$

where $\lambda < 1$ is the Lipschitz constant of $f^m$. Then, for $n \ge n_0$ and $x \in K(\bar{x}, \varepsilon)$ we have

$$\begin{aligned}d(f_n^m(x), \bar{x}) &= d(f_n^m(x), f^m(\bar{x})) \\ &\le d(f_n^m(x), f^m(x)) + d(f^m(x), f^m(\bar{x})) \\ &\le \varepsilon(1 - \lambda) + \lambda d(x, \bar{x}) \le \varepsilon.\end{aligned}$$

Hence $f_n^m(K(\bar{x}, \varepsilon)) \subset K(\bar{x}, \varepsilon)$ for all $n \ge n_0$. Since the maps $f_n^{m_n}$ are contractions, it follows that, for $n \ge n_0$, the fixed points $\bar{x}_n$ of $f_n$ belong to $K(\bar{x}, \varepsilon)$, that is, $d(\bar{x}_n, \bar{x}) \le \varepsilon$. □

Theorems 6.1 and 6.2 are due to Nadler [54] and Fraser and Nadler [28]. Uniform convergence in Theorem 6.1 cannot in general be replaced by pointwise convergence. For instance, in every infinite-dimensional separable or reflexive Banach space there

is a pointwise convergent sequence of contractions whose sequence of fixed points has no convergent subsequences (see, e.g., [36], Chap. 7). The following is an example along this direction for the (separable) Banach space $\ell^1$ of summable sequences.

**Example** Consider the Banach space $\ell^1$ normed by

$$\|x\| = \sum_{n=0}^{\infty} |x_n|, \qquad x = (x_0, x_1, x_2, \ldots).$$

For every $n \geq 1$, define the map $f_n : \ell^1 \to \ell^1$ as

$$f_n(x_0, \ldots, x_{n-1}, x_n, x_{n+1}, \ldots) = (0, \ldots, 0, (1 - 1/n)x_n + 1/n, 0, \ldots).$$

Every $f_n$ is a contraction with Lipschitz constant $\lambda_n = (1 - 1/n)$, with a unique fixed point $\bar{x}^n$ given by

$$\bar{x}^n = (0, \ldots, 0, 1, 0, \ldots).$$

For every $x \in \ell^1$, we have the convergence $f_n(x) \to f(x)$, where $f \equiv 0$ (however, $f_n \not\to f$ uniformly). Besides, for every $n, m \geq 1$ with $n \neq m$,

$$\|\bar{x}^n - \bar{x}^m\| = 2.$$

This implies that $\bar{x}^n$ has no convergent subsequences, and in particular $\bar{x}^n \not\to 0$, which is the unique fixed point of $f$.

# Chapter 7
# Fixed Points of Non-expansive Maps




Let $X$ be a Banach space, $C \subset X$ nonempty, closed, bounded and convex, and let $f : C \to C$ be a non-expansive map, namely, such that

$$d(f(x), f(y)) \leq d(x, y), \qquad \forall x, y \in C.$$

The problem is whether $f$ admits a fixed point in $C$. The answer, in general, is false.

**Example** Consider the Banach space $c_0$ of sequences vanishing at infinity, normed by

$$\|x\| = \max_{n \geq 0} |x_n|, \qquad x = (x_0, x_1, x_2, \ldots).$$

Setting $C = \overline{B}_{c_0}(0, 1)$, define the map $f : C \to C$ by

$$f(x) = (1, x_0, x_1, \ldots).$$

Then $f$ is non-expansive, it but clearly admits no fixed points in $C$.

Things are quite different in uniformly convex Banach spaces.

**Theorem 7.1** (Browder-Kirk) *Let $X$ be a uniformly convex Banach space and $C \subset X$ be nonempty, closed, bounded and convex. If $f : C \to C$ is a non-expansive map, then $f$ has a fixed point in $C$.*

Theorem 7.1 has been proved by Kirk [42], for the more general case when $X$ is a reflexive Banach spaces and $C$ has normal structure, and by Browder [11]. Here, we provide the proof in the particular case when $X$ is a Hilbert space, taken from a paper of Browder and Petryshyn [12] (see also [36], Chap. 6.4).

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*Proof* Let $x_* \in C$ be fixed, and consider a sequence $r_n \in (0, 1)$ converging to 1. For each $n \in \mathbb{N}$, define the map $f_n : C \to C$ as

$$f_n(x) = r_n f(x) + (1 - r_n)x_*.$$

Observe that $f_n$ is a contraction on $C$, hence there is a unique $x_n \in C$ such that $f_n(x_n) = x_n$. Since $C$ is weakly compact, $x_n$ has a subsequence (still denoted by $x_n$) weakly convergent to some $\bar{x} \in C$. We shall prove that $\bar{x}$ is a fixed point of $f$. Note first that

$$\lim_{n\to\infty} \left( \|f(\bar{x}) - x_n\|^2 - \|\bar{x} - x_n\|^2 \right) = \|f(\bar{x}) - \bar{x}\|^2.$$

Since $f$ is non-expansive we have

$$\begin{aligned} \|f(\bar{x}) - x_n\| &\le \|f(\bar{x}) - f(x_n)\| + \|f(x_n) - x_n\| \\ &\le \|\bar{x} - x_n\| + \|f(x_n) - x_n\| \\ &= \|\bar{x} - x_n\| + (1 - r_n)\|f(x_n) - x_*\|. \end{aligned}$$

But $r_n \to 1$ as $n \to \infty$ and $C$ is bounded, so we conclude that

$$\limsup_{n\to\infty} \left( \|f(\bar{x}) - x_n\|^2 - \|\bar{x} - x_n\|^2 \right) \le 0,$$

which yields the equality $f(\bar{x}) = \bar{x}$. □

**Proposition 7.1** *In the hypotheses of Theorem 7.1, the set $\mathcal{F}$ of fixed points of $f$ is closed and convex.*

*Proof* The first assertion is trivial. Assume then $x_0, x_1 \in \mathcal{F}$, with $x_0 \neq x_1$, and denote $x_t = (1 - t)x_0 + tx_1$, with $t \in (0, 1)$. We have

$$\begin{aligned} \|f(x_t) - x_0\| &= \|f(x_t) - f(x_0)\| \le \|x_t - x_0\| = t\|x_1 - x_0\|, \\ \|f(x_t) - x_1\| &= \|f(x_t) - f(x_1)\| \le \|x_t - x_1\| = (1 - t)\|x_1 - x_0\|, \end{aligned}$$

that imply the equalities

$$\begin{aligned} \|f(x_t) - x_0\| &= t\|x_1 - x_0\|, \\ \|f(x_t) - x_1\| &= (1 - t)\|x_1 - x_0\|. \end{aligned}$$

The proof is completed if we show that $f(x_t) = (1 - t)x_0 + tx_1$. This follows from a general fact about uniform convexity, which is recalled in the next lemma. □

**Lemma 7.1** *Let $X$ be a uniformly convex Banach space, and let $z, x, y \in X$ be such that*

$$\|z - x\| = t\|x - y\|, \qquad \|z - y\| = (1 - t)\|x - y\|,$$

*for some* $t \in [0, 1]$. *Then* $z = (1 - t)x + ty$.

*Proof* Without loss of generality, we can assume $t \geq 1/2$. We have

$$\begin{aligned}\|(1-t)(z-x) - t(z-y)\| &= \|(1-2t)(z-x) - t(x-y)\| \\ &\geq t\|x-y\| + (1-2t)\|z-x\| \\ &= 2t(1-t)\|x-y\|.\end{aligned}$$

Since the reverse inequality holds as well, and

$$(1-t)\|z-x\| = t\|z-y\| = t(1-t)\|x-y\|,$$

from the uniform convexity of $X$ (but strict convexity would suffice) we get

$$\|z - (1-t)x - ty\| = \|(1-t)(z-x) + t(z-y)\| = 0,$$

as claimed. □

# Chapter 8
# The Riesz Mean Ergodic Theorem




If $T$ is a non-expansive linear map of a uniformly convex Banach space, then *all* the fixed points of $T$ are recovered by means of a limit procedure.

**Definition 8.1** Let $X$ be a vector space. A linear operator $P : X \to X$ is called a *projection* in $X$ if

$$P^2x = PPx = Px, \qquad \forall x \in X.$$

It is easy to check that $P$ is the identity operator on $\mathrm{ran}(P)$ and

$$\ker(P) = \mathrm{ran}(I - P), \qquad \mathrm{ran}(P) = \ker(I - P), \qquad \ker(P) \cap \mathrm{ran}(P) = \{0\}.$$

Moreover, every $x \in X$ admits a unique decomposition $x = y + z$ with $y \in \ker(P)$ and $z \in \mathrm{ran}(P)$.

**Proposition 8.1** *If $X$ is a Banach space, then a projection $P$ is continuous if and only if $X = \ker(P) \oplus \mathrm{ran}(P)$.*

The notation $X = Y \oplus Z$ is used to mean that $Y$ and $Z$ are closed subspaces of $X$ such that $Y \cap Z = \{0\}$ and $Y + Z = X$.

*Proof* If $P$ is continuous, so is $I - P$. Hence $\ker(P)$ and $\mathrm{ran}(P) = \ker(I - P)$ are closed. Conversely, let $x_n \to x$, and $Px_n \to y$. Since $\mathrm{ran}(P)$ is closed, it follows that $y \in \mathrm{ran}(P)$, and therefore $Py = y$. But $Px_n - x_n \in \ker(P)$, and $\ker(P)$ is closed. So we have $x - y \in \ker(P)$, which implies $Py = Px$. Summarizing, we learned that $P$ is a closed operator, and the closed graph theorem implies that $P$ is continuous. □

**Theorem 8.1** (F. Riesz) *Let $X$ be a uniformly convex Banach space. Let $T : X \to X$ be a linear operator such that*

$$\|Tx\| \le \|x\|, \qquad \forall x \in X.$$

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*Then for every $x \in X$ the limit*

$$p_x = \lim_{n\to\infty} \frac{x + Tx + \cdots + T^n x}{n+1}$$

*exists. Moreover, the operator $P : X \to X$ defined by $Px = p_x$ is a continuous projection onto the vector space $\mathcal{M} = \{y \in X : Ty = y\}$.*

*Proof* Fix $x \in X$, and set

$$C = \overline{\text{co}}\big(\{x, Tx, T^2x, T^3x, \dots\}\big).$$

$C$ is a closed nonempty convex set, and from the uniform convexity of $X$ there is a unique $p_x \in C$ such that

$$\mu \doteq \|p_x\| = \inf\{\|z\| : z \in C\}.$$

Select $\varepsilon > 0$. Then, for $p_x \in C$, there exist $m \in \mathbb{N}$ and nonnegative constants $\alpha_0, \alpha_1, \dots, \alpha_m$ with $\sum_{j=0}^m \alpha_j = 1$ such that, setting

$$z = \sum_{j=0}^m \alpha_j T^j x,$$

we have

$$\|p_x - z\| < \varepsilon.$$

In particular, for every $n \in \mathbb{N}$,

$$\left\| \frac{z + Tz + \cdots + T^n z}{n+1} \right\| \le \|z\| < \mu + \varepsilon.$$

Note that

$$\begin{aligned} z + Tz + \cdots + T^n z = {} & (\alpha_0 x + \cdots + \alpha_m T^m x) + (\alpha_0 Tx + \cdots + \alpha_m T^{m+1} x) \\ & + \cdots + (\alpha_0 T^n x + \cdots + \alpha_m T^{m+n} x). \end{aligned}$$

Thus, assuming $n \gg m$, we get

$$z + Tz + \cdots + T^n z = x + Tx + \cdots + T^n x + r,$$

where

$$\begin{aligned} r = {} & (\alpha_0 - 1)x + \cdots + (\alpha_0 + \alpha_1 + \cdots + \alpha_{m-1} - 1)T^{m-1}x \\ & + (1 - \alpha_0)T^{1+n}x + \cdots + (1 - \alpha_0 - \alpha_1 - \cdots - \alpha_{m-1})T^{m+n}x. \end{aligned}$$

Therefore

$$\frac{x + Tx + \cdots + T^n x}{n+1} = \frac{z + Tz + \cdots + T^n z}{n+1} - \frac{r}{n+1}.$$

Since

$$\left\| \frac{r}{n+1} \right\| \leq \frac{2m\|x\|}{n+1},$$

upon choosing $n$ enough large such that $2m\|x\| < \varepsilon(n+1)$ we have

$$\left\| \frac{x + Tx + \cdots + T^n x}{n+1} \right\| \leq \left\| \frac{z + Tz + \cdots + T^n z}{n+1} \right\| + \left\| \frac{r}{n+1} \right\| < \mu + 2\varepsilon.$$

On the other hand, it must be

$$\left\| \frac{x + Tx + \cdots + T^n x}{n+1} \right\| \geq \mu.$$

Then,

$$\mu \leq \liminf_{n\to\infty} \left\| \frac{x + Tx + \cdots + T^n x}{n+1} \right\| \leq \limsup_{n\to\infty} \left\| \frac{x + Tx + \cdots + T^n x}{n+1} \right\| < \mu + 2\varepsilon,$$

and from the arbitrariness of $\varepsilon$ we conclude that

$$\lim_{n\to\infty} \left\| \frac{x + Tx + \cdots + T^n x}{n+1} \right\| = \mu.$$

This says that the above is a minimizing sequence in $C$, and due to the uniform convexity of $X$, we gain the convergence

$$\lim_{n\to\infty} \frac{x + Tx + \cdots + T^n x}{n+1} = p_x.$$

We are left to show that the operator $Px = p_x$ is a continuous projection onto $\mathcal{M}$. Indeed, it is apparent that if $x \in \mathcal{M}$ then $p_x = x$. In general,

$$\begin{aligned} Tp_x &= \lim_{n\to\infty} \frac{Tx + T^2 x + \cdots + T^{n+1} x}{n+1} \\ &= p_x + \lim_{n\to\infty} \frac{T^{n+1} x - x}{n+1} = p_x. \end{aligned}$$

Finally,

$$P^2 x = PPx = Pp_x = p_x = Px.$$

The continuity is ensured by the relation $\|p_x\| \leq \|x\|$. □

Theorem 8.1 can be found in [63] (see also [64]).

When $X$ is a Hilbert space, $P$ is actually an orthogonal projection. This follows from the next proposition.

**Proposition 8.2** *Let $H$ be a Hilbert space, $P = P^2 : H \to H$ a bounded linear operator with $\|P\| \leq 1$. Then $P$ is an orthogonal projection.*

*Proof* Since $P$ is continuous, $\mathrm{ran}(P)$ is closed. Let then $E$ be the orthogonal projection having range $\mathrm{ran}(P)$. Then

$$P = E + P(I - E).$$

Let now $x \in \mathrm{ran}(P)^{\perp}$. For any $\varepsilon > 0$ we have

$$\|P(Px + \varepsilon x)\| \leq \|Px + \varepsilon x\|,$$

which implies that

$$\|Px\|^2 \leq \frac{\varepsilon}{2+\varepsilon}\|x\|^2.$$

Hence $Px = 0$, and the equality $P = E$ holds. □

The role played by uniform convexity in Theorem 8.1 is essential, as the following example shows.

**Example** Consider the Banach space $\ell^{\infty}$ of bounded sequences, normed by

$$\|x\| = \sup_{n \geq 0} |x_n|, \qquad x = (x_0, x_1, x_2, \ldots).$$

Let $T \in L(\ell^{\infty})$ defined by

$$Tx = (0, x_0, x_1, \ldots).$$

Then $T$ has a unique fixed point, namely, the zero element of $\ell^{\infty}$. Nonetheless, if $y = (1, 1, 1, \ldots)$, for every $n \in \mathbb{N}$ we have

$$\left\| \frac{y + Ty + \cdots + T^n y}{n+1} \right\| = \frac{\|(1, 2, \ldots, n, n+1, n+1, \ldots)\|}{n+1} = 1.$$

# Chapter 9
# The Brouwer Fixed Point Theorem




For every integer $n \geq 1$, we denote the $n$-dimensional unit disk by

$$\mathbb{D}^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\},$$

and the unit sphere of $\mathbb{R}^n$, which has dimension $n-1$, by

$$\mathbb{S}^{n-1} = \partial \mathbb{D}^n = \left\{x \in \mathbb{R}^n : \|x\| = 1\right\}.$$

**Definition 9.1** A subset $E \subset \mathbb{D}^n$ is called a *retract* of $\mathbb{D}^n$ if there exists a continuous map $r : \mathbb{D}^n \to E$, called *retraction*, such that $r(x) = x$ for every $x \in E$.

**Lemma 9.1** *The set $\mathbb{S}^{n-1}$ is not a retract of $\mathbb{D}^n$.*

The lemma can be easily proved by means of algebraic topology tools. Indeed, a retraction $r$ induces a homomorphism

$$r_* : H_{n-1}(\mathbb{D}^n) \to H_{n-1}(\mathbb{S}^{n-1}),$$

where $H_{n-1}$ denotes the $(n-1)$-dimensional *homology group* (see, e.g., [51]). The natural injection $j : \mathbb{S}^{n-1} \to \mathbb{D}^n$ induces in turn a homomorphism

$$j_* : H_{n-1}(\mathbb{S}^{n-1}) \to H_{n-1}(\mathbb{D}^n),$$

and the composition $rj$ is the identity map on $\mathbb{S}^{n-1}$. Hence $(rj)_* = r_* j_*$ is the identity map on $H_{n-1}(\mathbb{S}^{n-1})$. But since $H_{n-1}(\mathbb{D}^n) = 0$, it follows that $j_*$ is the null map. On the other hand, $H_{n-1}(\mathbb{S}^{n-1}) = \mathbb{Z}$ if $n \neq 1$, and $H_0(\mathbb{S}^0) = \mathbb{Z} \oplus \mathbb{Z}$, leading to a contradiction.

The analytic proof reported below is less evident, and makes use of exterior forms. Moreover, it provides a weaker result, namely, it shows that there exists no retraction of class $\mathcal{C}^2$ from $\mathbb{D}^n$ to $\mathbb{S}^{n-1}$. This will be however enough for our scopes.

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*Proof* Associate to a $\mathcal{C}^2$ function $h = h(x) : \mathbb{D}^n \to \mathbb{D}^n$ the exterior form

$$\omega_h = h_1\, dh_2 \wedge \cdots \wedge dh_n,$$

where, for $k = 1, \ldots, n$,

$$dh_k = \sum_{i=1}^{n} \partial_{x_i} h_k\, dx_i.$$

Then,

$$d\omega_h = dh_1 \wedge \cdots \wedge dh_n = \det[J_h]\, dx,$$

where $J_h$ is the $(n \times n)$-Jacobian matrix of $h$, and

$$dx = dx_1 \wedge \cdots \wedge dx_n$$

is the infinitesimal volume element. Denoting

$$\mathcal{D}_h = \int_{\mathbb{S}^{n-1}} \omega_h,$$

the Stokes theorem (cf. [67, Ch.10]) entails

$$\mathcal{D}_h = \int_{\mathbb{D}^n} d\omega_h = \int_{\mathbb{D}^n} \det[J_h(x)]\, dx.$$

Assume now that there is a retraction $r$ of class $\mathcal{C}^2$ from $\mathbb{D}^n$ to $\mathbb{S}^{n-1}$. From the above formula, we see that $\mathcal{D}_r$ is determined only by the values of $r$ on $\mathbb{S}^{n-1}$. But $r_{|\mathbb{S}^{n-1}} = i_{|\mathbb{S}^{n-1}}$, where $i : \mathbb{D}^n \to \mathbb{D}^n$ is the identity map. Thus $\mathcal{D}_r = \mathcal{D}_i = \text{vol}\,(\mathbb{D}^n)$. On the other hand $\|r\| \equiv 1$, and this implies that the vector $J_r(x)r(x)$ is null for every $x \in \mathbb{D}^n$. So 0 is an eigenvalue of $J_r(x)$ for every $x \in \mathbb{D}^n$, and therefore $\det[J_r] \equiv 0$ which implies $\mathcal{D}_r = 0$. □

*Remark* Another way to show that $\det[J_r] \equiv 0$ is to observe that the determinant is a *null lagrangian* (for more details see, e.g., [25, Chap. 8]).

**Theorem 9.1** (Brouwer) *Let $f : \mathbb{D}^n \to \mathbb{D}^n$ be a continuous function. Then $f$ has a fixed point $\bar{x} \in \mathbb{D}^n$.*

*Proof* Since we want to rely on the analytic proof, let $f : \mathbb{D}^n \to \mathbb{D}^n$ be of class $\mathcal{C}^2$. If $f$ had no fixed point, then

$$r(x) = t(x) f(x) + (1 - t(x))x,$$

where

$$t(x) = \frac{\|x\|^2 - \langle x, f(x)\rangle - \sqrt{(\|x\|^2 - \langle x, f(x)\rangle)^2 + (1 - \|x\|^2)\|x - f(x)\|^2}}{\|x - f(x)\|^2}$$

is a retraction of class $\mathcal{C}^2$ from $\mathbb{D}^n$ to $\mathbb{S}^{n-1}$, against the conclusion of Lemma 9.1. Graphically, $r(x)$ is the intersection with $\mathbb{S}^{n-1}$ of the line obtained extending the segment connecting $f(x)$ to $x$. Hence such an $f$ has a fixed point.

Turning to the general case, let now $f : \mathbb{D}^n \to \mathbb{D}^n$ be continuous. Appealing to the Stone-Weierstrass theorem, we find a sequence $f_j : \mathbb{D}^n \to \mathbb{D}^n$ of functions of class $\mathcal{C}^2$ converging uniformly to $f$ on $\mathbb{D}^n$. Denote $\bar{x}_j$ the fixed point of $f_j$. Then there is $\bar{x} \in \mathbb{D}^n$ such that, up to a subsequence, $\bar{x}_j \to \bar{x}$. Therefore,

$$\|f(\bar{x}) - \bar{x}\| \le \|f(\bar{x}) - f(\bar{x}_j)\| + \|f(\bar{x}_j) - f_j(\bar{x}_j)\| + \|\bar{x}_j - \bar{x}\| \to 0$$

as $j \to \infty$, which yields $f(\bar{x}) = \bar{x}$. □

*Remark* Brouwer Theorem 9.1 is in [10] (see also [22]). An alternative approach to prove Theorem 9.1 makes use of the concept of *topological degree* (see, e.g., [18], Ch.1). A different proof that requires only elementary notions of continuity and compactness has been given by Kuga [45].

We now extend Theorem 9.1 to a more general situation.

**Theorem 9.2** *Let $K$ be a nonempty compact convex subset of a finite-dimensional real Banach space $X$. Then every continuous function $f : K \to K$ has a fixed point $\bar{x} \in K$.*

*Proof* Since $X$ is homeomorphic to $\mathbb{R}^n$ for some $n \in \mathbb{N}$, we assume without loss of generality $X = \mathbb{R}^n$. Also, we can assume $K \subset \mathbb{D}^n$. For every $x \in \mathbb{D}^n$, let $p(x) \in K$ be the unique point of minimum norm of the set $x - K$. Note that $p(x) = x$ for every $x \in K$. Moreover, $p$ is continuous on $\mathbb{D}^n$. Indeed, given $x_n, x \in \mathbb{D}^n$, with $x_n \to x$,

$$\|x - p(x)\| \le \|x - p(x_n)\| \le \|x - x_n\| + \inf_{k \in K} \|x_n - k\| \to \|x - p(x)\|$$

as $n \to \infty$. Thus $x - p(x_n)$ is a minimizing sequence as $x_n \to x$ in $x - K$, and this implies the convergence $p(x_n) \to p(x)$. Define now

$$g(x) = f(p(x)).$$

Then $g$ maps continuously $\mathbb{D}^n$ onto $K$. From Theorem 9.1 there is $\bar{x} \in K$ such that $g(\bar{x}) = \bar{x} = f(\bar{x})$. □

As an immediate application, consider the example of the introduction. If there is a compact and convex set $K \subset \mathbb{R}^n$ such that $f(K) \subset K$, then $f$ has a fixed point $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_n) \in K$. Accordingly, $\bar{x}$ is a (possibly nonunique) solution to the system of equations

$$g_j(x_1, \ldots, x_n) = 0, \qquad j = 1, \ldots, n.$$

Another quite direct application is the renowned Frobenius theorem.

**Theorem 9.3** (Frobenius) *Let $\mathbb{A}$ be a $n \times n$ matrix whose entries are all strictly positive. Then $\mathbb{A}$ has a strictly positive eigenvalue, whose corresponding eigenvector exhibits strictly positive entries only.*

*Proof* The matrix $\mathbb{A}$ can be viewed as a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. Introduce the compact convex set

$$K = \left\{ x \in \mathbb{R}^n : \sum_{j=1}^{n} x_j = 1, \;\; x_j \geq 0 \text{ for } j = 1, \ldots, n \right\},$$

and define $f(x) = \mathbb{A}x / \|\mathbb{A}x\|_1$ (where $\|\cdot\|_1$ is the 1-norm). Note that if $x \in K$, then all the entries of $x$ are nonnegative and at least one is strictly positive, hence all the entries of $\mathbb{A}x$ are strictly positive. Then $f$ is a continuous function mapping $K$ into $K$, and therefore there exists $\bar{x} \in K$ such that $\mathbb{A}\bar{x} = \|\mathbb{A}\bar{x}\|_1 \, \bar{x}$. Such an equality also tells that all the entries of $\bar{x}$ are strictly positive. □

We conclude the chapter by showing a famous misuse of Theorem 9.2 in connection with the fundamental theorem of algebra.

**Theorem 9.4** *Let $p(z) = a_0 + a_1 z + \cdots + a_n z^n$ be a complex polynomial of degree $n \geq 1$. Then there exists $z_0 \in \mathbb{C}$ such that $p(z_0) = 0$.*

In 1949, Arnold [2] proposed the following proof to Theorem 9.4.

*Proof* Let us identify $\mathbb{C}$ with $\mathbb{R}^2$. Suppose without loss of generality $a_n = 1$. Let $r = 2 + |a_0| + \cdots + |a_{n-1}|$. Define now the function $f : \mathbb{C} \to \mathbb{C}$ as

$$f(z) = \begin{cases} z - \dfrac{p(z)}{r} \, e^{i(1-n)\vartheta\varrho} & |z| \leq 1, \\ z - \dfrac{p(z)}{r} \, z^{(1-n)} & |z| > 1, \end{cases}$$

where $z = \varrho e^{i\vartheta}$ with $\vartheta \in [0, 2\pi)$. For $z = 0$ we have $\varrho = 0$, so that $i(1-n)\vartheta\varrho = 0$ no matter what values $\vartheta$ has. Consider then the compact convex set $K = \left\{ z : |z| \leq r \right\}$. Aiming to apply Theorem 9.2 we need to show that $f(K) \subset K$. Indeed, if $|z| \leq 1$,

$$|f(z)| \leq |z| + \frac{|p(z)|}{r} \leq 1 + \frac{1 + |a_0| + \cdots + |a_{n-1}|}{r} < 2 \leq r.$$

Conversely, if $1 < |z| \leq r$ we have

$$\begin{aligned} |f(z)| &\leq \left| z - \frac{p(z)}{r z^{n-1}} \right| = \left| z - \frac{z}{r} - \frac{a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}}{r z^{n-1}} \right| \\ &\leq r - 1 + \frac{|a_0| + \cdots + |a_{n-1}|}{r} = r - 1 + \frac{r-2}{r} < r. \end{aligned}$$

Hence $K$ is invariant for $f$, and so $f$ has a fixed point $z_0 \in K$, which is clearly a root of $p$. □

Unfortunately, although rather elegant, the proof contains an unrecoverable mistake preventing the application of Theorem 9.2. Indeed, the function $f$ above is *not* continuous along the positive real axis. This was pointed out by Arnold himself, who wrote a retraction note two years later [3]. In spite of that, a number of works in the literature still report this proof as correct.

# Chapter 10
# The Schauder-Tychonoff Fixed Point Theorem




In order to prove the main result of this chapter, the Schauder-Tychonoff fixed point theorem, we first need a definition and a lemma.

**Definition 10.1** Let $U_1, \ldots, U_n$ be open subsets of a locally compact Hausdorff space $X$, and let $K \subset X$ be a compact set such that

$$K \subset U_1 \cup \cdots \cup U_n.$$

A family of continuous functions $\varphi_j : X \to [0, 1]$, for $j = 1, \ldots, n$, is said to be a *partition of the unity* for $K$ subordinate to the open cover $\{U_1, \ldots, U_n\}$ if each $\varphi_j$ is supported on $U_j$ and

$$\varphi_1(x) + \cdots + \varphi_n(x) = 1, \qquad \forall x \in K.$$

**Lemma 10.1** *Let $K$ be a compact subset of a locally compact Hausdorff space $X$, and let $U_1, \ldots, U_n$ be an open cover of $K$. Then there exists a partition of the unity $\varphi_1, \ldots, \varphi_n$ for $K$ subordinate to $\{U_j\}$.*

*Proof* The two main ingredients of the proof are the following:

- In every locally compact Hausdorff space $X$, given a point $x$ and an open neighborhood $V$ of $x$, it is always possible to find an open neighborhood $W$ of $x$ with the property that $\overline{W}$ is compact and $\overline{W} \subset V$.
- The *Urysohn lemma*: if $A$ and $B$ are closed disjoint subsets of $X$ and $A$ is compact, then there is a continuous function $\psi : X \to [0, 1]$ such that $\psi(A) = \{1\}$ and $\psi(B) = \{0\}$ (see for instance [41, 52]).

Let then $\{U_1, \ldots, U_n\}$ be an open cover of the compact set $K$. The idea is to shrink the cover $\{U_j\}$ to a smaller open cover $\{V_j\}$ satisfying the property

$$\overline{V}_j \subset U_j, \qquad \forall j = 1, \ldots, n.$$

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For any $x \in K$, we have that $x \in U_i$ for some $i$ (possibly more than one). Choose an open neighborhood $O_x$ of $x$ with $\overline{O}_x$ contained in any such $U_i$. By compactness, $K$ is covered by a finite collection $\{O_{x_1}, \ldots, O_{x_k}\}$. Thus, let

$$V_j = \bigcup O_{x_i},$$

where the union runs over all $i$ for which $x_i \in U_j$. It is clear that the collection $\{V_j\}$ does the job. By the same token, we select a further open cover $\{W_j\}$ of $K$ such that $\overline{W}_j$ is compact and $\overline{W}_j \subset V_j$. At this point, we apply the Urysohn lemma, and we find for each $j$ a continuous function $\psi_j : X \to [0, 1]$ such that

$$\psi_j(\overline{W}_j) = \{1\} \quad \text{and} \quad \psi_j(V_j^C) = \{0\}.$$

Since $\psi_j^{-1}((0, 1]) \subset V_j$, we conclude that

$$\text{support}(\psi_j) \subset \overline{V}_j \subset U_j.$$

Besides, the sum $\psi(x) = \sum_{j=1}^n \psi_j(x)$ is strictly positive on the compact set

$$C = \bigcup_{j=1}^n \overline{W}_j \supset K.$$

Define then the functions $\hat{\varphi}_j : X \to [0, 1]$ as

$$\hat{\varphi}_j(x) = \begin{cases} \dfrac{\psi_j(x)}{\psi(x)} & \text{if } \psi(x) \neq 0, \\ 0 & \text{if } \psi(x) = 0. \end{cases}$$

Each $\hat{\varphi}_j$ is supported on $U_j$ and

$$\hat{\varphi}_1(x) + \cdots + \hat{\varphi}_n(x) = 1, \qquad \forall x \in K.$$

These $\hat{\varphi}_j$ are almost the functions we are looking for. Indeed, the only possible problem is the continuity of $\hat{\varphi}_j$ at the points where $\psi$ vanishes. To this end, we slightly modify each $\hat{\varphi}_j$ as follows. Since $\psi$ is continuous (and equals at least 1 on $C$), the set $Q = \psi^{-1}((1/2, \infty))$ is open in $X$. By a further application of the Urysohn lemma, we produce a continuous function $\xi : X \to [0, 1]$ such that $\xi(C) = \{1\}$ and $\xi(Q^C) = \{0\}$. Therefore, the modified functions

$$\varphi_j(x) = \xi(x)\hat{\varphi}_j(x)$$

are continuous on the whole space $X$ and satisfy the thesis. □

We are often interested to find partitions of the unity for a compact set $K \subset X$ whose members are continuous functions defined on $K$. Clearly, in this case $X$ need not be locally compact.

**Theorem 10.1** (Schauder-Tychonoff) *Let $X$ be a locally convex space, let $K \subset X$ be nonempty and convex (not necessarily closed), and let $K_0 \subset K$ be a compact set. Given a continuous map $f : K \to K_0$, there exists $\bar{x} \in K_0$ such that $f(\bar{x}) = \bar{x}$.*

Schauder's original proof of Theorem 10.1 for Banach spaces can be found in [70], whereas Tychonoff's generalization to locally convex spaces is in [72].

*Proof* Denote by $\mathcal{B}$ the local base for the topology of $X$ generated by the separating family of seminorms $\mathcal{P}$ on $X$. Recall that the members of $\mathcal{B}$ are convex sets. Given any $U \in \mathcal{B}$, from the compactness of $K_0$ there exist $x_1, \ldots, x_n \in K_0$ such that

$$K_0 \subset \bigcup_{j=1}^{n} (x_j + U).$$

Let $\varphi_1, \ldots, \varphi_n \in C(K_0)$ be a partition of the unity for $K_0$ subordinate to the open cover $\{x_j + U\}$, and define

$$f_U(x) = \sum_{j=1}^{n} \varphi_j(f(x))x_j, \qquad \forall x \in K.$$

Then,

$$f_U(K) \subset K_U \doteq \mathrm{co}\big(\{x_1, \ldots, x_n\}\big) \subset K,$$

and since $K_U$ is compact (being a bounded closed set in a finite-dimensional space), Theorem 9.2 yields the existence of $x_U \in K_U$ such that

$$f_U(x_U) = x_U.$$

In particular,

$$x_U - f(x_U) \in U. \tag{10.1}$$

Indeed,

$$x_U - f(x_U) = f_U(x_U) - f(x_U) = \sum_{j=1}^{n} \varphi_j(f(x_U))(x_j - f(x_U)),$$

where $\varphi_j(f(x_U)) = 0$ whenever $x_j - f(x_U)$ does not belong to the (convex) set $U$. Appealing again to the compactness of $K_0$, by the finite intersection property, there exists

$$\bar{x} \in \bigcap_{W \in \mathcal{B}} \overline{\{f(x_U) : U \in \mathcal{B},\ U \subset W\}} \subset K_0.$$

Select now an arbitrary $V \in \mathcal{B}$. From the continuity of $f$ on $K$, there is $W \in \mathcal{B}$, $W \subset V$, such that for every $x \in K$ we have the implication

$$x - \bar{x} \in 2W \qquad \Rightarrow \qquad f(x) - f(\bar{x}) \in V.$$

For this particular $W$, we know that

$$\bar{x} \in \overline{\{f(x_U) : U \in \mathcal{B},\ U \subset W\}}.$$

Hence, there exists $U \in \mathcal{B}$, $U \subset W$, for which

$$\bar{x} - f(x_U) \in W. \tag{10.2}$$

Collecting (10.1) and (10.2) we get

$$x_U - \bar{x} = x_U - f(x_U) + f(x_U) - \bar{x} \in U + W \subset W + W = 2W,$$

yielding in turn

$$f(x_U) - f(\bar{x}) \in V. \tag{10.3}$$

Therefore, (10.2)–(10.3) entail

$$\bar{x} - f(\bar{x}) = \bar{x} - f(x_U) + f(x_U) - f(\bar{x}) \in W + V \subset V + V = 2V.$$

Being $V$ arbitrary, we conclude that $\bar{x} - f(\bar{x})$ belongs to any neighborhood of zero, which readily implies the equality $f(\bar{x}) = \bar{x}$. □

The following two propositions are quite direct consequences of Theorem 10.1, and deal with the existence of zeros of maps on Banach spaces. For a Banach space $X$ and $r > 0$, let

$$B_r = \bar{B}_X(0, r),$$

and consider a continuous map $g : B_r \to X$, such that $g(B_r)$ is relatively compact.

**Proposition 10.1** *Let $g(x) \notin \{\lambda x : \lambda > 0\}$ for every $x \in \partial B_r$. Then there is $x_0 \in B_r$ such that $g(x_0) = 0$.*

*Proof* If not so, the function $f(x) = rg(x)/\|g(x)\|$ is continuous from $B_r$ to $B_r$ and $f(B_r)$ is relatively compact. From Theorem 10.1, $f$ admits a fixed point $\bar{x} \in B_r$. Then $g(\bar{x}) = \|g(\bar{x})\|\bar{x}/r$ with $\|\bar{x}\| = r$, against the hypotheses.

**Proposition 10.2** *Assume that for every $x \in \partial B_r$ there exists $\Lambda_x \in X^*$ such that $\Lambda_x x = r$ and $\Lambda_x g(x) \geq 0$. Then there is $x_0 \in B_r$ such that $g(x_0) = 0$.*

*Proof* Define $f(x) = -rg(x)/\|g(x)\|$. If $g$ has no zeros, reasoning as above, $f$ has a fixed point $\bar{x} \in B_r$, and the relation $-g(\bar{x}) = \|g(\bar{x})\|\bar{x}/r$, with $\|\bar{x}\| =$

$r$, holds. Taking $\Lambda \in X^*$ such that $\Lambda \bar{x} = r$, we obtain $\Lambda g(\bar{x}) = -\|g(\bar{x})\| < 0$. Contradiction. □

Let us see an interesting corollary about the surjectivity of maps from $\mathbb{R}^n$ to $\mathbb{R}^n$, that extends a well-known result for matrices to the more general situation of continuous maps.

**Corollary 10.1** *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous function satisfying*

$$\lim_{\|x\| \to \infty} \frac{\langle f(x), x \rangle}{\|x\|} = \infty.$$

*Then $f(\mathbb{R}^n) = \mathbb{R}^n$.*

*Proof* Fix $y_0 \in \mathbb{R}^n$, and set $g(x) = f(x) - y_0$. Then, for $r > 0$ big enough,

$$\langle g(x), x/\|x\| \rangle \geq 0, \qquad \forall x \in \partial B_r.$$

Hence for every $x \in \partial B_r$, the functional $\Lambda_x = \langle \cdot, x/\|x\| \rangle$ fulfills the hypotheses of Proposition 10.2. Therefore there is $x_0 \in B_r$ such that $g(x_0) = 0$, i.e., $f(x_0) = y_0$. □

*Remark* These kind of conditions concerning the behavior of $g$ on $\partial B_r$ are known as *Leray-Schauder boundary conditions* [47]. The above results can be generalized to continuous functions defined on the closure of open subsets of $X$ with values in $X$, and satisfying certain compactness properties. See for instance [18, pp. 204–205] and [36, Chap. 5.3].

# Chapter 11
# Further Consequences of the Schauder-Tychonoff Theorem




In concrete applications it is somehow easier to work with functions defined on the whole space $X$, and rather ask more restrictive conditions, such as compactness, on the maps.

**Definition 11.1** Let $X, Y$ be Banach spaces, and $C \subset X$. A map $f : C \to Y$ is said to be *compact* provided it transforms bounded sets into relatively compact sets.

*Remark* In terms of sequences, $f$ is compact if, for every bounded sequence $x_n$, the sequence $f(x_n)$ has a convergent subsequence. If $f \in L(X, Y)$, then $f$ compact is the same as saying that the image of the unit closed ball under $f$ is relatively compact.

In this direction, we have the following result [69].

**Theorem 11.1** (Schaefer) *Let $X$ be a Banach space, $f : X \to X$ continuous and compact. Assume further that the set*

$$\mathcal{F} = \left\{x \in X \,:\, x = \lambda f(x), \quad \text{for some } \lambda \in [0, 1]\right\}$$

*is bounded. Then $f$ has a fixed point $\bar{x} \in X$.*

Note that the set $\mathcal{F}$ is nonempty, since it always contains the point $x = 0$ (just take $\lambda = 0$).

*Remark* Theorem 11.1 holds if we can prove a priori estimates on the set of all the *possible* fixed points of $\lambda f$. This technique is typical in partial differential equations, where one proves estimates on a possible solution to a certain equation, and then, in force of those estimates, concludes that such solution actually exists.

*Proof* Let $r > \sup_{x \in \mathcal{F}} \|x\|$, and define the map

$$g(x) = \begin{cases} f(x) & \text{if } \|f(x)\| \le r, \\ \dfrac{rf(x)}{\|f(x)\|} & \text{if } \|f(x)\| > r. \end{cases}$$

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Then $g$ maps continuously $\overline{B}_X(0,r)$ into $\overline{B}_X(0,r)$, and the closure of the image $g(\overline{B}_X(0,r))$ is compact and contained in $\overline{B}_X(0,r)$. Hence Theorem 10.1 yields the existence of $\bar{x} \in \overline{B}_X(0,r)$ for which $g(\bar{x}) = \bar{x}$. Note that $\|f(\bar{x})\| \leq r$, for otherwise

$$\bar{x} = \lambda_0 f(\bar{x}) \qquad \text{with} \qquad \lambda_0 = \frac{r}{\|f(\bar{x})\|} < 1,$$

which forces the equality $\|\bar{x}\| = r$, against the fact that $\bar{x} \in \mathcal{F}$. We conclude that

$$g(\bar{x}) = f(\bar{x}) = \bar{x},$$

as claimed. □

A second result concerns with fixed points of a sum of maps [44].

**Theorem 11.2** (Krasnosel'skiĭ) *Let $X$ be a Banach space, $C \subset X$ nonempty, closed and convex. Let $f, g : C \to X$ be such that:*

- $f(y) + g(x) \in C, \quad \forall x, y \in C.$
- *$f$ is continuous and $f(C)$ is relatively compact.*
- *$g$ is a contraction (from $C$ into $X$).*

*Then there is $\bar{x} \in C$ such that $f(\bar{x}) + g(\bar{x}) = \bar{x}$.*

*Proof* Observe first that $I - g$ maps homeomorphically $C$ onto $(I - g)(C)$. Indeed, denoting by $\lambda < 1$ the Lipschitz constant of $g$, the map $I - g$ is continuous and

$$\|(I - g)(x) - (I - g)(y)\| \geq \|x - y\| - \|g(x) - g(y)\| \geq (1 - \lambda)\|x - y\|.$$

Hence $(I - g)^{-1}$ is continuous. For any $y \in C$, the map $x \mapsto f(y) + g(x)$ is a contraction on $C$, and by Theorem 1.1 there is a unique $z = z(y) \in C$ such that

$$z = f(y) + g(z).$$

Thus

$$z = (I - g)^{-1}(f(y)) \in C.$$

On the other hand, the map $(I - g)^{-1} \circ f$ is continuous from $C$ to $C$ and the set $(I - g)^{-1}(f(C))$ is relatively compact. Then Theorem 10.1 entails the existence of $\bar{x} \in C$ such that

$$(I - g)^{-1}(f(\bar{x})) = \bar{x},$$

that is, $f(\bar{x}) + g(\bar{x}) = \bar{x}$. □

In general, it is not possible to extend Theorem 10.1 to noncompact settings. This fact was already envisaged in our previous discussion about non-expansive maps. Let us recall another famous example in Hilbert spaces.

**Example** (Kakutani) Consider the Hilbert space $\ell^2$ of square summable sequences, normed by

$$\|x\| = \left( \sum_{n=0}^{\infty} |x_n|^2 \right)^{1/2}, \qquad x = (x_0, x_1, x_2, \ldots).$$

For a fixed $\varepsilon \in (0, 1]$, let

$$f_\varepsilon : \overline{B}_{\ell^2}(0, 1) \to \overline{B}_{\ell^2}(0, 1)$$

be given by

$$f_\varepsilon(x) = (\varepsilon(1 - \|x\|), x_0, x_1, \ldots).$$

Then $f_\varepsilon$ has no fixed points in $\overline{B}_{\ell^2}(0, 1)$, but it is Lipschitz continuous with Lipschitz constant slightly greater than 1. Indeed,

$$\|f_\varepsilon(x) - f_\varepsilon(y)\| \leq \sqrt{1 + \varepsilon^2}\, \|x - y\|$$

for all $x, y \in \overline{B}_{\ell^2}(0, 1)$.

It is therefore a natural question to ask when a continuous map $f : C \to C$, where $C$ is a closed, bounded and convex subset of a Banach space $X$, admits fixed points. We know from Theorem 10.1 that if $X$ is finite-dimensional the answer is positive, since finite-dimensional Banach spaces have the Heine-Borel property (i.e., closed and bounded sets are compact). The analogous result in infinite-dimensional Banach spaces turns out to be false, without a compactness assumption.

We first report a characterization of noncompact closed bounded sets.

**Lemma 11.1** *Let $X$ be an infinite-dimensional Banach space, $C \subset X$ a closed, bounded noncompact set. Then there are $\varepsilon > 0$ and a sequence $x_n$ of elements of $C$ such that*

$$\operatorname{dist}\big(x_{n+1}, \operatorname{span}\big(\{x_0, \ldots, x_n\}\big)\big) \geq \varepsilon. \tag{11.1}$$

Observe that if $X$ is finite-dimensional no such $C$ can exist.

*Proof* We first show that there is $\varepsilon > 0$ such that, for any finite set $F \subset X$,

$$C \setminus \big[\operatorname{span}(F) + B_X(0, \varepsilon)\big] \neq \emptyset.$$

If not, for every $\varepsilon > 0$ we find a finite set $F \subset X$ such that

$$C \subset \operatorname{span}(F) + B_X(0, \varepsilon).$$

Since $C$ is bounded, $C \subset B_X(0, r)$ for some $r > 0$. Therefore,

$$C \subset \big[\mathrm{span}(F) + B_X(0,\varepsilon)\big] \cap B_X(0,r) \subset \big[\mathrm{span}(F) \cap B_X(0,r+\varepsilon)\big] + B_X(0,\varepsilon).$$

But $\mathrm{span}(F) \cap B_X(0, r+\varepsilon)$ is totally bounded, and so it admits a finite cover of balls of radius $\varepsilon$, which in turn implies that $C$ admits a finite cover of balls of radius $2\varepsilon$, that is, $C$ is totally bounded (hence compact, being closed), contradicting the hypotheses.

To construct the required sequence $x_n$, we proceed inductively. First we select an arbitrary $x_0 \in C$. Then, given $x_0, \ldots, x_{n+1}$ satisfying (11.1), we choose

$$x_{n+2} \in C \setminus \big[\mathrm{span}\big(\{x_0, \ldots, x_{n+1}\}\big) + B_X(0,\varepsilon)\big].$$

This finishes the inductive argument. □

**Theorem 11.3** (Klee) *Let $X$ be an infinite-dimensional Banach space, $C \subset X$ a closed, bounded, convex noncompact set. Then there exists a continuous map $f : C \to C$ which is fixed point free.*

Theorem 11.3 has been proved by Klee [43]. The result actually holds for metrizable locally convex spaces. In the subsequent paper [48], Lin and Sternfeld have shown that the map $f$ of Theorem 11.3 can be taken to be Lipschitz continuous, with Lipschitz constant greater than (but arbitrarily close to) 1.

*Proof* Let $x_n$ be a sequence in $C$ satisfying (11.1). Without loss of generality, we may assume that $0 \in C$ and $\|x_0\| \geq \varepsilon$. We construct a piecewise linear curve joining the points $x_0, x_1, x_2, \ldots$, by setting

$$\Gamma = \bigcup_{n=0}^{\infty} [x_n, x_{n+1}],$$

having defined the segment

$$[x_n, x_{n+1}] = \mathrm{co}\big(\{x_n, x_{n+1}\}\big).$$

Then $\Gamma \subset C$ is closed and can be parameterized by a function $\gamma : [0,\infty) \to \Gamma$, given by

$$\gamma(t) = (1-s)x_n + s x_{n+1},$$

where $n = \lfloor t \rfloor$ and $s = t - n$. By means of (11.1), $\gamma$ is one-to-one, onto and open since, for every open set $O \subset [0,\infty)$, $\gamma(O)$ contains the intersection of $\Gamma$ with an open ball of $X$ of radius $\nu$ for some $\nu < \varepsilon$. Hence its inverse $\gamma^{-1} : \Gamma \to [0,\infty)$ is a continuous function. Applying a slightly modified version of the Tietze extension theorem (see, e.g., [41]), we can extend $\gamma^{-1}$ to a continuous function $g : C \to [0,\infty)$. Now a fixed point free continuous map $f : C \to C$ can be defined as

$$f(x) = \gamma(g(x) + 1).$$

Indeed, if $f(\bar{x}) = \bar{x}$, then $\bar{x} \in \Gamma$. Hence,

$$\bar{x} = \gamma(\gamma^{-1}(\bar{x}) + 1) \qquad \Rightarrow \qquad \gamma^{-1}(\bar{x}) = \gamma^{-1}(\bar{x}) + 1,$$

which is a contradiction. □

*Remark* Note however that Klee's map $f : C \to C$ cannot be uniformly continuous, since the restriction of $g$ on $\Gamma$ is easily seen to be uniformly continuous, whereas $g \circ f$ maps the bounded convex set $C$ onto $[1, \infty)$.

In particular, Theorem 11.3 says that in an infinite-dimensional Banach space $X$ there is a continuous map $f : \overline{B}_X(0, 1) \to \overline{B}_X(0, 1)$ fixed point free. This allow us to provide another interesting characterization of infinite-dimensional Banach spaces.

**Corollary 11.1** *Let $X$ be an infinite-dimensional Banach space. Then the closed unit sphere is a retract of the closed unit ball.*

*Proof* Let $f : \overline{B}_X(0, 1) \to \overline{B}_X(0, 1)$ be a continuous fixed point free map. Extend the map to the doubled ball $\overline{B}_X(0, 2)$ by defining

$$f_1(x) = \begin{cases} f(x) & \text{if} \quad \|x\| \leq 1, \\ (2 - \|x\|) f(x/\|x\|) & \text{if} \quad 1 < \|x\| \leq 2, \end{cases}$$

and consider the new map $f_2 : \overline{B}_X(0, 1) \to \overline{B}_X(0, 1)$ as

$$f_2(x) = \frac{1}{2} f_1(2x).$$

Observe that $f_2$ is fixed point free, and for all $x \in \partial B_X(0, 1)$ we have $f_2(x) = 0$. Then the function $r : \overline{B}_X(0, 1) \to \partial B_X(0, 1)$ given by

$$r(x) = \frac{x - f_2(x)}{\|x - f_2(x)\|}$$

is the desired retraction. □

# Chapter 12
# The Markov-Kakutani Theorem




This result is concerned with common fixed points of a family of linear maps.

**Theorem 12.1** (Markov-Kakutani) *Let $X$ be a locally convex space, and let $K \subset X$ be a nonempty, compact and convex set. Assume $\mathcal{G}$ is a family of bounded linear operators from $X$ into $X$ such that:*

- *$\mathcal{G}$ is abelian, that is, $TS = ST$ for every $T, S \in \mathcal{G}$.*
- *$TK \subset K$ for every $T \in \mathcal{G}$.*

*Then there exists $\bar{x} \in K$ such that $T\bar{x} = \bar{x}$ for every $T \in \mathcal{G}$.*

*Proof* For any $T \in \mathcal{G}$ and any $n \in \mathbb{N}$, define the operator

$$T_n = \frac{I + T + \cdots + T^n}{n+1}.$$

Note that the convexity of $K$ implies that

$$T_n K \subset K.$$

Given $T^{(1)}, \ldots, T^{(k)} \in \mathcal{G}$ and $n_1, \ldots, n_k \in \mathbb{N}$, it follows that

$$\bigcap_{j=1}^{k} T_{n_j}^{(j)} K \neq \emptyset.$$

Indeed, for any $T, S \in \mathcal{G}$ and any $n, m \in \mathbb{N}$,

$$T_n K \supset T_n S_m K = S_m T_n K \neq \emptyset,$$

and

$$S_m K \supset S_m T_n K = T_n S_m K \neq \emptyset.$$

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Accordingly,

$$T_n K \cap S_m K \supset T_n S_m K = S_m T_n K \neq \emptyset.$$

Invoking the compactness of $K$, the finite intersection property yields

$$\mathcal{F} = \bigcap_{T \in \mathcal{G},\, n \in \mathbb{N}} T_n K \neq \emptyset.$$

We claim that every $x \in \mathcal{F}$ is a fixed point for all $T \in \mathcal{G}$. Let then $\bar{x} \in \mathcal{F}$, and select any $T \in \mathcal{G}$. Then, for every $n \in \mathbb{N}$ there is $y = y(n) \in K$ such that $\bar{x} = T_n y$. Hence

$$\begin{aligned} T\bar{x} - \bar{x} &= \frac{Ty + T^2 y + \cdots + T^{n+1} y}{n+1} - \frac{y + Ty + \cdots + T^n y}{n+1} \\ &= \frac{T^{n+1} y - y}{n+1} \\ &\in \frac{1}{n+1}\big(K - K\big). \end{aligned}$$

Thus

$$T\bar{x} - \bar{x} \in \bigcap_{n \in \mathbb{N}} \frac{1}{n+1}\big(K - K\big).$$

The set $K - K$ is clearly compact, being the image of $K \times K$ under the continuous map $\Phi(\xi, \eta) = \xi - \eta$. Let then $p$ be a seminorm on $X$. Then, for any $\varepsilon > 0$, there is $n \in \mathbb{N}$ such that

$$\frac{1}{n+1}\big(K - K\big) \subset \big\{x \in X : p(x) < \varepsilon\big\}.$$

We conclude that $p(T\bar{x} - \bar{x}) = 0$ for every seminorm on $X$, which entails the equality $T\bar{x} = \bar{x}$. □

*Remark* With no changes in the proof, the result holds more generally if $\mathcal{G}$ is an abelian family of continuous affine maps from $K$ to $K$. A map $f : C \to C, C$ convex, is said to be *affine* if, for all $x, y \in C$ and $t \in [0, 1]$,

$$f(tx + (1-t)y) = tf(x) + (1-t)f(y).$$

Theorem 12.1 has been proved with different techniques by Markov [50] and Kakutani [39]. The same kind of result does not hold in general for commuting nonlinear continuous maps. Indeed Boyce [7] and Huneke [35] have provided an example of two commuting nonlinear continuous selfmaps of $[0, 1]$ without a common fixed point.

# Chapter 13
# The Kakutani-Ky Fan Theorem




Along this chapter, let $X$ be a locally convex space.

**Definition 13.1** Let $C \subset X$ be a convex set. A function $f : C \to (-\infty, \infty]$ is said to be *convex* if

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y),$$

for all $\forall x, y \in C$ and every $\lambda \in [0, 1]$. A function $g : C \to [-\infty, \infty)$ is said to be *concave* if $-g$ is convex (it is understood that $-(\infty) = -\infty$).

We now recall the well-known definition of a lower semicontinuous real function.

**Definition 13.2** Let $Y$ be a topological space. A function $f : Y \to (-\infty, \infty]$ is said to be *lower semicontinuous* if $f^{-1}((\alpha, \infty])$ is open for every $\alpha \in \mathbb{R}$. Similarly, a function $g : Y \to [-\infty, \infty)$ is said to be *upper semicontinuous* if $-g$ is lower semicontinuous.

It is an immediate consequence of the definition that the supremum of any collection of lower semicontinuous functions is lower semicontinuous.

*Remark* If $f$ is lower semicontinuous and $Y$ is compact, then $f$ attains its minimum on $Y$. Indeed, if it is not so, denoting

$$m = \inf_{y \in Y} f(y) \in [-\infty, \infty),$$

the sets $f^{-1}((\alpha, \infty])$ with $\alpha > m$ form an open cover of $Y$ that admits no finite subcover.

The next result is the so-called Ky Fan inequality.

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**Lemma 13.1** (Ky Fan) *Let $K \subset X$ be a nonempty, compact and convex set. Let $\Phi : K \times K \to \mathbb{R}$ be a map such that:*

- $\Phi(\cdot, y)$ *is lower semicontinuous* $\forall y \in K$.
- $\Phi(x, \cdot)$ *is concave and bounded above* $\forall x \in K$.

*Then there exists $x_0 \in K$ such that*

$$\sup_{y \in K} \Phi(x_0, y) \leq \sup_{y \in K} \Phi(y, y).$$

*Remark* Observe that a concave function on a compact convex set is not automatically bounded above. Indeed, on infinite-dimensional spaces it is possible to find linear functionals (hence, in particular, concave) which are unbounded on compact convex sets (e.g., a unit ball of a reflexive Banach space with the weak topology).

*Proof* Fix $\varepsilon > 0$. For any given $x \in K$ there is $y_x \in K$ such that

$$\Phi(x, y_x) > \sup_{y \in K} \Phi(x, y) - \frac{\varepsilon}{2}.$$

Since $\Phi(\cdot, y_x)$ is lower semicontinuous, there exists an open neighborhood $U_x$ of $x$ such that

$$\Phi(z, y_x) > \Phi(x, y_x) - \frac{\varepsilon}{2}, \qquad \forall z \in U_x \cap K.$$

Therefore,

$$\Phi(z, y_x) > \sup_{y \in K} \Phi(x, y) - \varepsilon, \qquad \forall z \in U_x \cap K.$$

Being $K$ compact, for some $x_1, \ldots, x_n \in K$ we have the inclusion

$$K \subset U_{x_1} \cup \cdots \cup U_{x_n}.$$

Let now $\varphi_1, \ldots, \varphi_n \in C(K)$ be a partition of the unity for $K$ subordinate to the open cover $\{U_{x_j}\}$ (see Lemma 10.1), and define

$$f(x) = \sum_{j=1}^{n} \varphi_j(x) y_{x_j}, \qquad \forall x \in K.$$

The map $f$ is clearly continuous, and

$$f\big(\mathrm{co}(\{y_{x_1}, \ldots, y_{x_n}\})\big) \subset \mathrm{co}(\{y_{x_1}, \ldots, y_{x_n}\}).$$

Hence by Theorem 9.2 $f$ admits a fixed point $\bar{x} \in K$. Using the fact that $\Phi(\bar{x}, \cdot)$ is concave,

$$\begin{aligned}\sup_{y\in K}\Phi(y,y) &\geq \Phi(\bar{x},\bar{x})\\ &= \Phi(\bar{x},f(\bar{x}))\\ &\geq \sum_{j=1}^{n}\varphi_j(\bar{x})\Phi(\bar{x},y_{x_j}).\end{aligned}$$

Since $\varphi_j(\bar{x})=0$ whenever $\bar{x}\notin U_{x_j}$, we readily get

$$\begin{aligned}\sum_{j=1}^{n}\varphi_j(\bar{x})\Phi(\bar{x},y_{x_j}) &\geq \sum_{j=1}^{n}\varphi_j(\bar{x})\Big[\sup_{y\in K}\Phi(x_j,y)-\varepsilon\Big]\\ &\geq \inf_{x\in K}\Big[\sup_{y\in K}\Phi(x,y)\Big]-\varepsilon.\end{aligned}$$

At this point, the lower semicontinuity of the map

$$x\mapsto\Big[\sup_{y\in K}\Phi(x,y)\Big]$$

ensures the existence of $x_0\in K$ such that

$$\inf_{x\in K}\Big[\sup_{y\in K}\Phi(x,y)\Big]=\sup_{y\in K}\Phi(x_0,y).$$

In conclusion, we found $x_0\in K$, independent of $\varepsilon$, such that

$$\sup_{y\in K}\Phi(y,y)\geq\sup_{y\in K}\Phi(x_0,y)-\varepsilon.$$

Letting finally $\varepsilon\to 0$, the desired conclusion follows. □

**Example** Let us show that the Brouwer fixed point Theorem 9.1 can be recovered from Lemma 13.1. Indeed, given a continuous function $f:\mathbb{D}^n\to\mathbb{D}^n$, define the map $\Phi:\mathbb{D}^n\times\mathbb{D}^n\to\mathbb{R}$ as

$$\Phi(x,y)=\|f(x)-x\|-\|f(x)-y\|.$$

The set $\mathbb{D}^n$ is nonempty, compact and convex, and $\Phi$ is continuous, as $f$ is continuous and so is the norm. In particular, $\Phi(\mathbb{D}^n\times\mathbb{D}^n)$ is bounded, due to the compactness of $\mathbb{D}^n\times\mathbb{D}^n$. Let now $x\in\mathbb{D}^n$ be fixed, and set $a=\|f(x)-x\|$. Since $y\mapsto\|f(x)-y\|$ is convex (by the triangle inequality), it follows that the map $y\mapsto a-\|f(x)-y\|$ is concave. Therefore, we meet the hypotheses

of Lemma 13.1. Accordingly, there exists $\bar{x} \in \mathbb{D}^n$ such that

$$\sup_{y \in \mathbb{D}^n} \left[ \|f(\bar{x}) - \bar{x}\| - \|f(\bar{x}) - y\| \right] \leq 0.$$

Choosing $y = f(\bar{x}) \in \mathbb{D}^n$, we conclude that

$$\|f(\bar{x}) - \bar{x}\| \leq 0 \qquad \Rightarrow \qquad f(\bar{x}) = \bar{x}.$$

We point out that the above is just a nice exercise, and it cannot be considered in any way a "proof" of the Brouwer theorem, since the Brouwer theorem itself is the foundational argument in the proof of Lemma 13.1.

The aim of this chapter is to consider a fixed point theorem for maps carrying points into sets. Let $K \subset X$, and consider a map

$$f : K \to 2^K \doteq \{Y : Y \subset K\}.$$

**Definition 13.3** A fixed point for $f : K \to 2^K$ is a point $x \in K$ such that

$$x \in f(x).$$

**Definition 13.4** The map $f$ is *upper semicontinuous* if for every $x \in K$ and every open set $U \supset f(x)$, there exists a neighborhood $V$ of $x$ such that if $y \in V$ then $U \supset f(y)$.

Upper semicontinuous point-closed maps from a compact set $K$ with values in $2^K$ can be characterized in terms of graphs.

**Proposition 13.1** *Let $K \subset X$ be compact, and let $f : K \to 2^K$ be such that $f(x)$ is closed for every $x \in K$. Then $f$ is upper semicontinuous if and only if the set*

$$\mathcal{G}(f) = \{(x, y) \in K \times K : y \in f(x)\}$$

*is closed in $K \times K$.*

*Proof* Let $f$ be upper semicontinuous, and let

$$(x_0, y_0) \in (K \times K) \setminus \mathcal{G}(f).$$

Then $y_0 \notin f(x_0)$. Since $K$ is compact, we find two disjoint open sets $U_1, U_2$ such that $y_0 \in U_1$ and $f(x_0) \subset U_2$. By the upper semicontinuity of $f$, there is an open set $V \ni x_0$ such that $f(x) \subset U_2$ for all $x \in V$. Thus the neighborhood $V \times U_1$ of $(x_0, y_0)$ does not intersect $\mathcal{G}(f)$, which is henceforth closed.

Conversely, let $\mathcal{G}(f)$ be closed. Let $x$ be an arbitrary point of $K$, and $U$ be an arbitrary open set containing $f(x)$. If $f$ is not upper semicontinuous at $x$, for every

neighborhood $V \ni x$ there is $y \in V$ such that $f(y) \not\subset U$. So we have

$$\mathcal{G}(f) \cap (\overline{V} \times U^C) \neq \emptyset.$$

But $\mathcal{G}(f)$ and $\overline{V} \times U^C$ are both compact, being closed in $K \times K$, and since the finite intersection property holds, we conclude that there is $(x_0, y_0) \in K \times K$ such that

$$(x_0, y_0) \in \bigcap_{V \ni x} \mathcal{G}(f) \cap (\overline{V} \times U^C).$$

This implies that $x_0 = x$, and since $y_0 \in f(x_0) = f(x)$, it follows that $y_0 \in U$. Contradiction. □

**Theorem 13.1** (Kakutani-Ky Fan) *Let $K$ be a nonempty, compact and convex subset of a locally convex space $X$. Let $f : K \to 2^K$ be upper semicontinuous, such that $f(x)$ is nonempty, convex and closed for every $x \in K$. Then $f$ has a fixed point $\bar{x} \in K$.*

*Proof* If we assume the theorem false, from the Hahn-Banach separation theorem (cf. Introduction) for every $x \in K$ there are $\alpha_x \in \mathbb{R}$ and $\Lambda_x \in X^*$ such that

$$\sup_{z \in f(x)} \operatorname{Re} \Lambda_x z < \alpha_x < \operatorname{Re} \Lambda_x x.$$

The set

$$U_x = \left\{ z \in X : \operatorname{Re} \Lambda_x z < \alpha_x \right\}$$

is open, and contains $f(x)$. Hence there is an open neighborhood $V_x$ of $x$ such that $f(y) \subset U_x$ whenever $y \in V_x \cap K$. We conclude that there exists an open neighborhood $W_x \subset V_x$ of $x$ such that

$$\operatorname{Re} \Lambda_x y > \alpha_x, \qquad \forall y \in W_x \cap K,$$

and

$$\operatorname{Re} \Lambda_x z < \alpha_x, \qquad \forall y \in W_x \cap K, \quad \forall z \in f(y).$$

From the compactness of $K$, there exist $x_1, \ldots, x_n \in K$ such that

$$K \subset W_{x_1} \cup \cdots \cup W_{x_n}.$$

Let $\varphi_1, \ldots, \varphi_n \in C(K)$ be a partition of the unity for $K$ subordinate to the open cover $\{W_{x_j}\}$ (see Lemma 10.1), and define $\Phi : K \times K \to \mathbb{R}$ as

$$\Phi(x, z) = \sum_{j=1}^{n} \varphi_j(x) \operatorname{Re} \Lambda_{x_j} x - \sum_{j=1}^{n} \varphi_j(x) \operatorname{Re} \Lambda_{x_j} z.$$

Clearly, $\Phi$ fulfills the hypotheses of Lemma 13.1. Hence there is $x_0 \in K$ such that

$$\sup_{z\in K} \Phi(x_0, z) \leq \sup_{z\in K} \Phi(z, z) = 0.$$

Therefore, for $z \in f(x_0)$,

$$\begin{aligned}\sum_{j=1}^{n} \varphi_j(x_0)\alpha_{x_j} &< \sum_{j=1}^{n} \varphi_j(x_0)\mathrm{Re}\ \Lambda_{x_j} x_0 \\ &\leq \sum_{j=1}^{n} \varphi_j(x_0)\mathrm{Re}\ \Lambda_{x_j} z \\ &< \sum_{j=1}^{n} \varphi_j(x_0)\alpha_{x_j},\end{aligned}$$

which is a contradiction. Note that in the above sums we have nonzero contributions only for those $j$ for which $x_0 \in W_{x_j}$. □

Theorem 13.1 for finite-dimensional spaces is due to Kakutani [40]. The generalization to infinite-dimensional spaces has been given by Fan [26].

# Part II
# Some Applications

# Chapter 14
# The Implicit Function Theorem




Let $X, Y$ be Banach spaces, let $x_0 \in U \subset X$ with $U$ open, and let $f : U \to Y$.

**Definition 14.1** $f$ is *Fréchet differentiable* at $x_0$ is there exist $T \in L(X, Y)$ and a function $\sigma : X \to Y$, with

$$\lim_{\|x\|_X \to 0} \frac{\|\sigma(x)\|_Y}{\|x\|_X} = 0,$$

such that

$$f(x) - f(x_0) = T(x - x_0) + \sigma(x - x_0), \qquad \forall x \in U.$$

The operator $T$ is called the *Fréchet derivative* of $f$ at $x_0$, and is denoted by $f'(x_0)$. The function $f$ is said to be Fréchet differentiable in $U$ if it is Fréchet differentiable at every $x_0 \in U$.

It is straightforward to verify that the Fréchet derivative at one point, if it exists, is unique.

**Definition 14.2** If $f$ is Fréchet differentiable at $x_0$, we call *tangent* of $f$ at the point $x_0$ the affine map $\tau : X \to Y$

$$\tau(x) = f'(x_0)(x - x_0) + f(x_0).$$

**Lemma 14.1** *Let $X, Y$ be Banach spaces, and let $\lambda \geq 0$ be a given constant. Assume that $f : B_X(0, r) \to Y$ is Fréchet differentiable with*

$$\|f'(x)\|_{L(X,Y)} \leq \lambda, \qquad \forall x \in B_X(0, r).$$

*Then $f$ is Lipschitz continuous with Lipschitz constant less than or equal to $\lambda$.*

*Proof* Let $x_1, x_2 \in B_X(0, r)$. By the Hahn-Banach theorem (cf. Introduction), there is $\Lambda \in Y^*$ of unit norm such that

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$$\|f(x_1) - f(x_2)\|_Y = |\Lambda(f(x_1) - f(x_2))|.$$

For $t \in [0, 1]$ set

$$\Phi(t) = \Lambda f(tx_1 + (1-t)x_2).$$

Applying the Lagrange mean value theorem to $\Phi$, there is $\tau \in (0, 1)$ such that

$$|\Phi(1) - \Phi(0)| \leq |\Phi'(\tau)|,$$

where we have equality if $X$ is a real Banach space. Since the chain rule holds as in the classical case, we get

$$|\Lambda f(x_1) - \Lambda f(x_2)| \leq |\Lambda f'(\tau x_1 + (1-\tau)x_2)(x_1 - x_2)|.$$

Hence

$$\|f(x_1) - f(x_2)\|_Y \leq \|f'(\tau x_1 + (1-\tau)x_2)(x_1 - x_2)\|_Y \leq \lambda \|x_1 - x_2\|_X,$$

as claimed. □

Given two Banach spaces $X$ and $Y$, the product space $X \times Y$ is a Banach space with any of the (equivalent) norms

$$\|(x, y)\|_p = \begin{cases} \left(\|x\|_X^p + \|y\|_Y^p\right)^{1/p} & \text{if } p \in [1, \infty), \\ \max\left\{\|x\|_X, \|y\|_Y\right\} & \text{if } p = \infty. \end{cases}$$

**A word of warning**

In the sequel, we will always use the $\infty$-norm of $X \times Y$, so that

$$B_{X\times Y}((x_0, y_0), r) = B_X(x_0, r) \times B_Y(y_0, r),$$

and

$$\overline{B}_{X\times Y}((x_0, y_0), r) = \overline{B}_X(x_0, r) \times \overline{B}_Y(y_0, r).$$

Given $X, Y, Z$ Banach spaces and two operators $T \in L(X, Z)$ and $R \in L(Y, Z)$, the sum operator $S : X \times Y \to Z$ defined by

$$S(x, y) = Tx + Ry$$

belongs to $L(X \times Y, Z)$. Conversely, any $S \in L(X \times Y, Z)$, has the above representation with

$$Tx = S(x, 0) \qquad \text{and} \qquad Ry = S(0, y).$$

It is then immediate to see that $L(X, Z) \times L(Y, Z)$ and $L(X \times Y, Z)$ are isomorphic Banach spaces. Given

$$f : U \subset X \times Y \to Z,$$

with $U$ open set, if $f$ Fréchet differentiable at $u_0 = (x_0, y_0) \in U$, one easily checks that the partial derivatives

$$D_x f(u_0) \qquad \text{and} \qquad D_y f(u_0)$$

exist, namely, the Fréchet derivatives of $f(\cdot, y_0) : X \to Z$ in $x_0$ and of $f(x_0, \cdot) : Y \to Z$ in $y_0$. Moreover,

$$f'(u_0)(x, y) = D_x f(u_0)x + D_y f(u_0)y.$$

We are now in the position to state the famous result due to Dini [21].

**Theorem 14.1** (Implicit function theorem) *Let $X$, $Y$, $Z$ be three Banach spaces, let $u_0 = (x_0, y_0) \in U \subset X \times Y$ with $U$ open set, and let $F : U \to Z$. Moreover, let the following assumptions hold:*

- *$F$ is continuous and $F(u_0) = 0$.*
- *$D_y F(u)$ exists for every $u = (x, y) \in U$.*
- *$D_y F$ is continuous at $u_0$ and $D_y F(u_0)$ is invertible.*

*Then there exist $\alpha, \beta > 0$ for which $\overline{B}_X(x_0, \alpha) \times \overline{B}_Y(y_0, \beta) \subset U$ and a unique continuous function $f : \overline{B}_X(x_0, \alpha) \to \overline{B}_Y(y_0, \beta)$ such that the relation*

$$F(x, y) = 0 \qquad \Leftrightarrow \qquad y = f(x)$$

*holds for all $(x, y) \in \overline{B}_X(x_0, \alpha) \times \overline{B}_Y(y_0, \beta)$.*

*Proof* Without loss of generality, we can assume $x_0 = 0$ and $y_0 = 0$. Define the continuous function $G : U \to Y$ as

$$G(x, y) = y - [D_y F(0, 0)]^{-1} F(x, y).$$

Since

$$D_y G(x, y) = [D_y F(0, 0)]^{-1} \left( D_y F(0, 0) - D_y F(x, y) \right)$$

and $D_y F$ is continuous at $(0, 0)$, there is $\gamma > 0$ small enough such that

$$\| D_y G(x, y) \|_{L(Y)} \leq \frac{1}{2},$$

for every $(x, y) \in B_X(0, \gamma) \times B_Y(0, \gamma) \subset U$. From Lemma 14.1 we draw the inequality

$$\|G(x, y_1) - G(x, y_2)\|_Y \leq \frac{1}{2}\|y_1 - y_2\|_Y,$$

for every $x, y_1, y_2$ such that

$$\max\left\{\|x\|_X, \|y_1\|_Y, \|y_2\|_Y\right\} \leq \beta < \gamma.$$

Besides, using the continuity of $G$ at $(0, 0)$, we find $0 < \alpha < \beta$ such that

$$\|G(x, 0)\|_Y \leq \frac{\beta}{2},$$

whenever $\|x\|_X \leq \alpha$. Then, for $\|x\|_X \leq \alpha$ and $\|y\|_Y \leq \beta$,

$$\|G(x, y)\|_Y \leq \|G(x, 0)\|_Y + \|G(x, y) - G(x, 0)\|_Y \leq \frac{1}{2}\big(\beta + \|y\|_Y\big) \leq \beta.$$

Therefore the continuous map

$$G : \overline{B}_X(0, \alpha) \times \overline{B}_Y(0, \beta) \to \overline{B}_Y(0, \beta)$$

is a contraction on $\overline{B}_Y(0, \beta)$ uniformly in $\overline{B}_X(0, \alpha)$. From Corollary 1.2, there exists a unique continuous function

$$f : \overline{B}_X(0, \alpha) \to \overline{B}_Y(0, \beta)$$

such that $G(x, f(x)) = f(x)$, that is, $F(x, f(x)) = 0$. □

It is clear that the conclusions of Theorem 14.1 remain valid if one replaces closed balls with open balls.

**Corollary 14.1** *Let the hypotheses of Theorem 14.1 hold. If in addition $F$ is Fréchet differentiable at $u_0 = (x_0, y_0)$, then $f$ is Fréchet differentiable at $x_0$, and*

$$f'(x_0) = -[D_y F(u_0)]^{-1} D_x F(u_0).$$

*Proof* Applying the definition of Fréchet differentiability to $F(x, f(x))$ at the point $(x_0, f(x_0))$, we get

$$0 = D_x F(u_0)(x - x_0) + D_y F(u_0)(f(x) - f(x_0)) + \sigma(x - x_0, f(x) - f(x_0)).$$

Note that the above relation implies that $f$ is locally Lipschitz at $x_0$. Hence,

$$\frac{\|[D_y F(u_0)]^{-1}\sigma(x - x_0,\, f(x) - f(x_0))\|_Y}{\|x - x_0\|_X}$$
$$\leq \|[D_y F(u_0)]^{-1}\|_{L(Z,Y)} \frac{\|\sigma(x - x_0,\, f(x) - f(x_0))\|_Z}{\|x - x_0\|_X} \to 0$$

as $\|x - x_0\|_X \to 0$, which yields the thesis. □

A consequence of Theorem 14.1 is the inverse function theorem.

**Theorem 14.2** (Inverse function theorem) *Let $X, Y$ be two Banach spaces, and let $y_0 \in V \subset Y$ with $V$ open. Assume that $g : V \to X$ is a Fréchet differentiable function in a neighborhood of $y_0$ satisfying the following assumptions:*

- $g(y_0) = x_0$.
- $g'$ *is continuous at* $y_0$.
- $g'(y_0)$ *is invertible.*

*Then there are $\alpha, \beta > 0$ and a unique continuous function $f : B_X(x_0, \alpha) \to B_Y(y_0, \beta)$ such that $x = g(f(x))$ for every $x \in B_X(x_0, \alpha)$. Moreover, $f$ is Fréchet differentiable at $x_0$ and*

$$f'(x_0) = [g'(y_0)]^{-1}.$$

*Proof* Apply Theorem 14.1 and the subsequent corollary to $F(x, y) = g(y) - x$. □

Theorem 14.1 can also be exploited to provide an alternative proof to the well-known fact that the set of invertible bounded linear operators between Banach spaces is open.

**Theorem 14.3** *Let $X, Y$ be Banach spaces, and let $L_{\text{inv}}(X, Y) \subset L(X, Y)$ be the set of invertible bounded linear operators from $X$ onto $Y$. Then $L_{\text{inv}}(X, Y)$ is open in $L(X, Y)$. Moreover, if $T \in L_{\text{inv}}(X, Y)$, then the map $T \mapsto T^{-1}$ is continuous.*

*Proof* In what follows, $I_X$ and $I_Y$ denote the identity maps on $X$ and $Y$, respectively. Let $F : L(X, Y) \times L(Y, X) \to L(Y)$ defined by

$$F(T, S) = I_Y - TS.$$

Let $T_0 \in L_{\text{inv}}(X, Y)$, and set $S_0 = T_0^{-1}$. Note that

$$D_S F(T, S)R = -TR.$$

In particular,

$$D_S F(T_0, S_0)R = -T_0 R.$$

The hypotheses of Theorem 14.1 are then satisfied, and there exist $\alpha > 0$ and a continuous function

$$f : B_{L(X,Y)}(T_0, \alpha) \to L(Y, X)$$

such that

$$I_Y - Tf(T) = 0 \qquad \Rightarrow \qquad Tf(T) = I_Y.$$

Analogously (perhaps for a smaller $\alpha$), we can find a continuous function

$$g : B_{L(X,Y)}(T_0, \alpha) \to L(Y, X)$$

such that

$$g(T)T = I_X.$$

It is straightforward to verify that $f \equiv g$, that is,

$$f(T) = T^{-1}$$

for all $T \in B_{L(X,Y)}(T_0, \alpha)$. □

# Chapter 15
# Location of Zeros




Let $X$, $Y$ be Banach spaces, and $f : B_X(x_0, r) \to Y$ be a Fréchet differentiable map. In order to find a zero for $f$, the idea is to apply an iterative method constructing a sequence $x_n$ starting from $x_0$ so that $x_{n+1}$ is the zero of the tangent $\tau$ of $f$ at the point $x_n$ (see Definition 14.2). Assuming then $[f'(x)]^{-1} \in L(Y, X)$ on $B_X(x_0, r)$, one has

$$x_{n+1} = x_n - [f'(x_n)]^{-1} f(x_n), \tag{15.1}$$

provided that $x_n \in B_X(x_0, r)$ for every $n$. This procedure is known as the *Newton-Kantorovich method* (see [38]). However, for practical purposes, it might be complicated to invert $f'$ at each step. So one can try the modification

$$x_{n+1} = x_n - [f'(x_0)]^{-1} f(x_n), \tag{15.2}$$

which presents the further advantage that $f'(x)$ is required to be invertible only at the point $x = x_0$. At the same time, using (15.2) in place of (15.1), a lower convergence rate of $x_n$ to the zero of $f$ is to be expected.

The following result is based on (15.2).

**Theorem 15.1** *Let $X$, $Y$ be Banach spaces, and $f : B_X(x_0, r) \to Y$ be a Fréchet differentiable map such that $f'(x_0)$ is invertible with $[f'(x_0)]^{-1} \in L(Y, X)$. Assume that, for some $\lambda > 0$ and every $x \in B_X(x_0, r)$,*

$$\|f'(x) - f'(x_0)\|_{L(X,Y)} \leq \lambda \|x - x_0\|_X.$$

*Besides, let*

$$s \doteq 2\|[f'(x_0)]^{-1} f(x_0)\|_X < r,$$

*and*

$$\mu \doteq 2\lambda s \|[f'(x_0)]^{-1}\|_{L(Y,X)} \leq 1.$$

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*Then there exists a unique* $\bar{x} \in \overline{B}_X(x_0, s)$ *such that* $f(\bar{x}) = 0$.

*Proof* Define the function $F : \overline{B}_X(x_0, s) \to X$ as

$$F(x) = x - [f'(x_0)]^{-1} f(x).$$

Then, for every $x \in \overline{B}_X(x_0, s)$,

$$\begin{aligned} \|F'(x)\|_{L(X)} &\le \|[f'(x_0)]^{-1}\|_{L(Y,X)} \|f'(x_0) - f'(x)\|_{L(X,Y)} \\ &\le \lambda s \|[f'(x_0)]^{-1}\|_{L(Y,X)} \\ &= \frac{\mu}{2}. \end{aligned}$$

By Lemma 14.1, we infer that $F$ is Lipschitz, with Lipschitz constant less than or equal to $\mu/2 \le 1/2$. Moreover,

$$\|F(x_0) - x_0\|_X = \|[f'(x_0)]^{-1} f(x_0)\|_X = \frac{s}{2},$$

which in turn gives

$$\begin{aligned} \|F(x) - x_0\|_X &\le \|F(x) - F(x_0)\|_X + \|F(x_0) - x_0\|_X \\ &\le \frac{\mu}{2} \|x - x_0\|_X + \frac{s}{2} \\ &\le s. \end{aligned}$$

Hence $F$ is a contraction on $\overline{B}_X(x_0, s)$. From the BCP Theorem 1.1 there exists a unique $\bar{x} \in \overline{B}_X(x_0, s)$ such that $F(\bar{x}) = \bar{x}$, which implies $f(\bar{x}) = 0$. □

Concerning the convergence speed of the sequence $x_n = F^n(x_0)$ to $\bar{x}$, by virtue of Corollary 1.1, we get

$$\|x_n - \bar{x}\|_X \le \frac{s\mu^n}{(2 - \mu)2^n}.$$

Also, since

$$\begin{aligned} x_{n+1} - \bar{x} &= F(x_n) - F(\bar{x}) \\ &= F'(\bar{x})(x_n - \bar{x}) + \sigma(x_n - \bar{x}) \\ &= [f'(x_0)]^{-1}(f'(x_0) - f'(\bar{x}))(x_n - \bar{x}) + \sigma(x_n - \bar{x}), \end{aligned}$$

where

$$\lim_{\|x\|_X \to 0} \frac{\|\sigma(x)\|_X}{\|x\|_X} = 0,$$

it follows that

$$\|x_{n+1} - \bar{x}\|_X \leq \lambda s \|[f'(x_0)]^{-1}\|_{L(Y,X)} \|x_n - \bar{x}\|_X + \|\sigma(x_n - \bar{x})\|_X$$
$$= \frac{\mu}{2} \|x_n - \bar{x}\|_X + \|\sigma(x_n - \bar{x})\|_X.$$

Hence

$$\|x_{n+1} - \bar{x}\|_X \leq c \|x_n - \bar{x}\|_X$$

for some $c < 1$ and all $n$ large. This is usually referred to as *linear convergence* of the method.

*Remark* If in addition $f : B_X(x_0, r) \to Y$ is a map of class $\mathcal{C}^1$, with $x \mapsto f'(x)$ Lipschitz with Lipschitz constant equal to $\lambda$, then the result can be proved (in fact for $\mu < 2$) also by exploiting the iterative scheme (15.1), with a completely different proof (see, e.g., [18], pp. 157–159). In this case, the convergence speed of the sequence $x_n$ defined iteratively in formula (15.1) to $\bar{x}$ turns out to be faster, as expected. Indeed, one obtains the estimates

$$\|x_n - \bar{x}\|_X \leq \frac{s}{2^n} \left(\frac{\mu}{2}\right)^{2^n - 1},$$

and

$$\|x_{n+1} - \bar{x}\|_X \leq c \|x_n - \bar{x}\|_X^2,$$

for some $c < 1$ and all $n$ large. In this case, we speak of *quadratic convergence* of the method.

# Chapter 16
# Ordinary Differential Equations in Banach Spaces




Let $X$ be a Banach space, and let $[a, b]$ be a closed bounded interval of the real line. The notion of Riemann integral and the related properties can be extended with no differences from the case of real-valued functions to $X$-valued functions on $[a, b]$. In particular, if $f \in C([a, b], X)$ then $f$ is Riemann integrable on $[a, b]$, and the following hold:

$$\left\| \int_a^b f(t)\, dt \right\|_X \leq \int_a^b \|f(t)\|_X\, dt,$$

and

$$\frac{d}{dt} \int_a^t f(s)\, ds = f(t), \qquad \forall t \in [a, b].$$

Recall that a function $h : [a, b] \to X$ is differentiable at $t_0 \in [a, b]$ if the limit

$$\lim_{t \to t_0} \frac{h(t) - h(t_0)}{t - t_0}$$

exists in $X$ (if $t_0 = a$ or $t_0 = b$ we take right or left limits, respectively). The value of the limit is the derivative of $h$ at $t_0$, denoted by

$$h'(t_0) \qquad \text{or} \qquad \frac{d}{dt} h(t_0).$$

If $t_0 \in (a, b)$ we recover the definition of Fréchet differentiability.

**Proposition 16.1** *If $h : [a, b] \to X$ is differentiable in $[a, b]$ and $h'(t) = 0$ for all $t \in [a, b]$, then $h$ is constant.*

*Proof* For every $\Lambda \in X^*$, we have

$$(\Lambda \circ h)'(t) = \Lambda h'(t) = 0,$$

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in turn implying

$$\Lambda(h(t) - h(a)) = 0.$$

But then, from the Hahn-Banach theorem (cf. Introduction), there is $\Lambda \in X^*$ such that

$$\Lambda(h(t) - h(a)) = \|h(t) - h(a)\|.$$

In conclusion, $h(t) = h(a)$ for every $t \in [a, b]$. □

**The Cauchy Problem**

Let $X$ be a Banach space, $U \subset \mathbb{R} \times X$ be an open set, and let $f : U \to X$ be a continuous function. Given $u_0 = (t_0, x_0) \in U$, the problem is to find a closed interval $I$, with $t_0$ belonging to the interior of $I$, along with a differentiable function $x : I \to X$ such that

$$\begin{cases} x'(t) = f(t, x(t)) & t \in I, \\ x(t_0) = x_0. \end{cases} \tag{16.1}$$

It is apparent that such an $x$ is automatically of class $C^1$ on $I$. Also, it is readily seen that (16.1) is equivalent to the integral equation

$$x(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds, \qquad t \in I. \tag{16.2}$$

Namely, $x$ is a solution to (16.1) if and only if it is a solution to (16.2).

As we already did before, we will use in the sequel the $\infty$-norm on the product Banach space $\mathbb{R} \times X$.

**Theorem 16.1** (Local solution) *Let there exists $r > 0$ such that*

$$B_r \doteq \overline{B}_{\mathbb{R}\times X}(u_0, r) \subset U$$

*and the following hold:*

1. *There exists a nonnegative function $k \in L^1(t_0 - r, t_0 + r)$ such that*

$$\|f(t, x_1) - f(t, x_2)\|_X \le k(t)\|x_1 - x_2\|_X, \qquad \forall (t, x_1), (t, x_2) \in B_r.$$

2. *There exists $m \ge 0$ such that*

$$\|f(t, x)\|_X \le m, \qquad \forall (t, x) \in B_r.$$

*Then there is $\tau > 0$ such that, setting $I_\tau = [t_0 - \tau, t_0 + \tau]$, there exists a unique solution $x \in C^1(I_\tau, X)$ to the Cauchy problem (16.1).*

*Remark* Before going to the proof, we observe that if the function $k \geq 0$ of point 1 is constant, then point 2 is automatically satisfied. Indeed, for $(x, t) \in B_r$, we get

$$\begin{aligned} \|f(t,x)\|_X &\leq \|f(t,x) - f(t,x_0)\|_X + \|f(t,x_0)\|_X \\ &\leq kr + \max_{|t-t_0|\leq r} \|f(t,x_0)\|_X, \end{aligned}$$

since the map $t \mapsto \|f(t,x_0)\|_X : [t_0 - r, t_0 + r] \to \mathbb{R}$ is continuous. Also, point 2 is always true if $X$ is finite-dimensional, for $f$ is continuous and closed balls are compact.

*Proof* Let us set

$$\tau = \min\left\{r, \frac{r}{m}\right\}.$$

Consider then the complete metric space

$$Z = \overline{B}_{C(I_\tau, X)}(x_0, r),$$

with the metric induced by the norm of $C(I_\tau, X)$. Here, $x_0$ is understood to be the constant function equal to $x_0$. Since $\tau < r$, for all $t \in I_\tau$ we have the implication

$$z \in Z \qquad \Rightarrow \qquad (t, z(t)) \in B_r.$$

Hence, for $z \in Z$, define

$$[F(z)](t) = x_0 + \int_{t_0}^{t} f(s, z(s))\, ds, \qquad t \in I_\tau.$$

Due to point 2,

$$\sup_{t\in I_\tau} \|[F(z)](t) - x_0\|_X \leq \sup_{t\in I_\tau} \left| \int_{t_0}^{t} \|f(s,z(s))\|_X\, ds \right| \leq m\tau \leq r.$$

We conclude that $F$ maps $Z$ into $Z$. The last step is showing that $F^n$ is a contraction on $Z$ for some integer $n \geq 1$. To this end, by induction on $n$, we prove that for every $t \in I_\tau$

$$\|[F^n(z_1)](t) - [F^n(z_2)](t)\|_X \leq \frac{1}{n!} \left| \int_{t_0}^{t} k(s)\, ds \right|^n \|z_1 - z_2\|_{C(I_\tau, X)}. \tag{16.3}$$

For $n = 1$, this follows immediately from point 1. So, assume it is true for $n - 1$, with $n \geq 2$. Then, taking $t > t_0$ (the argument for $t < t_0$ is analogous), and using again point 1,

$$
\begin{aligned}
&\|[F^n(z_1)](t) - [F^n(z_2)](t)\|_X \\
&\quad = \|[F(F^{n-1}(z_1))](t) - [F(F^{n-1}(z_2))](t)\|_X \\
&\quad \le \int_{t_0}^t \|f\big(s, [F^{n-1}(z_1)](s)\big) - f\big(s, [F^{n-1}(z_2)](s)\big)\|_X \, ds \\
&\quad \le \int_{t_0}^t k(s)\|[F^{n-1}(z_1)](s) - [F^{n-1}(z_2)](s)\|_X \, ds \\
&\quad \le \frac{1}{(n-1)!}\left[\int_{t_0}^t k(s)\left(\int_{t_0}^s k(\sigma)\, d\sigma\right)^{n-1} ds\right]\|z_1 - z_2\|_{C(I_\tau, X)} \\
&\quad = \frac{1}{n!}\left(\int_{t_0}^t k(s)\, ds\right)^n \|z_1 - z_2\|_{C(I_\tau, X)}.
\end{aligned}
$$

Therefore, from (16.3) we learn that

$$
\|F^n(z_1) - F^n(z_2)\|_{C(I_\tau, X)} \le \frac{1}{n!}\|k\|^n_{L^1(I_\tau)}\|z_1 - z_2\|_{C(I_\tau, X)},
$$

which shows that $F^n$ is a contraction, up to choosing $n$ large enough. By means of Corollary 1.3, we conclude that $F$ admits a unique fixed point, which is clearly the unique solution to the integral equation (16.2), and so the unique solution to the Cauchy problem (16.1). □

**Theorem 16.2** (Global solution) *Assume that $U = (a, b) \times X$, and let there exists a nonnegative function $k \in L^1(a, b)$ such that*

$$
\|f(t, x_1) - f(t, x_2)\|_X \le k(t)\|x_1 - x_2\|_X, \qquad \forall (t, x_1), (t, x_2) \in U.
$$

*Then, for every $t_0 \in (a, b)$, every closed interval $I \subset [a, b]$ containing $t_0$, and every initial datum $x_0 \in X$, the Cauchy problem (16.1) admits a unique solution $x \in C^1(I, X)$.*

*Proof* Recast word by word the proof of Theorem 16.1, taking now

$$
Z = C(I, X).
$$

The details are left to the reader. □

Theorems 16.1 and 16.2 are refined versions of the so-called "method of successive approximations", devised by Cauchy and Liouville, and developed in the most general form by Picard [62].

The next step is showing that the solution depends with continuity on the initial data. To this end, the crucial tool is the celebrated Gronwall lemma [31]. Actually, the proof reported here differs from the original one in [31], and is a particular case of the proof of a generalized version of the lemma (see [58]).

**Lemma 16.1** (Gronwall) *Let $\varphi : [a, b] \to [0, \infty)$ be a bounded measurable function, and let $k \in L^1(a, b)$ be nonnegative. For a fixed $t_0 \in [a, b]$ and $c \geq 0$, assume to have the inequality*

$$\varphi(t) \leq c + \left| \int_{t_0}^{t} k(s)\varphi(s)\, ds \right|,$$

*for every $t \in [a, b]$. Then the inequality*

$$\varphi(t) \leq c\, \exp\left[ \left| \int_{t_0}^{t} k(s)\, ds \right| \right]$$

*holds for every $t \in [a, b]$.*

*Proof* We suppose $t_0 < b$ and we prove the claim for every $t \in [t_0, b]$. The case $t_0 > a$ and $t \in [a, t_0]$ is completely analogous and left to the reader. For $\mu > 0$ arbitrarily fixed, define then the function $\psi : [t_0, b] \to \mathbb{R}$ as

$$\psi(t) = (c + \mu)\, \exp\left[ \int_{t_0}^{t} k(s)\, ds \right].$$

Note that $\psi$ is absolutely continuous, hence differentiable almost everywhere with $\psi' \in L^1(t_0, b)$. By a direct calculation, we have the equality

$$\psi'(s) = k(s)\psi(s),$$

which, integrated on $[t_0, t]$, yields

$$\psi(t) = c + \mu + \int_{t_0}^{t} k(s)\psi(s)\, ds, \qquad \forall t \in [t_0, b].$$

Therefore, the difference

$$\zeta(t) = \varphi(t) - \psi(t)$$

satisfies

$$\zeta(t) \leq -\mu + \int_{t_0}^{t} k(s)\zeta(s)\, ds, \qquad \forall t \in [t_0, b].$$

Since $\zeta(t_0) \leq -\mu$ and $k$ is nonnegative, this tells that $\zeta$ remains negative in the whole interval $[t_0, b]$. If not, calling $t_* = \sup\left\{ t \geq t_0\,;\ \zeta(t) < 0 \right\}$ and using the fact that $\zeta$ is bounded, there exists a sequence $t_n \downarrow t_*$ such that

$$0 \leq \zeta(t_n) \leq -\mu + \int_{t_*}^{t_n} k(s)\zeta(s)\, ds \to -\mu$$

as $n \to \infty$, which is a contradiction. Thus,

$$\varphi(t) < (c+\mu)\,\exp\left[\int_{t_0}^{t} k(s)\,ds\right], \qquad \forall t \in [t_0, b],$$

and the desired conclusion follows by letting $\mu \to 0$.

**Proposition 16.2** (Continuous dependence) *Let the hypotheses of Theorem 16.2 hold. Given any two solutions $x_1, x_2 \in C(I, X)$ to (16.1) with initial data $x_{01}, x_{02} \in X$ assigned at $t = t_0$, we have the continuous dependence estimate*

$$\|x_1(t) - x_2(t)\|_X \le \|x_{01} - x_{02}\|_X\,\exp\left[\left|\int_{t_0}^{t} k(s)\,ds\right|\right], \qquad \forall\, t \in I.$$

*Proof* The difference $x = x_1 - x_2$ fulfills the integral equation

$$x(t) = x_0 + \int_{t_0}^{t} \big[f(s, x_1(s)) - f(s, x_2(s))\big]\,ds, \qquad t \in I,$$

having set $x_0 = x_{01} - x_{02}$. Then, denoting

$$\varphi(t) = \|x(t)\|_X,$$

we readily obtain the inequality

$$\varphi(t) \le \|x_0\|_X + \left|\int_{t_0}^{t} k(s)\varphi(s)\,ds\right|, \qquad \forall t \in I,$$

and the claim follows from the Gronwall Lemma 16.1. □

*Remark* It is clear from the proof that Proposition 16.2 applies also in the hypotheses of Theorem 16.1. In this case, we have to assume that the two solutions are defined on a common interval $I_\tau$, and that the Lipschitz condition stated in point 1 holds wherever needed.

When $f$ is merely continuous and fulfills a compactness property, it is still possible to provide the existence of solutions by exploiting the Schauder-Tychonoff theorem. The result, that commonly goes by the name of Peano theorem, has been actually proved by Peano [60] for the case of a finite-dimensional Banach space $X$.

**Theorem 16.3** (Peano) *Let $f$ be a continuous function, and $f(V)$ be relatively compact in $X$ for some open neighborhood $V \subset U$ of $u_0$. Then there exists $\tau > 0$ such that there is a (possibly nonunique) solution $x \in C^1(I_\tau, X)$ to the Cauchy problem (16.1), with $I_\tau = [t_0 - \tau, t_0 + \tau]$.*

*Proof* Choose $r > 0$ such that $B_r = \overline{B}_{\mathbb{R}\times X}(u_0, r) \subset V$, and let

$$m = \sup\big\{\|f(t,x)\|_X \,:\, (x,t) \in B_r\big\}.$$

Finally, set $\tau = r/m$. Then the map $z \mapsto F(z)$, defined as

$$[F(z)](t) = x_0 + \int_{t_0}^{t} f(s, z(s))\, ds, \qquad t \in I_\tau,$$

is a selfmap of the closed and convex set

$$Z = \overline{B}_{C(I_\tau, X)}(x_0, r).$$

Moreover, $F$ is continuous, since for $z_n, z \in Z$,

$$\sup_{t \in I_\tau} \|[F(z_n)](t) - [F(z)](t)\|_X \le \int_{t_0-\tau}^{t_0+\tau} \|f(s, z_n(s)) - f(s, z(s))\|_X\, ds,$$

which vanishes as $z_n \to z$ in $C(I_\tau, X)$, due to the dominated convergence theorem. The Cauchy problem (16.1) has a solution if $F$ has a fixed point, which is guaranteed by Theorem 10.1 once we show that $F(Z)$ is relatively compact. To establish this fact, we appeal to the Ascoli theorem for X-valued continuous functions (see, e.g., Theorem 0.4.11 in [23]).

• First we observe that $F(Z)$ is an equicontinuous family, since for any $z \in Z$ and $t_1, t_2 \in I_\tau$,

$$\|[F(z)](t_1) - [F(z)](t_2)\|_X = \left\| \int_{t_2}^{t_1} f(s, z(s))\, ds \right\|_X \le m|t_1 - t_2|.$$

• Next we prove that, for every fixed $t \in I_\tau$, the set

$$\bigcup_{z \in Z} [F(z)](t)$$

is relatively compact. From the hypotheses, there is a compact set $K \subset X$ such that $f(s, z(s)) \in K$ for all $s \in I_\tau$ and all $z \in Z$. Let $H = \mathrm{co}(K)$. It is well-known that in a Banach space the convex hull of a compact set is relatively compact (see, e.g., [66], Theorem 3.25). Then, for a fixed $t \in I_\tau$ and every $z \in Z$ we have

$$[F(z)](t) = x_0 + \int_{t_0}^{t} f(s, z(s))\, ds \in x_0 + (t - t_0)\overline{H},$$

which is a compact set. Indeed,

$$\int_{t_0}^{t} f(s, z(s))\, ds$$

is the limit of the Riemann sums

$$(t - t_0) \sum_i f(s_i, z(s_i)) \frac{s_i - s_{i-1}}{t - t_0}.$$

In summary, the assumptions of the Ascoli theorem are satisfied, and we conclude that $F(Z)$ is a relatively compact subset of $C(I_\tau, X)$, hence of the closed set $Z$. At this point, Theorem 10.1 applies, providing the existence of a (possibly nonunique) fixed point of the map $F$, which is nothing but the sought solution to the Cauchy problem (16.1). □

**Definition 16.1** The Peano theorem is said to hold in a Banach space $X$ if, for an arbitrary continuous function $f$ and an arbitrary $x_0 \in X$, the Cauchy problem (16.1) admits a solution $x(t)$ in some neighborhood of $t_0$.

It is clear from Theorem 16.3 that if $X$ is finite-dimensional the Peano theorem holds, since finite-dimensional Banach spaces have the Heine-Borel property, so that $f(V)$ is relatively compact whenever $\overline{V}$ is (bounded and) contained in $U$. What about infinite-dimensional Banach spaces? Let us examine a famous counterexample to the existence problem (see [20]).

**Example** (Dieudonné) Take the Banach space $c_0$ of real-valued sequences vanishing at infinity with the supremum norm, and define the function $f : c_0 \to c_0$ by

$$(f(x))_n = \sqrt{|x_n|} + \frac{1}{1+n}, \qquad x = (x_0, x_1, x_2, \ldots).$$

Consider the Cauchy problem

$$\begin{cases} x'(t) = f(x(t)) & t \in (-\varepsilon, \varepsilon), \\ x(0) = 0. \end{cases}$$

Due to uniform continuity of the real map $s \mapsto \sqrt{|s|}$, $f$ is uniformly continuous on $c_0$. Assume now that the Cauchy problem admits a solution $y \in C^1((-\varepsilon, \varepsilon), c_0)$ for some $\varepsilon > 0$. In particular, each component $y_n$ is differentiable in $(-\varepsilon, \varepsilon)$ and fulfills the Cauchy problem

$$\begin{cases} y_n'(t) = \sqrt{|y_n(t)|} + \dfrac{1}{1+n} & t \in (-\varepsilon, \varepsilon), \\ y_n(0) = 0. \end{cases}$$

Then $y_n(t) > 0$ for $t \in (0, \varepsilon)$, and

$$2\sqrt{y_n(t)} - \frac{2}{1+n}\log\left(\sqrt{y_n(t)} + \frac{1}{1+n}\right) + \frac{2}{1+n}\log\left(\frac{1}{1+n}\right) = t, \qquad \forall t \in (0, \varepsilon).$$

On the other hand, $y(t) \in c_0$, that is, $\lim_{n\to\infty} y_n(t) = 0$. Therefore the limit as $n \to \infty$ of the left-hand side of the equality above must be 0 for all $t \in (0, \varepsilon)$. This is a contradiction, and we conclude that no such solution $y$ exists.

In fact, the Peano theorem provides a characterization of finite-dimensional Banach spaces. This is the last result of the chapter.

**Theorem 16.4** (Yorke-Cellina-Godunov) *If the Peano theorem holds in a Banach space $X$, then $X$ is finite-dimensional.*

Theorem 16.4 has been first proved by Yorke [75] for separable Hilbert spaces, and then by Cellina [15] for nonreflexive Banach spaces. The result in its full generality, is due to Godunov [29]. Here, we report the proof of Yorke, namely, we restrict ourselves to the case of a separable Hilbert space $X$. Since every (infinite-dimensional) separable Hilbert space is isomorphic to the space of summable sequences $\ell^2$, we may (and do) assume $X = \ell^2$.

*Proof* We construct a continuous function $f : \mathbb{R} \times \ell^2 \to \ell^2$ such that the Cauchy problem

$$\begin{cases} x'(t) = f(t, x(t)) \quad t \in I, \\ x(0) = 0, \end{cases} \tag{16.4}$$

has no solution on any open interval $I$ containing 0.

• For every $x = (x_0, x_1, x_2, \ldots) \in \ell^2$, let us introduce the projection $P_n$, with $n \in \mathbb{N}$, by

$$P_n x = (0, \ldots, 0, x_n, x_{n+1}, x_{n+2}, \ldots).$$

For $t \in \mathbb{R}$ and $x \in \ell^2$, define

$$P(t)x = \begin{cases} 0 & \text{if } t \le 0, \\ x & \text{if } t \ge 1, \\ (2 - 2^{n+1}t)P_{n+1}x + (2^{n+1}t - 1)P_n x & \text{if } t \in [2^{-n-1}, 2^{-n}). \end{cases}$$

We claim that $P$ is continuous on $\mathbb{R} \times \ell^2$. Indeed, $P$ is clearly continuous at $(t, x)$ if $t \neq 0$. Let then $t_k \to 0$, and let $x^k = (x_0^k, x_1^k, x_2^k, \ldots) \in \ell^2$ be a sequence converging to $x = (x_0, x_1, x_2, \ldots)$. For each fixed $n$,

$$\|P(t_k)x^k\|^2 \le \sum_{i=n}^{\infty} |x_i^k|^2 \qquad \text{for} \qquad t_k \le 2^{-n}.$$

Accordingly

$$\limsup_{k\to\infty} \|P(t_k)x^k\|^2 \le \sum_{i=n}^{\infty} |x_i|^2.$$

Letting $n \to \infty$ the claim follows.

• Define now

$$G(x) = \begin{cases} x/\sqrt{\|x\|} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

and

$$A(x) = (|x_0|, |x_1|, |x_2|, \ldots).$$

Finally, calling $\xi = (2^{-0}, 2^{-1}, 2^{-2}, \ldots) \in \ell^2$, set

$$f(t, x) = G(P(t)A(x)) + \max\left\{0\,,\ \tfrac{1}{4}t^2 - \|x\|\right\} P(t/2)\xi,$$

and write $f = (f_0, f_1, f_2, \ldots)$. Note that $f : \mathbb{R} \times \ell^2 \to \ell^2$ is continuous.

• Arguing by contradiction, let now $x(t) = (x_0(t), x_1(t), x_2(t), \ldots)$ be a solution to the Cauchy problem (16.4). Since $P(t) = 0$ for $t \le 0$, we have

$$x(t) = 0 \qquad \text{for} \qquad t \le 0.$$

From the definition of $A(\cdot)$, we readily see that $f_n(t, x) \ge 0$ for all $n, t, x$. Hence each $x_n(t)$ is nondecreasing and $x_n(t) \ge 0$ for $t \in I$, implying that

$$A(x(t)) = x(t).$$

Besides, for $t > 0$,

$$f(t, 0) = \frac{1}{4}t^2 P(t/2)\xi \neq 0,$$

so $x(t) \neq 0$ for $t > 0$.

• For all $t \le 2^{-n-1}$, we have that

$$f_n(t, x) = 0 \qquad \Rightarrow \qquad x_n'(t) = 0 \qquad \Rightarrow \qquad x_n(t) = 0.$$

Moreover,

$$x_n'(t) = \frac{(2^{n+1}t - 1)x_n(t)}{\sqrt{\|x(t)\|}}, \qquad \forall t \in [2^{-n-1}, 2^{-n}].$$

Since the map $t \mapsto 1/\sqrt{\|x(t)\|}$ is continuous and $x_n(2^{-n-1}) = 0$, we conclude that $x_n = 0$ on the whole interval $[0, 2^{-n}]$. This yields the equality

$$P(t)A(x(t)) = x(t), \qquad \forall t \in I.$$

Setting then $\varrho(t) = \|x(t)\|^2$, we obtain

$$\frac{d}{dt}\varrho(t) = 2\langle x'(t), x(t)\rangle \geq 2\langle G(x(t)), x(t)\rangle = 2\varrho^{3/4}(t), \qquad \forall t > 0,\ t \in I.$$

This implies that $\varrho(t) \geq (t/2)^4$, that is, $\|x(t)\| \geq t^2/4$ for $t \geq 0$. Therefore,

$$\max\left\{0\,,\ \tfrac{1}{4}t^2 - \|x\|\right\} = 0,$$

so that $x(t)$ satisfies the equation

$$x'(t) = x(t)\varrho^{-1/4}(t), \qquad \forall t > 0,\ t \in I.$$

Componentwise,

$$x_n'(t) = x_n(t)\varrho^{-1/4}(t).$$

Due to the fact that $\varrho^{-1/4}(t)$ is strictly positive and continuous for $t > 0$, the latter equation, complemented with the initial condition $x_n(2^{-n}) = 0$, has the unique solution $x_n(t) \equiv 0$ for $t > 0$. Since this is true for all $n$, we conclude that $x(t) \equiv 0$ for $t > 0$, in contradiction with the previous point. □

*Remark* This proof can be extended to many Banach spaces, such as $\ell^p$ for $1 \leq p \leq \infty$, in which

$$\|(x_0, x_1, x_2, \ldots)\| \leq \|(y_0, y_1, y_2, \ldots)\|$$

is implied by

$$|x_n| \leq |y_n|, \qquad \forall n \in \mathbb{N}.$$

This fact, rather than the existence of a scalar product, is what really needed in the proof. Besides, for $x \in \ell^p$, the function $f$ is continuous without any change. Accordingly, there is no solution to the Cauchy problem (16.4).

# Chapter 17
# The Lax-Milgram Lemma




Let $V$ be a *real* Hilbert space with scalar product $\langle \cdot, \cdot \rangle_V$ and norm $\| \cdot \|_V$. We denote the action of an element $v^*$ of the dual space $V^*$ on $v \in V$ by $\langle v^*, v \rangle$.

Let $a$ be a continuous bilinear form on $V$, namely, a function $a : V \times V \to \mathbb{R}$ such that:

- $a(\cdot, \cdot)$ is linear in both variables.
- $|a(u, v)| \leq m \|u\|_V \|v\|_V$, for all $u, v \in V$ and some $m \geq 0$.

We associate to $a$ a linear bounded operator $A : V \to V^*$ as follows: for every fixed $u \in V$, the application

$$v \mapsto a(u, v)$$

is continuous from $V$ to $\mathbb{R}$, so it is an element $w_u^* \in V^*$. Then we define

$$Au = w_u^*,$$

that is,

$$\langle Au, v \rangle = a(u, v), \qquad \forall v \in V. \tag{17.1}$$

The map $u \mapsto Au$ is linear from $V$ to $V^*$, and

$$\|A\|_{L(V,V^*)} \leq m.$$

Conversely, given an operator $A \in L(V, V^*)$, we can define a continuous bilinear form on $V$ by (17.1).

**Definition 17.1** A continuous bilinear form $a$ on $V$ is called *coercive* if there is $\alpha > 0$ such that

$$a(v, v) \geq \alpha \|v\|_V^2, \qquad \forall v \in V.$$

In this case, the associated operator $A$ is called *elliptic*.

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If $a$ is coercive, the associated operator $A$ is bounded below (hence injective). Indeed, if $0 \neq u \in V$,

$$\|Au\|_{V^*} = \sup_{\|v\|_V \leq 1} |\langle Au, v\rangle| \geq |\langle Au, \frac{u}{\|u\|_V}\rangle| = \frac{1}{\|u\|_V}\, a(u,u) \geq \alpha \|u\|_V.$$

Then $A$ is an isomorphism of $V$ into $V^*$. If the operator $A$ is also onto $V^*$, it follows that

$$A^{-1} \in L(V^*, V) \qquad \text{and} \qquad \|A^{-1}\|_{L(V^*,V)} \leq \frac{1}{\alpha}.$$

*Remark* If in addition $a$ is symmetric, i.e.,

$$a(u,v) = a(v,u), \qquad \forall u, v \in V,$$

then the associated operator $A$ is onto $V^*$. Indeed, $a(\cdot,\cdot)$ is in this case an equivalent scalar product on $V$. So if we fix $u^* \in V^*$, the functional $v \mapsto \langle u^*, v\rangle$ is continuous on $V$, and by the Riesz representation theorem there exists a unique $u \in V$ such that

$$\langle u^*, v\rangle = a(u,v), \qquad \forall\, v \in V.$$

We conclude that $u^* = Au$.

**Definition 17.2** In the particular case when $a(u,v) = \langle u, v\rangle_V$, then the associated operator $A$ is called the *canonical isomorphism* of $V$ onto $V^*$, and it is usually denoted by $\Lambda$, i.e.,

$$\langle \Lambda u, v\rangle = \langle u, v\rangle_V, \qquad \forall v \in V.$$

Observe that both $\Lambda$ and $\Lambda^{-1}$ are norm preserving, namely,

$$\|\Lambda u\|_{V^*} = \|u\|_V, \qquad \text{and} \qquad \|\Lambda^{-1} u^*\|_V = \|u^*\|_{V^*},$$

for all $u \in V$ and $u^* \in V^*$.

The remarkable fact is that $A$ is onto without requiring the symmetry of $a$. This is the content of the celebrated Lax-Milgram lemma [46].

**Lemma 17.1** (Lax-Milgram) *If $a$ is bilinear, continuous and coercive on $V$, then $A$ is an isomorphism from $V$ onto $V^*$. In other words, given any $u^* \in V^*$ there is a unique $u \in V$ such that*

$$a(u,v) = \langle u^*, v\rangle, \qquad \forall\, v \in V.$$

*Remark* Writing a posteriori $u = A^{-1}u^*$, we see that the unique solution $u$ to the equation above fulfills the estimate

$$\|u\|_V \leq \frac{1}{\alpha}\|u^*\|_{V^*}.$$

*Proof* Let $u^* \in V^*$ be given. We have to find the (unique) solution $u \in V$ to the equation

$$Au = u^*.$$

This is clearly equivalent to solve the equation

$$u = u - \varrho\Lambda^{-1}(Au - u^*),$$

for an arbitrarily fixed $\varrho > 0$, where $\Lambda$ is the canonical isomorphism of $V$ onto $V^*$. Let us define the map $f : V \to V$ as

$$f(u) = u - \varrho\Lambda^{-1}(Au - u^*).$$

For $u_1, u_2 \in V$, denoting $\bar{u} = u_1 - u_2$, we have

$$\begin{aligned}
\|f(u_1) - f(u_2)\|_V^2 &= \|\bar{u} - \varrho\Lambda^{-1}A\bar{u}\|_V^2 \\
&= \|\bar{u}\|_V^2 + \varrho^2\|\Lambda^{-1}A\bar{u}\|_V^2 - 2\varrho\langle\Lambda^{-1}A\bar{u}, \bar{u}\rangle_V \\
&= \|\bar{u}\|_V^2 + \varrho^2\|A\bar{u}\|_{V^*}^2 - 2\varrho\langle A\bar{u}, \bar{u}\rangle \\
&= \|\bar{u}\|_V^2 + \varrho^2\|A\bar{u}\|_{V^*}^2 - 2\varrho a(\bar{u}, \bar{u}) \\
&\leq \big(1 + \varrho^2 m^2 - 2\varrho\alpha\big)\|\bar{u}\|_V^2.
\end{aligned}$$

Here we used the equality

$$\langle A\bar{u}, \bar{u}\rangle = \langle\Lambda\Lambda^{-1}A\bar{u}, \bar{u}\rangle = \langle\Lambda^{-1}A\bar{u}, \bar{u}\rangle_V.$$

Therefore, upon choosing $\varrho$ such that $\varrho < 2\alpha/m^2$, the function $f$ is a contraction on $V$. Accordingly, by the BCP Theorem 1.1, it has a unique fixed point $u$, which is solution to $Au = u^*$. □

**Example** For a domain $\Omega \subset \mathbb{R}^n$ (a domain is an open connected set) with smooth boundary $\partial\Omega$, consider the elliptic boundary value problem in the unknown
$u : \overline{\Omega} \to \mathbb{R}$

$$\begin{cases} -\Delta u = g & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{17.2}$$

If $g \in C(\Omega)$, a classical solution is a function

$$u \in C^2(\Omega) \cap C(\overline{\Omega}) \qquad \text{with} \qquad u|_{\partial\Omega} = 0$$

such that $-\Delta u = g$ in $\Omega$. Aiming to extend the notion of solution to a wider class of functions, assuming $u$ sufficiently regular, we multiply both sides of the equation by a smooth function $\varphi$ with compact support in $\Omega$. Using the Gauss-Green formula, we end up with

$$\int_\Omega \nabla u(x) \cdot \nabla \varphi(x)\, dx = \int_\Omega g(x)\varphi(x)\, dx.$$

There are no boundary terms since $\varphi \equiv 0$ on $\partial\Omega$. By density, the above identity holds for every $\varphi$ belonging to the Sobolev space $V = H_0^1(\Omega)$, namely, the space of square summable functions $u$ on $\Omega$ with values in $\mathbb{R}$ such that $u|_{\partial\Omega} = 0$ (in the trace sense), and whose weak derivatives $\partial_{x_i} u$ are square summable as well for $i = 1, \dots, n$. This is a separable Hilbert space with respect to the scalar product

$$\langle u, v \rangle_V = \int_\Omega \nabla u(x) \cdot \nabla v(x)\, dx.$$

We address the reader to [1, 9, 25, 71] for more details on Sobolev spaces. This motivates the following definition of *weak solution*.

**Definition 17.3** A weak solution to problem (17.2) is a vector $u \in V$ such that

$$\langle u, \varphi \rangle_V = \langle g, \varphi \rangle, \qquad \forall \varphi \in V.$$

For any given $g \in V^*$, a direct application of Lemma 17.1 (where $a$ is the scalar product on $V$) entails the existence of a unique weak solution to the problem. In this case, since $a$ is symmetric, one could reach the same conclusion via the Riesz representation theorem. Nonetheless, the Lax-Milgram lemma becomes a crucial tool to handle more complicated elliptic equations, where the associated bilinear form is not symmetric.

**The Hilbert triple**

In view of the applications in partial differential equations, it is useful to introduce another real Hilbert space $H$, the so-called *pivot space*, with embedding $V \subset H$ continuous and dense. Then we can construct in a natural way a dense embedding $H^* \subset V^*$. Such an embedding is to be interpreted as follows: given any continuous linear functional on $H$, its restriction on $V$ determines a unique element of $V^*$. By means of the Riesz representation theorem, $H$ is identified with its dual space $H^*$ via the isometric isomorphism

$$h \leftrightarrow \langle h, \cdot \rangle_H.$$

On the other hand, for any given $h \in H$, the map

$$v \in V \mapsto \langle h, v\rangle_H$$

is a continuous linear functional on $V$. Indeed,

$$|\langle h, v\rangle_H| \le \|h\|_H \|v\|_H \le c_0 \|h\|_H \|v\|_V,$$

where $c_0 > 0$ is the embedding constant of $V \subset H$, that is,

$$\|v\|_H \le c_0 \|v\|_V, \qquad \forall v \in V.$$

So it defines an element $Jh \in V^*$ such that

$$\langle Jh, v\rangle = \langle h, v\rangle_H.$$

**Proposition 17.1** *The map $h \mapsto Jh$ belongs to $L(H, V^*)$ and*

$$\|J\|_{L(H,V^*)} = \sup_{\|h\|_H \le 1} \|Jh\|_{V^*} = \sup_{\|h\|_H \le 1} \sup_{\|v\|_V \le 1} |\langle h, v\rangle_H| \le c_0.$$

*Moreover, $J$ is one-to-one on $H$ and $JH$ is dense in $V^*$.*

*Proof* The facts that $J$ belongs to $L(H, V^*)$ and its norm is given by the formula above are apparent from the very definition of $J$. To prove the injectivity, simply note that $Jh = 0$ means

$$\langle h, v\rangle_H = 0, \qquad \forall v \in V.$$

Since $V$ is dense in $H$, this yields $h = 0$. A little bit more delicate is showing that $JH$ is dense in $V^*$. Indeed, if not, by a well-known corollary of the Hahn-Banach theorem (cf. Introduction) there exists a continuous linear functional $v^{**}$ on $V^*$ (i.e., an element of the bidual space $V^{**}$) such that

$$v^{**} \ne 0 \qquad \text{and} \qquad v^{**}|_{JH} = 0.$$

On the other hand, since $V$ is reflexive, there exists $v \in V$ such that

$$v^{**}(v^*) = \langle v^*, v\rangle, \qquad \forall v^* \in V^*.$$

Therefore,

$$\langle Jh, v\rangle = \langle h, v\rangle_H = 0, \qquad \forall h \in H,$$

so that $v = 0$, implying in turn $v^{**} = 0$. □

Hence, identifying the subset $JH \subset V^*$ with $H^*$, we have the continuous and dense embeddings

$$V \subset H \equiv H^* \subset V^*.$$

Accordingly,

$$\langle h, v\rangle = \langle h, v\rangle_H, \qquad \forall h \in H,\ \forall v \in V.$$

To be more precise, in order to identify $JH$ with $H^*$, given any $h \in H$ one should extend in a unique way $Jh$ (which acts on $V$) on the whole space $H$. This can be done via the bounded extension theorem, due to the fact that $V$ is dense in $H$, and the desired extension is nothing but $\langle h, \cdot\rangle_H \in H^*$.

**Definition 17.4** In the literature, three real Hilbert spaces $V$, $H$, $V^*$ satisfying such a property are usually called a *Hilbert triple*.

At this point, starting from the elliptic operator $A : V \to V^*$ associated to a continuous and coercive bilinear form on $V$, we can construct a linear (unbounded) operator $A_0$ on $H$ of domain

$$\mathrm{dom}(A_0) = \big\{u \in V : Au \in H\big\},$$

acting as

$$A_0 u = Au, \qquad \forall u \in \mathrm{dom}(A_0).$$

The operator $A_0$ enjoys the following properties:

1. $A_0$ is bounded below (hence injective). Indeed, if $u \in \mathrm{dom}(A_0)$, by the very definition of $A_0$ we get

$$\langle A_0 u, u\rangle_H = \langle Au, u\rangle = a(u, u) \geq \alpha \|u\|_V^2 \geq \frac{\alpha}{c_0^2}\|u\|_H^2.$$

   Hence,

$$\|A_0 u\|_H = \sup_{\|v\|_H \leq 1} |\langle A_0 u, v\rangle|_H \geq |\langle A_0 u, \frac{u}{\|u\|_H}\rangle_H| \geq \frac{\alpha}{c_0^2}\|u\|_H.$$

2. $\mathrm{ran}(A_0) = H$. Indeed, if $v \in H$, then there is $u \in V$ such that $Au = v$, and by definition this $u$ belongs to $\mathrm{dom}(A_0)$.

Collecting properties 1 and 2, we conclude that $A_0$ is invertible, and

$$A_0^{-1} \in L(H) \qquad \text{with} \qquad \|A_0\|_{L(H)}^{-1} \leq \frac{c_0^2}{\alpha}.$$

We now make the further assumption that $V$ is separable (hence $H$ and $V^*$ are separable) and that the embedding $V \subset H$ is compact, and we write

$$V \Subset H.$$

Besides, let the bilinear form $a$ be symmetric. In this case, as already mentioned, it defines an equivalent scalar product on $V$, and there is no loss of generality in

assuming

$$a(u, v) = \langle u, v \rangle_V, \qquad u, v \in V.$$

The symmetry of the bilinear form transfers to $A_0$, which in turn becomes a symmetric operator, for

$$\langle Au, v \rangle = \langle A_0 u, v \rangle_H, \qquad \forall u, v \in \mathrm{dom}(A_0).$$

By a well-known result of operator theory [66, Theorem 13.11], a linear operator which is simultaneously symmetric and onto is selfadjoint. Accordingly, $A_0$ is selfadjoint, strictly positive (being bounded below), and with compact inverse (as $V \Subset H$). In particular, by the spectral theorem, $A_0$ has countably many eigenvalues $\lambda_n$, having the property that

$$0 < \lambda_1 < \lambda_2 \le \lambda_3 \le \dots \qquad \text{and} \qquad \lambda_n \to \infty.$$

The first assertion means that each eigenvalue $\lambda_n$ has finite multiplicity, and the smallest eigenvalue $\lambda_1$ has multiplicity equal to one. Besides, the corresponding eigenvectors $w_n \in \mathrm{dom}(A_0)$ can be chosen in a way to form an orthonormal basis of $H$. In other words, $A_0$ has the spectral decomposition

$$A_0 u = \sum_n \lambda_n \langle u, w_n \rangle_H w_n, \qquad \forall u \in \mathrm{dom}(A_0).$$

Then, for every $r \ge 0$, we can define in a natural manner the powers $A_0^r$ as

$$A_0^r u = \sum_n \lambda_n^r \langle u, w_n \rangle_H w_n,$$

of domain

$$\mathrm{dom}(A_0^r) = \Big\{ u \in H \,:\, \sum_n \lambda_n^{2r} \langle u, w_n \rangle_H^2 < \infty \Big\}.$$

Setting $H_0 = H$, the spaces

$$H_r = \mathrm{dom}(A_0^{r/2})$$

turn out to be Hilbert spaces endowed with the scalar products and norms

$$\langle \cdot, \cdot \rangle_r = \langle A_0^{r/2} \cdot, A_0^{r/2} \cdot \rangle_H \qquad \text{and} \qquad \| \cdot \|_r = \| A_0^{r/2} \cdot \|_H,$$

and, for every $r$, the vectors $w_n / \lambda_n^{r/2}$ form an orthonormal basis of $H_r$. Moreover these spaces are compactly nested, namely,

$$H_r \Subset H_s \qquad \text{whenever} \qquad r > s.$$

We finally show that $H_1$ coincides with $V$. Indeed, the equality

$$\langle w_n, w_k\rangle_V = \langle Aw_n, w_k\rangle = \langle A_0 w_n, w_k\rangle_H = \lambda_n \langle w_n, w_k\rangle_H = \lambda_n \delta_{nk}$$

implies that the vectors $\tilde{w}_n = w_n/\lambda_n^{1/2}$ form an orthonormal family of $V$. This family is complete as well. Indeed, if $v \in V$ is orthogonal to all $\tilde{w}_n$, then

$$0 = \langle \tilde{w}_n, v\rangle_V = \langle A\tilde{w}_n, v\rangle = \langle A_0 \tilde{w}_n, v\rangle_H = \lambda_n^{1/2} \langle w_n, v\rangle_H.$$

Since $\{w_n\}$ is complete in $H$, this implies that $v = 0$ in $H$, and so in $V$. On the other hand, we know that $\{\tilde{w}_n\}$ is also a complete orthonormal family of $H_1$. In summary, $V$ and $H_1$, both continuously embedded into the same space $H$, share the same orthonormal basis $\{\tilde{w}_n\}$. Thus a vector $u \in H$ belongs to either space if and only if

$$u = \sum_n a_n \tilde{w}_n \qquad \text{with} \qquad \sum_n a_n^2 < \infty.$$

This establishes the equality $V = H_1$. In particular, for any $u \in V$ we have

$$\|u\|_V^2 = \sum_n \lambda_n \langle u, w_n\rangle_H^2 \geq \lambda_1 \sum_n \langle u, w_n\rangle_H^2 = \lambda_1 \|u\|_H^2,$$

and equality holds for $u = w_1$. We conclude that the best embedding constant $c_0 > 0$ of $V \subset H$ is given by

$$c_0 = \frac{1}{\sqrt{\lambda_1}}.$$

This value usually goes by the name of *Poincaré constant*.

With reference to the elliptic problem (17.2) of the previous example, let us set

$$V = H_0^1(\Omega) \qquad \text{and} \qquad H = L^2(\Omega).$$

By the Rellich-Kondrachov compactness theorem (see, e.g., [25]), it is known that $V \Subset H$. Then, the linear operator $-\Delta$ of (17.2) can be viewed either way as a continuous operator $A$ from $V$ onto $V^*$, or as selfadjoint strictly positive operator $A_0$ on $H$, sometimes called the *Laplace-Dirichlet operator*. In which case, we have

$$\mathrm{dom}(A_0) = H^2(\Omega) \cap V \Subset H,$$

where $H^2(\Omega)$ is the (Hilbert) space of square summable functions $u$ on $\Omega$ whose weak derivatives $\partial_{x_i} u$ of order one and $\partial_{x_i x_k} u$ of order two are square summable as well (see [1, 9, 25, 71] for more details). In light of the discussion above, we also know that

$$\mathrm{dom}(A_0^{1/2}) = V.$$

# Chapter 18
# An Abstract Elliptic Problem




Let $(V, \langle\cdot,\cdot\rangle_V, \|\cdot\|_V)$ be a *real* Hilbert space. Again, we denote the action of an element $v^*$ of the dual space $V^*$ on $v \in V$ by $\langle v^*, v\rangle$. Assume we are given an elliptic operator $A \in L(V, V^*)$ associated to a continuous and coercive bilinear form $a$ on $V$, that is, $A$ complies with (17.1). Thus,

$$\langle Av, v\rangle \geq \alpha\|v\|_V^2, \qquad \forall v \in V, \tag{18.1}$$

for some $\alpha > 0$. Besides, let $B : V \to V^*$ be a continuous and compact map (see Definition 11.1), satisfying

$$\langle B(v), v\rangle \geq -\kappa - \beta\|v\|_V^2, \qquad \forall v \in V, \tag{18.2}$$

for some $\kappa \geq 0$ and $0 \leq \beta < \alpha$. We will prove the following result.

**Proposition 18.1** *For an arbitrarily given $g \in V^*$, the abstract equation*

$$Au + B(u) = g \tag{18.3}$$

*admits a (possibly nonunique) solution $u \in V$.*

Observe that (18.3) is the same as saying that

$$a(u, v) + \langle B(u), v\rangle = \langle g, v\rangle, \qquad \forall v \in V.$$

*Proof* In light of Lemma 17.1, the operator $A$ is invertible and $A^{-1} \in L(V^*, V)$. Define then the map $f : V \to V$ as

$$f(u) = A^{-1}(g - B(u)).$$

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Due to the assumptions on $B$, the reader can easily verify that the map $f$ is continuous and compact. Suppose now that, for some $\lambda \in [0, 1]$, there is $u_\lambda \in V$ such that

$$u_\lambda = \lambda f(u_\lambda).$$

This means that $u_\lambda$ solves

$$Au_\lambda + \lambda B(u_\lambda) = \lambda g.$$

Taking the duality pairing of the above equation and $u_\lambda$, and using (18.1)–(18.2) and the fact that $\lambda \leq 1$, we obtain

$$\alpha \|u_\lambda\|_V^2 \leq \kappa + \beta \|u_\lambda\|_V^2 + \|g\|_{V^*} \|u_\lambda\|_V.$$

An application of the Young inequality yields

$$\|g\|_{V^*} \|u_\lambda\|_V \leq \omega \|u_\lambda\|_V^2 + \frac{1}{4\omega} \|g\|_{V^*}^2,$$

having set $\omega = (\alpha - \beta)/2 > 0$. Hence, we end up with the *a priori* estimate

$$\|u_\lambda\|_V^2 \leq \frac{\kappa}{\omega} + \frac{1}{4\omega^2} \|g\|_{V^*}^2.$$

In conclusion, the set

$$\mathcal{F} = \left\{ u_\lambda \in V \,:\, u_\lambda = \lambda f(u_\lambda), \quad \text{for some } \lambda \in [0, 1] \right\}$$

is bounded in $V$. An application of the Schaefer Theorem 11.1 entails the existence of a fixed point $\bar{u}$ for $f$, which is clearly a solution to (18.3). □

**Example** Let $\Omega \subset \mathbb{R}^3$ be a bounded domain with smooth boundary $\partial\Omega$. We consider the nonlinear elliptic boundary value problem in the unknown $u : \overline{\Omega} \to \mathbb{R}$

$$\begin{cases} -\Delta u + B(u) = g & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{18.4}$$

where

$$B(u) = |u|^{p-1} u \qquad \text{with} \qquad p \in [1, 5).$$

Setting $V = H_0^1(\Omega)$, we recall that in three-dimensional bounded domains the continuous and compact Sobolev embeddings

$$V \Subset L^{p+1}(\Omega) \qquad \text{and} \qquad L^{(p+1)/p}(\Omega) \Subset V^* \tag{18.5}$$

hold for every $p \in [1, 5)$ (see [1, 9, 25, 71]). The second embedding is to be interpreted as follows: if $\psi \in L^{(p+1)/p}(\Omega)$, then it defines a continuous linear functional on $V$ (i.e., $\psi$ is an element of $V^*$) by the formula

$$\langle \psi, v \rangle = \int_\Omega \psi(x) v(x)\, dx, \qquad \forall v \in V.$$

Indeed, by the the Hölder inequality with conjugate exponents $\{(p+1)/p,\ p+1\}$ we get

$$|\langle \psi, v \rangle| \le \|\psi\|_{L^{(p+1)/p}(\Omega)} \|v\|_{L^{p+1}(\Omega)},$$

which in turn gives

$$\|\psi\|_{V^*} \le c_p \|\psi\|_{L^{(p+1)/p}(\Omega)},$$

where $c_p > 0$ is the embedding constant of $V \Subset L^{p+1}(\Omega)$. In fact, here one has to recast the proof of Proposition 17.1, using the well-known fact that

$$\left[L^{p+1}(\Omega)\right]^* = L^{(p+1)/p}(\Omega).$$

Then, taking $g \in V^*$, and observing that

$$\|B(u)\|_{L^{(p+1)/p}(\Omega)} = \left[\int_\Omega |u(x)|^{p+1}\, dx\right]^{p/(p+1)} = \|u\|^p_{L^{p+1}(\Omega)}, \tag{18.6}$$

by the very same argument leading to Definition 17.3, we call weak solution to this problem a vector $u \in V$ such that

$$\langle u, \varphi \rangle_V + \langle B(u), \varphi \rangle = \langle g, \varphi \rangle, \qquad \forall \varphi \in V.$$

Thus $A = -\Delta$, with associated bilinear form $a(\cdot, \cdot) = \langle \cdot, \cdot \rangle_V$.

We now show that the map $u \mapsto B(u)$ meets the hypotheses of Proposition 18.1. It is readily seen that (18.2) holds with $\kappa = \beta = 0$. Concerning the compactness, from (18.5)–(18.6) we learn that $B$ carries bounded sets of $V$ into bounded sets of $L^{(p+1)/p}(\Omega)$, hence relatively compact in $V^*$. Finally, in order to prove the continuity of $B$, for $u = u(x)$ and $v = v(x)$ in $V$ we write

$$B(u) - B(v) = w \int_0^1 b(\lambda u + (1 - \lambda)v)\, d\lambda,$$

having set

$$w = u - v \qquad \text{and} \qquad b(s) = \frac{d}{ds} B(s) = p|s|^{p-1}.$$

Then (in what follows $c > 0$ stands for a *generic* constant),

$$\begin{aligned}
&\|B(u) - B(v)\|_{L^{(p+1)/p}(\Omega)} \\
&\le c\left( \int_\Omega |w|^{(p+1)/p} \left( |u|^{(p+1)(p-1)/p} + |v|^{(p+1)(p-1)/p} \right) dx \right)^{p/(p+1)} \\
&\le c\|w\|_{L^{p+1}(\Omega)} \left( \|u\|^{p-1}_{L^{p+1}(\Omega)} + \|v\|^{p-1}_{L^{p+1}(\Omega)} \right).
\end{aligned}$$

In the latter passage we exploited the Hölder inequality with conjugate exponents $\{p,\, p/(p-1)\}$. In light of the previous embeddings, we conclude that

$$\|B(u) - B(v)\|_{V^*} \le c\|w\|_V \left( \|u\|^{p-1}_V + \|v\|^{p-1}_V \right),$$

yielding the sought continuity. Accordingly, by Proposition 18.1 we established the existence of a solution to our problem (18.4).

For this particular example, the solution is unique as well. To see that, let $u$ and $v$ be two solutions to (18.4). Then, the difference $w$ solves

$$\langle w, \varphi \rangle_V + \langle B(u) - B(v), \varphi \rangle = 0, \qquad \forall \varphi \in V.$$

Choosing $\varphi = w$,

$$\|w\|^2_V + \langle B(u) - B(v), w \rangle = 0.$$

But in this case,

$$B(u) - B(v) = \ell w,$$

where

$$\ell = p \int_0^1 |\lambda u + (1-\lambda)v|^{p-1} \, d\lambda \ge 0.$$

Therefore,

$$0 = \|w\|^2_V + \langle B(u) - B(v), w \rangle \ge \|w\|^2_V,$$

meaning that $w = 0$.

# Chapter 19
# Semilinear Evolution Equations




Let $X$ be a Banach space. Assume we are given an evolution equation in $X$ of the form

$$x'(t) = Tx(t),$$

where $T : \mathrm{dom}(T) \to X$ is a densely defined linear operator on $X$. Such an equation is *autonomous*, meaning that there is no explicit dependence on the variable $t$, referred to as time. We assume that, for any initial datum $x_0 \in X$, the equation has a unique solution

$$x \in C([0, \infty), X),$$

satisfying the initial condition $x(0) = x_0$. By "solution" we mean a solution in some weak sense. Also, we suppose that the solution continuously depends on the initial data. Let us then denote by

$$S(t)x_0$$

the solution at time $t \geq 0$ with initial datum $x_0$ given at the initial time $t = 0$. So, we built a one-parameter family $\{S(t)\}_{t\geq 0}$ of (linear) maps from $X$ into $X$. Obviously, this family has to reflect the results we know about the equation. Therefore, existence and uniqueness of global solutions for all initial data translates into requiring that $S(t) : X \to X$ is well defined for every $t \geq 0$. Since the system is autonomous, the solution at time $t + \tau$ with initial datum $x_0$ is the same of the solution at time $t$ with initial datum $x(\tau)$, where $x(\tau)$ is the solution at time $\tau$ with initial datum $x_0$. In terms of $S(t)$, we have the equality

$$S(t + \tau) = S(t)S(\tau).$$

Since $x(0) = x_0$ this means that

$$S(0) = I.$$

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For every $x_0 \in X$, the solution $x(t)$ is continuous in $X$, that is,

$$t \mapsto S(t)x_0 \in C([0,\infty), X).$$

Finally, if $x_0 \to \hat{x}_0$ then we have the convergence $x(t) \to \hat{x}(t)$ of the corresponding solutions for every fixed $t > 0$. This reads

$$S(t) \in L(X).$$

We summarize this discussion into a definition.

**Definition 19.1** Let $X$ be a Banach space. A *strongly continuous semigroup* of bounded linear operators, or $C_0$-*semigroup* for short, acting on $X$ is a family of maps

$$S(t) \in L(X),$$

depending on the parameter $t \geq 0$, satisfying the following axioms:

1. $S(0) = I$.
2. $S(t+\tau) = S(t)S(\tau)$, for all $t, \tau \geq 0$.
3. $\lim_{t\to 0} S(t)x = x$ for all $x \in X$.

Note that axiom 2 implies that $S(t)$ and $S(\tau)$ commute for all $t, \tau \geq 0$.

As a quite direct application of the uniform boundedness principle, we find a (time-dependent) bound on the norm of $S(t)$.

**Theorem 19.1** *Let $S(t)$ be a strongly continuous semigroup on $X$. Then there exist $M \geq 1$ and $\omega \in \mathbb{R}$ such that*

$$\|S(t)\|_{L(X)} \leq Me^{\omega t}, \quad \forall t \geq 0.$$

*Proof* We first show that there is $\vartheta > 0$ such that

$$\sup_{t\in[0,\vartheta]} \|S(t)\|_{L(X)} \leq K,$$

for some positive $K$. If not, there is a sequence $t_n \to 0$ for which

$$\|S(t_n)\|_{L(X)} \geq n.$$

By the uniform boundedness principle we infer that

$$\|S(t_n)x\|_X \to \infty,$$

for some $x \in X$, contrary to axiom 3. Since $S(0) = I$, it must be $K \geq 1$.

Next, we set

$$\varepsilon = \|S(\vartheta)\|_{L(X)}.$$

If $\varepsilon = 0$, it follows that $\|S(t)\|_{L(X)} = 0$ for all $t \geq \vartheta$, and the result trivially holds. Suppose then $\varepsilon > 0$. Given any $t \geq 0$, we write

$$t = n\vartheta + \tau, \qquad \text{with} \qquad n \in \mathbb{N},\ \tau \in [0, \vartheta).$$

Hence,

$$\|S(t)\|_{L(X)} = \|S(n\vartheta + \tau)\|_{L(X)} \leq K\varepsilon^n.$$

On the other hand,

$$\varepsilon^n = \varepsilon^{-\tau/\vartheta}\varepsilon^{t/\vartheta} \leq \max\left\{1, \varepsilon^{-1}\right\}\varepsilon^{t/\vartheta} = \max\left\{1, 1/\varepsilon\right\}e^{(\log\varepsilon/\vartheta)t},$$

and the claim follows with $M = K\max\{1, 1/\varepsilon\}$ and $\omega = \log\varepsilon/\vartheta$. □

Exploiting this result, we can easily improve the continuity of $S(t)$.

**Corollary 19.1** *The map $(t, x) \mapsto S(t)x$ is jointly continuous, namely,*

$$(t, x) \mapsto S(t)x \in C([0, \infty) \times X, X).$$

*Proof* Let us first prove the continuity of the map $t \mapsto S(t)x$ for every fixed $x \in X$. For $h > 0$, using the semigroup axioms and Theorem 19.1 we have

$$\begin{aligned}\|S(t + h)x - S(t)x\|_X &= \|S(t)[S(h)x - x]\|_X \\ &\leq Me^{\omega t}\|S(h)x - x\|_X,\end{aligned}$$

and (if $t > h$)

$$\begin{aligned}\|S(t - h)x - S(t)x\|_X &= \|S(t - h)[x - S(h)x]\|_X \\ &\leq Me^{\omega(t-h)}\|S(h)x - x\|_X.\end{aligned}$$

Both quantities converge to zero as $h \to 0^+$. At this point, the claim easily follows from the inequality

$$\begin{aligned}\|S(t)x - S(\tau)y\|_X &\leq \|S(t)x - S(\tau)x\|_X + \|S(\tau)x - S(\tau)y\|_X \\ &\leq \|S(t)x - S(\tau)x\|_X + Me^{\omega\tau}\|x - y\|_X,\end{aligned}$$

valid for all $x, y \in X$ and $t, \tau \geq 0$. □

**Definition 19.2** The linear operator $T$ of domain

$$\mathrm{dom}(T) = \left\{x \in X : \lim_{t\to 0}\frac{S(t)x - x}{t} \ \text{ exists}\right\},$$

defined by

$$Tx = \lim_{t\to 0} \frac{S(t)x - x}{t},$$

is called the *infinitesimal generator* of $S(t)$.

Roughly speaking, $T$ can be interpreted as the derivative of $S(t)$ at zero. Formally, one writes

$$S(t) = e^{tT}.$$

*Remark* There are cases, though, for which $e^{tT}$ makes sense, and the latter equality is not only formal, for instance when $T$ is selfadjoint or $T \in L(X)$ (see, e.g., [59]).

We recall without proof three important properties of the infinitesimal generator $T$ (see, e.g., [59]):

- $T$ is a closed operator.
- $T$ has dense domain in $X$.
- $T$ uniquely determines the semigroup.

**Theorem 19.2** *For every fixed $x \in \mathrm{dom}(T)$, we have that $S(t)x \in \mathrm{dom}(T)$, the map $t \mapsto S(t)x$ belongs to $C^1([0,\infty), X)$, and*

$$\frac{d}{dt} S(t)x = TS(t)x = S(t)Tx.$$

*Proof* Let $x \in \mathrm{dom}(T)$ and $t \geq 0$. Then,

$$\lim_{h\to 0} \frac{S(h)S(t)x - S(t)x}{h} = \lim_{h\to 0} S(t)\,\frac{S(h)x - x}{h} = S(t)Tx.$$

This proves that

$$S(t)x \in \mathrm{dom}(T) \qquad \text{and} \qquad TS(t)x = S(t)Tx.$$

Moreover, it sets the value of the right derivative of $t \mapsto S(t)x$ to be $S(t)Tx$. Let then $0 < h < t$. We have

$$\frac{S(t-h)x - S(t)x}{-h} = S(t-h)\left[\frac{S(h)x - x}{h}\right].$$

Taking the limit $h \to 0$, exploiting Corollary 19.1 we see that the left derivative exists and equals $S(t)Tx$. The continuity of the derivative is then obvious. □

As an immediate consequence, the function $x(t) = S(t)x_0$ is a solution to the linear homogeneous Cauchy problem

$$\begin{cases} x'(t) = Tx(t) & t > 0, \\ x(0) = x_0, \end{cases} \tag{19.1}$$

for any given $x_0 \in \mathrm{dom}(T)$. By definition of solution, we mean a differentiable function $x$ such that $x(t) \in \mathrm{dom}(T)$ for all $t \geq 0$. Actually, such a solution is unique as well. Indeed, if $y(t)$ is another solution, for any fixed $\tau > 0$ and every $t \in [0, \tau]$, set

$$\Lambda(t) = S(\tau - t)y(t).$$

We get

$$\Lambda'(t) = -TS(\tau - t)y(t) + S(\tau - t)y'(t) = 0, \qquad \forall t \in (0, \tau].$$

Hence

$$y(\tau) = S(0)y(\tau) = \Lambda(\tau) = \Lambda(0) = S(\tau)y(0) = S(\tau)x_0 = x(\tau),$$

and the claim follows from the arbitrariness of $\tau$.

Nonetheless, $S(t)x_0$ makes sense for *any* $x_0 \in X$. This motivates the following definition.

**Definition 19.3** The function $t \mapsto S(t)x_0$ is called the *mild solution* to (19.1) with initial datum $x_0$ assigned at the initial time $t = 0$.

### A semilinear Cauchy problem

For a fixed $t_0 > 0$, we now consider the following Cauchy problem in $X$:

$$\begin{cases} x'(t) = Tx(t) + f(t, x(t)) & 0 < t \leq t_0, \\ x(0) = x_0 \in X, \end{cases} \tag{19.2}$$

where $T$ is the infinitesimal generator of a strongly continuous semigroup $S(t)$, and $f : [0, t_0] \times X \to X$ is a continuous function, uniformly (as $t \leq t_0$) Lipschitz continuous on $X$ with Lipschitz constant $\lambda \geq 0$, that is,

$$\|f(t, x) - f(t, y)\|_X \leq \lambda \|x - y\|_X,$$

for all $x, y \in X$ and $t \in [0, t_0]$.

**Definition 19.4** A function $x : [0, t_0] \to X$ is said to be a *classical solution* to (19.2) if it is differentiable on $[0, t_0]$, $x(t) \in \mathrm{dom}(T)$ for every $t \in [0, t_0]$, and (19.2) is satisfied on $[0, t_0]$.

If $x$ is a classical solution, it is necessarily unique, and it is given by

$$x(t) = S(t)x_0 + \int_0^t S(t - s) f(s, x(s))\, ds. \tag{19.3}$$

This can be easily proved by integrating in $ds$ on $[0, t]$ the derivative with respect to $s$ of the (differentiable) function $S(t-s)x(s)$ and using (19.2). Note that the (Riemann) integral above is well defined, since if

$$x \in C([0, t_0], X) \qquad \Rightarrow \qquad t \mapsto f(t, x(t)) \in C([0, t_0], X).$$

Of course, there is no reason why there should exist a classical solution for a certain initial value $x_0$. But again, as in the linear homogeneous case, formula (19.3) makes sense for *any* $x_0 \in X$, provided that $x \in C([0, t_0], X)$. Accordingly, we give the following definition.

**Definition 19.5** A function $x : [0, t_0] \to X$ is said to be a *mild solution* to (19.2) if it is continuous on $[0, t_0]$ and fulfills the integral equation (19.3).

**Theorem 19.3** *For every $x_0 \in X$ the Cauchy problem* (19.2) *has a unique mild solution. Moreover the map $x_0 \mapsto x(t)$ is Lipschitz continuous from $X$ into $C([0, t_0], X)$.*

*Proof* For a given $x_0 \in X$, we define the map $F : C([0, t_0], X) \to C([0, t_0], X)$ by

$$[F(x)](t) = S(t)x_0 + \int_0^t S(t-s) f(s, x(s))\, ds.$$

The fact that $F(x) \in C([0, t_0], X)$ is guaranteed by Corollary 19.1. We set

$$m = \sup_{t \in [0, t_0]} \|S(t)\|_{L(X)},$$

which is finite by Theorem 19.1. Given now $x, y \in C([0, t_0], X)$, by means of an inductive argument analogous to the one used in the proof of Theorem 16.1, we show that

$$\|[F^n(x)](t) - [F^n(y)](t)\|_X \le \frac{(\lambda m t)^n}{n!} \|x - y\|_{C([0, t_0], X)}, \tag{19.4}$$

for every $t \in [0, t_0]$. Indeed, for $n = 1$ we have

$$\begin{aligned} \|[F(x)](t) - [F(y)](t)\|_X &\le \lambda m \int_0^t \|x(s) - y(s)\|_X\, ds \\ &\le \lambda m t \|x - y\|_{C([0, t_0], X)}. \end{aligned}$$

If it is true for $n - 1$, with $n \ge 2$, then

$$
\begin{aligned}
&\|[F^n(x)](t) - [F^n(y)](t)\|_X \\
&\quad = \|[F(F^{n-1}(x))](t) - [F(F^{n-1}(y))](t)\|_X \\
&\quad \le \lambda m \int_0^t \|[F^{n-1}(x)](s) - [F^{n-1}(y)](s)\|_X \, ds \\
&\quad \le \frac{(\lambda m)^n}{(n-1)!} \left( \int_0^t s^{n-1} \|x - y\|_{C([0,t_0],X)} \, ds \right) \\
&\quad = \frac{(\lambda m t)^n}{n!} \|x - y\|_{C([0,t_0],X)}.
\end{aligned}
$$

Hence, we deduce from (19.4) the estimate

$$
\|F^n(x) - F^n(y)\|_{C([0,t_0],X)} \le \frac{(\lambda m t_0)^n}{n!} \|x - y\|_{C([0,t_0],X)},
$$

telling that for $n$ large enough $F^n$ is a contraction. Accordingly, by Corollary 1.3 we conclude that $F$ has a unique fixed point in $C([0, t_0], X)$, which is clearly a mild solution to the Cauchy problem (19.2).

To complete the proof, let $y$ be a mild solution corresponding to the initial value $y_0 \in X$. Then, for every $t \in [0, t_0]$,

$$
\|x(t) - y(t)\|_X \le m\|x_0 - y_0\|_X + \lambda m \int_0^t \|x(s) - y(s)\|_X \, ds.
$$

By the Gronwall Lemma 16.1 we infer the continuous dependence estimate

$$
\|x(t) - y(t)\|_X \le m e^{\lambda m t_0} \|x_0 - y_0\|_X, \qquad \forall t \in [0, t_0],
$$

which gives in one shot both the uniqueness of the mild solution and the Lipschitz continuity of the map $x_0 \mapsto x(t)$. □

We conclude with an application to a concrete partial differential equation.

**Example** Let $\Omega \subset \mathbb{R}^3$ be a bounded domain with smooth boundary $\partial\Omega$. We consider the semilinear problem in the unknown $u = u(x, t) : \overline{\Omega} \times [0, \infty) \to \mathbb{R}$

$$
\begin{cases}
\partial_t u = \Delta u + f(u) & \text{in } \Omega \times (0, \infty), \\
u = 0 & \text{on } \partial\Omega \times (0, \infty),
\end{cases}
$$

where $f : \mathbb{R} \to \mathbb{R}$ is a Lipschitz function. The equation is supplemented with the initial condition

$$
u(x, 0) = u_0(x),
$$

for a given $u_0 \in L^2(\Omega)$. This problem can be given the abstract form (19.2) by setting $X = L^2(\Omega)$ and

$$T = -A_0,$$

where $A_0$ is the Laplace-Dirichlet operator on $X$ defined at Chap. 17, and viewing $f$ as a (Lipschitz continuous) function from $X$ to $X$. Indeed, it is possible to show that such a $T$ is the infinitesimal generator of a strongly continuous semigroup $S(t)$ acting on the Hilbert space $X$ (see, e.g., [59]). Then, we meet the hypotheses of Theorem 19.3, and we have a unique mild solution defined for all times.

# Chapter 20
# An Abstract Parabolic Problem




Let $(V, H, V^*)$ be a Hilbert triple, with $V$ separable and $V \Subset H$ (cf. Chap. 17), and let $A : V \to V^*$ be an elliptic operator associated to a continuous and coercive symmetric bilinear form on $V$ (which can be assumed to be equal to the scalar product of $V$). Finally, let $A_0$ be the (unbounded) linear operator generated by $A$ on $H$ of domain

$$W = \{u \in V : Au \in H\}.$$

As we saw at Chap. 17, $W$ is a Hilbert space normed by $\|\cdot\|_W = \|A_0 \cdot\|_H$, whereas the norm on $V$ can be equivalently written as $\|\cdot\|_V = \|A_0^{1/2} \cdot\|_H$, for we have the equality

$$V = \mathrm{dom}(A_0^{1/2}).$$

For further purpose, we introduce for every $t_0 > 0$ the space

$$X_{t_0} = C([0, t_0], V) \cap L^2(0, t_0; W),$$

being $L^2(0, t_0; W)$ the Hilbert space of square summable $W$-valued functions on $[0, t_0]$. It is well known that $X_{t_0}$ is a Banach space with the norm

$$\|u\|_{X_{t_0}} = \|u\|_{C([0,t_0],V)} + \|u\|_{L^2(0,t_0;W)}.$$

### A semilinear Cauchy problem

For a fixed $t_0 > 0$, we consider the Cauchy problem in $V$:

$$\begin{cases} \partial_t u(t) + A_0 u(t) = f(u(t)) \quad 0 < t \le t_0, \\ u(0) = \xi \in V, \end{cases} \tag{20.1}$$

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where $f$ is a nonlinear function and $\partial_t u$ is the weak derivative of $u$ with respect to $t$, namely, it is defined by the equality

$$\int_0^\infty \langle \partial_t u(t), \varphi(t) \rangle \, dt = -\int_0^\infty \langle u(t), \varphi'(t) \rangle_H \, dt,$$

for every smooth $\varphi : \mathbb{R}^+ \to V$ compactly supported on $\mathbb{R}^+$ (see, e.g., [25]).

Before going to the notion of solution, specifying the assumptions on $f$, some comments are in order. Let us first dwell on the linear homogeneous version of (20.1), namely,

$$\begin{cases} \partial_t v(t) + A_0 v(t) = 0 \quad 0 < t \le t_0, \\ v(0) = \xi \in V. \end{cases} \tag{20.2}$$

In this case, we fall in the framework of the previous chapter. Indeed, it can be shown that if $A_0$ is a selfadjoint elliptic operator, then $-A_0$ is the infinitesimal generator of a strongly continuous semigroup $S(t)$ on $H$ (see, e.g., [59]). The same is true for the restriction $\hat{A}_0$ of $A_0$ on $\mathrm{dom}(A_0^{3/2})$, which is still selfadjoint elliptic, so that $-\hat{A}_0$ is the infinitesimal generator of a strongly continuous semigroup $\hat{S}(t)$ on $V$. Besides, $\hat{S}(t)$ is nothing but the restriction of $S(t)$ on $V$. Hence, for every $t_0 > 0$ and every initial datum $\xi \in V$, the Cauchy problem (20.2) has a unique mild solution $v \in C([0, t_0], V)$. On the other hand, since the operator $A_0$ is elliptic, the solution $v$ turns out to have a further regularity. This can be seen by taking (formally at this stage) the product in $H$ of the equation with $A_0 v$. Indeed, if

$$v \in L^2(0, t_0; W) \qquad \text{and} \qquad \partial_t v \in L^2(0, t_0; H),$$

then the map $t \mapsto \|v(t)\|_V^2$ is absolutely continuous, and the equality

$$\frac{1}{2} \frac{d}{dt} \|v\|_V^2 = \langle \partial_t v, A_0 v \rangle$$

holds almost everywhere (see, e.g., [25, 71]). In which case, we get

$$\frac{d}{dt} \|v\|_V^2 + 2\|v\|_W^2 = 0,$$

and an integration in time yields

$$\|v(t)\|_V^2 + 2 \int_0^t \|v(s)\|_W^2 \, ds = \|\xi\|_V^2. \tag{20.3}$$

The latter identity holds assuming in advance that $v$ is regular enough to justify the calculations. The strategy, as commonly done in partial differential equations, is working within a suitable regularization scheme (see, e.g., [25, 71]), where the

equality above is attained for an approximate solution. Then, passing to the limit, one remains with an inequality for the actual solution $v$ of the form

$$\|v(t)\|_V^2 + 2\int_0^t \|v(s)\|_W^2\, ds \le \|\xi\|_V^2.$$

This suffices to tell that the solution $v$ undergoes an instantaneous regularization, for it belongs to the space $L^2(0, t_0; W)$. This phenomenon is sometimes referred to as *parabolicity*, and that's why Eq. (20.2), as well as Eq. (20.1), are named *parabolic*. As a byproduct, one discovers *a posteriori* that the solution $v$ possesses the necessary regularity in order for (20.3) to hold, as one reads from the equation that $\partial_t v \in L^2(0, t_0; H)$. Incidentally, this also tells that the weak derivative $\partial_t v$ is actually a derivative in the classical sense almost everywhere in time.

Summarizing, the linear homogeneous Cauchy problem (20.2) admits a unique solution $v \in X_{t_0}$ (in fact, for every $t_0 > 0$). The very same picture occurs if we consider the linear nonhomogeneous (and nonautonomous) problem

$$\begin{cases} \partial_t v(t) + A_0 v(t) = g(t) \quad 0 < t \le t_0, \\ v(0) = \xi \in V, \end{cases} \tag{20.4}$$

with $g \in L^2(0, t_0; H)$. In this case, arguing as above, we obtain the estimate

$$\frac{d}{dt}\|v\|_V^2 + 2\|v\|_W^2 \le 2\|v\|_W \|g\|_H \le \|v\|_W^2 + \|g\|_H^2,$$

leading to the inequality, valid for every $t \le t_0$,

$$\|v(t)\|_V^2 + \int_0^t \|v(s)\|_W^2\, ds \le \|\xi\|_V^2 + \int_0^t \|g(s)\|_H^2\, ds.$$

Again, the Cauchy problem (20.4) admits a unique solution $v \in X_{t_0}$.

**Definition 20.1** Let $f : W \to H$. We call a solution to (20.1) a function $u \in X_{t_0}$, with

$$\partial_t u \in L^2(0, t_0; H) \qquad \text{and} \qquad u(0) = \xi,$$

that solves (20.1) almost everywhere.

**Theorem 20.1** *Let the nonlinearity $f$ satisfy $f(0) = 0$ and, for every $u, v \in W$,*

$$\|f(u) - f(v)\|_H \le \|u - v\|_V (1 + \|u\|_W + \|v\|_W)^{1-\varepsilon} Q_0(\|u\|_V + \|v\|_V),$$

*for some $\varepsilon \in (0, 1]$ and some increasing positive function $Q_0$. Then, for every $R \ge 0$ there exists $t_0 = t_0(R) > 0$ such that the Cauchy problem (20.1) admits a unique solution $u \in X_{t_0}$, for every initial datum $\xi \in V$ with $\|\xi\|_V \le R$.*

*Proof* For $t_0 > 0$ to be fixed in a later moment, let us consider the complete metric space

$$B = \overline{B}_{X_{t_0}}(0, 4R).$$

Define the map $F : B \to X_{t_0}$ by

$$w \mapsto F(w) = u,$$

where $u$ solves the problem

$$\begin{cases} \partial_t u(t) + A_0 u(t) = f(w(t)) \quad 0 < t \le t_0, \\ u(0) = \xi \in V. \end{cases}$$

It is clear from the assumptions of $f$ that $f(w) \in L^2(0, t_0; H)$, so we are exactly in the situation of the Cauchy problem (20.4), where the existence and uniqueness of the solution in $X_{t_0}$ is guaranteed.

First we show that $F$ is Lipschitz continuous with Lipschitz constant $\lambda < 1$, provided that $t_0$ is small enough. To see that, let us consider $w_1, w_2 \in B$, with $u_j = F(w_j)$. Then, the difference $\hat{u} = u_1 - u_2$ solves

$$\begin{cases} \partial_t \hat{u}(t) + A_0 \hat{u}(t) = f(w_1(t)) - f(w_2(t)) \quad 0 < t \le t_0, \\ \hat{u}(0) = 0. \end{cases}$$

Taking the product in $H$ of the equation above with $A_0 \hat{u}$, and calling $\hat{w} = w_1 - w_2$, we get

$$\frac{d}{dt}\|\hat{u}\|_V^2 + 2\|\hat{u}\|_W^2 \le 2\|\hat{u}\|_W \|f(w_1) - f(w_2)\|_H \le \|\hat{u}\|_W^2 + \|f(w_1) - f(w_2)\|_H^2.$$

Since $w_j \in B$, we obtain

$$\|f(w_1) - f(w_2)\|_H^2 \le \|\hat{w}\|_{X_{t_0}}^2 (1 + \|w_1\|_W + \|w_2\|_W)^{2-2\varepsilon} Q_1(R),$$

for some other increasing positive function $Q_1$. Therefore

$$\frac{d}{dt}\|\hat{u}\|_V^2 + \|\hat{u}\|_W^2 \le \|\hat{w}\|_{X_{t_0}}^2 (1 + \|w_1\|_W + \|w_2\|_W)^{2-2\varepsilon} Q_1(R),$$

and an integration in time gives

$$\begin{aligned} &\|\hat{u}(t)\|_V^2 + \int_0^t \|\hat{u}(s)\|_W^2 \, ds \\ &\quad \le \|\hat{w}\|_{X_{t_0}}^2 Q_1(R) \int_0^{t_0} (1 + \|w_1(s)\|_W + \|w_2(s)\|_W)^{2-2\varepsilon} \, ds. \end{aligned}$$

Exploiting the Hölder inequality with conjugate exponents $\{1/(1-\varepsilon), 1/\varepsilon\}$, and using again the fact that $w_j \in B$, it is readily seen that

$$Q_1(R) \int_0^{t_0} (1 + \|w_1(s)\|_W + \|w_2(s)\|_W)^{2-2\varepsilon}\, ds \le Q_2(R + t_0) t_0^{\varepsilon},$$

where $Q_2$ is another increasing positive function, which can be explicitly computed. Accordingly, calling

$$\lambda = \lambda(t_0) = \sqrt{4 Q_2(R + t_0) t_0^{\varepsilon}}\,,$$

we end up with

$$\|\hat{u}(t)\|_V^2 + \int_0^t \|\hat{u}(s)\|_W^2\, ds \le \frac{\lambda^2}{4} \|\hat{w}\|_{X_{t_0}}^2,$$

from which we obtain

$$\|\hat{u}\|_{X_{t_0}} \le \lambda \|\hat{w}\|_{X_{t_0}}.$$

It is then clear that, up to choosing $t_0$ small enough (and the smallness depends only on $R$), we can render $\lambda < 1$, as claimed. Actually, in view of the next point, we shall further reduce $\lambda$ to be $\lambda \le 1/2$.

Lastly, we show that $F$ is a map from $B$ to $B$. To this end, we merely apply the previous argument, this time with $w_1 = w$, and $w_2 = 0$. Then $u_1 = F(w) = u$, while $u_2 = F(0) = v$, where $v$ is the solution to (20.2). Here we use the assumption $f(0) = 0$. In particular, (20.3) tells that

$$\|v\|_{X_{t_0}} \le 2R, \qquad \forall t_0 > 0.$$

Therefore,

$$\|u\|_{X_{t_0}} \le \|\hat{u}\|_{X_{t_0}} + \|v\|_{X_{t_0}} \le \lambda \|w\|_{X_{t_0}} + 2R \le 4R.$$

In summary, we proved that $F : B \to B$ is a contraction. Hence, in light of the BCP Theorem 1.1, there is a unique fixed point $\bar{u} = F(\bar{u})$, which is the sought solution to the Cauchy problem (20.1) on the time interval $[0, t_0]$, with $t_0$ depending on $R$. □

*Remark* Recasting word by word the scheme above, we can also consider the nonautonomous Cauchy problem

$$\begin{cases} \partial_t u(t) + A_0 u(t) = f(u(t)) + g(t) & 0 < t \le t_0, \\ u(0) = \xi \in V, \end{cases} \tag{20.5}$$

for which the following result holds.

**Theorem 20.2** *Let $f$ be as in Theorem 20.1, and let $g \in L^2(0, t_*; H)$ for some $t_* > 0$. Then, for every $R \geq 0$ there exists*

$$t_0 = t_0(R) > 0 \qquad \text{with} \qquad t_0 \leq t_*$$

*such that the Cauchy problem (20.5) admits a unique solution $u \in X_{t_0}$, for every initial datum $\xi \in V$ with $\|\xi\|_V \leq R$.*

The proof is left as an exercise.

**Example** Let $\Omega \subset \mathbb{R}^3$ be a bounded domain with smooth boundary $\partial\Omega$. We consider the semilinear parabolic problem in the unknown $u = u(x, t) : \overline{\Omega} \times [0, t_0] \to \mathbb{R}$

$$\begin{cases} \partial_t u - \Delta u = f(u) & \text{in } \Omega \times (0, t_0], \\ u = 0 & \text{on } \partial\Omega \times (0, t_0], \end{cases}$$

where

$$f(u) = \pm |u|^{p-1} u \qquad \text{with} \qquad p \in [1, 5).$$

The problem is supplemented with the initial condition

$$u(x, 0) = \xi(x),$$

for a given $\xi \in H_0^1(\Omega)$. This problem can be given the abstract form (20.1) by defining the spaces

$$H = L^2(\Omega), \qquad V = H_0^1(\Omega), \qquad W = H^2(\Omega) \cap H_0^1(\Omega),$$

and

$$-\Delta = A_0 \qquad \text{with} \qquad \text{dom}(A_0) = W,$$

that is, the Laplace-Dirichlet operator on $H$ defined at Chap. 17. We are left to show that the nonlinearity $f$ complies with the hypotheses of Theorem 20.1. To this end, setting $w = u - v$ and $\varphi(s) = \pm p|s|^{p-1}$, we write

$$f(u) - f(v) = w \int_0^1 \varphi(\lambda u + (1 - \lambda)v)\, d\lambda.$$

Without loss of generality, we may take $p \geq 3$. Then, from the Hölder inequality with conjugate exponents $\{\infty, 3, 3/2\}$ we get (in what follows $c > 0$ stands for a *generic* constant),

$$
\begin{aligned}
&\|f(u) - f(v)\|_H^2 \\
&\quad \le c \int_\Omega |w|^2 \big(|u|^{2(p-1)} + |v|^{2(p-1)}\big)\, dx \\
&\quad \le c\|w\|_{L^6(\Omega)}^2 \big(1 + \|u\|_{L^\infty(\Omega)} + \|v\|_{L^\infty(\Omega)}\big)^{2(p-3)} \big(1 + \|u\|_{L^6(\Omega)} + \|v\|_{L^6(\Omega)}\big)^4.
\end{aligned}
$$

Using the continuous Sobolev embedding $V \subset L^6(\Omega)$ and the *Agmon inequality* (see, e.g., [71])

$$
\|\zeta\|_{L^\infty(\Omega)}^2 \le c_\Omega \|\zeta\|_V \|\zeta\|_W, \qquad \forall \zeta \in W,
$$

for some $c_\Omega > 0$ depending only on $\Omega$, we conclude that

$$
\begin{aligned}
&\|f(u) - f(v)\|_H \\
&\quad \le c\|w\|_V \big(1 + \|u\|_W + \|v\|_W\big)^{(p-3)/2} \big(1 + \|u\|_V + \|v\|_V\big)^{(p+1)/2},
\end{aligned}
$$

which is exactly the required estimate with $\varepsilon = (5-p)/2$. Then, by Theorem 20.1, for every $R \ge 0$ there exists $t_0 = t_0(R) > 0$ such that the problem admits a unique solution $u \in X_{t_0}$, for every initial datum $\xi \in V$ with $\|\xi\|_V \le R$.

# Chapter 21
# The Invariant Subspace Problem




The invariant subspace problem is probably *the* problem of operator theory. The question, that attracted the attention of a great number of mathematicians, is quite simple to state:

> Given a *complex* Banach space $X$ of linear dimension $> 1$ and an operator $T \in L(X)$, find a closed nontrivial subspace $\mathcal{M}$ of $X$ (i.e., $\{0\} \neq \mathcal{M} \neq X$) for which $T\mathcal{M} \subset \mathcal{M}$.

Such an $\mathcal{M}$ is said to be an *invariant subspace* for $T$. It is known that not all continuous linear operators on Banach spaces have invariant subspaces [24]. The question is still open for Hilbert spaces.

*Remark* Here, it is crucial to work in complex Banach spaces. Indeed, for real Banach spaces there are trivial counterexamples. For instance, for $X = \mathbb{R}^2$ the operator

$$T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

has no invariant subspaces.

The most general and at the same time most spectacular result on the subject, is the Lomonosov theorem, that provides the existence of hyperinvariant subspaces for a vast class of operators. The proof is relatively simple, and the role played by the Schauder-Tychonoff theorem is essential. In order to state the result, we first need two definitions.

**Definition 21.1** An invariant subspace $\mathcal{M}$ for $T \in L(X)$ is said to be *hyperinvariant* if it is invariant for all operators commuting with $T$, that is, for all $S \in L(X)$ such that $TS = ST$.

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**Definition 21.2** An operator $T \in L(X)$ is called *nonscalar* if it is not a multiple of the identity.

We anticipate two simple lemmas that will be used in the proof.

**Lemma 21.1** *Let $T \in L(X)$ be a nonscalar operator. If $T$ has a finite-dimensional invariant subspace, then it has a hyperinvariant subspace as well.*

*Proof* Let $\mathcal{M}$ be a finite-dimensional invariant subspace for $T$. Then the restriction $T|_{\mathcal{M}}$ of $T$ on $\mathcal{M}$ is a linear map from $\mathcal{M}$ to $\mathcal{M}$. Therefore $T|_{\mathcal{M}}$ has an eigenvalue $\lambda \in \mathbb{C}$, which is clearly also an eigenvalue of $T$. The corresponding eigenspace

$$\mathcal{E} = \{x \in X : Tx = \lambda x\}$$

is hyperinvariant for $T$. Indeed, $\mathcal{E} \neq X$, as $T$ is nonscalar. Moreover, if $x \in \mathcal{E}$ and $S$ commutes with $T$, we have that

$$\lambda Sx = STx = TSx,$$

meaning that $Sx \in \mathcal{E}$. □

**Lemma 21.2** *Let $K \in L(X)$ be a compact operator. If $\lambda \neq 0$ is an eigenvalue of $K$, then the eigenspace $\mathcal{E}$ relative to $\lambda$ has finite dimension.*

*Proof* Since $K$ is compact, it transforms bounded sets into relatively compact sets (cf. Definition 11.1). But the restriction of $K$ on $\mathcal{E}$ is a (nonzero) multiple of the identity, and the identity is compact if and only if the space is finite-dimensional. □

We are now in the position to state and prove the Lomonosov theorem.

**Theorem 21.1** (Lomonosov) *Let $X$ be a complex Banach space. Let $T \in L(X)$ be a nonscalar operator commuting with a nonzero compact operator $K \in L(X)$. Then $T$ has a hyperinvariant subspace.*

*Proof* We proceed by contradiction, assuming that the operator $T$ has no hyperinvariant subspaces.

Let $\mathcal{A}$ be the algebra of continuous linear operators commuting with $T$. Recall that an algebra $\mathcal{A}$ in $L(X)$ is a linear subspace of $L(X)$ such that the product (i.e., composition) of any two elements of $\mathcal{A}$ belongs to $\mathcal{A}$. It is immediate to see that if $T$ has no hyperinvariant subspaces, then

$$\overline{\mathcal{A}x} = X, \qquad \forall x \in X,\ x \neq 0.$$

Indeed, if $\overline{\mathcal{A}x} \neq X$ for some $x \neq 0$, it follows that $\overline{\mathcal{A}x}$ is a hyperinvariant subspace for $T$, since for every $S \in \mathcal{A}$ we have that

$$S\overline{\mathcal{A}x} \subset \overline{S\mathcal{A}x} \subset \overline{\mathcal{A}x}.$$

In the first inclusion we exploited the continuity of $S$ (cf. the remark at Chap. 4).

Without loss of generality, we can assume that $\|K\|_{L(X)} \leq 1$. Choose then $x_0 \in X$ such that $\|Kx_0\|_X > 1$ (which implies $\|x_0\|_X > 1$) and set

$$B = \overline{B}_X(x_0, 1).$$

For $x \in \overline{KB}$ (note that $x$ cannot be the zero vector), there is $S \in \mathcal{A}$ such that

$$\|Sx - x_0\|_X < 1.$$

Hence every $x \in \overline{KB}$ has an open neighborhood $U_x$ such that $SU_x \subset B$ for some $S \in \mathcal{A}$. Exploiting the compactness of $\overline{KB}$, we find open sets $U_1, \ldots, U_n$ such that

$$\overline{KB} \subset \bigcup_{j=1}^{n} U_j,$$

and $S_1, \ldots, S_n \in \mathcal{A}$ such that

$$S_j U_j \subset B, \qquad \forall j = 1, \ldots, n.$$

Let $\varphi_1, \ldots, \varphi_n \in C(\overline{KB})$ be a partition of the unity for $\overline{KB}$ subordinate to the open cover $\{U_j\}$ (see Lemma 10.1), and define for $x \in B$

$$f(x) = \sum_{j=1}^{n} \varphi_j(Kx) S_j K x.$$

Note that the only nonzero terms in the sum are those corresponding to the $j$ for which $Kx \in U_j$. Accordingly, the sum is a convex combination of elements of $B$, hence it belongs to $B$. This proves that $f$ is a map from $B$ into $B$, and it is clearly continuous. Next, we claim that $f(B)$ is relatively compact. To this end, it is enough showing that $\varphi_j(KB)S_j KB$ is relatively compact for every $j$. Let then $x_n \in B$ be any sequence. From the compactness of $K$, there is $\xi \in X$ such that $Kx_n \to \xi$ up to a subsequence. Hence, as $S_j$ and $\varphi_j$ are continuous,

$$\varphi_j(Kx_n) S_j K x_n \to \varphi_j(\xi) S_j \xi,$$

which proves the claim.

Summarizing, we are within the hypotheses of Theorem 10.1, so there exists $\bar{x} \in B$ such that $f(\bar{x}) = \bar{x}$, that is,

$$\sum_{j=1}^{n} \varphi_j(K\bar{x}) S_j K \bar{x} = \bar{x}.$$

Defining the operator $A \in \mathcal{A}$ as

$$A = \sum_{j=1}^{n} \varphi_j(K\bar{x})S_j,$$

the former equality reads

$$AK\bar{x} = \bar{x},$$

which tells that 1 is an eigenvalue of $AK$. But $AK$ is a compact operator, and by Lemma 21.2 the eigenspace $\mathcal{E}$ relative to 1 is finite-dimensional. Observe that $\mathcal{E}$ is invariant for $T$. Indeed, if $x \in \mathcal{E}$, using the fact that $AK$ commutes with $T$ we get

$$Tx = TAKx = AKTx \qquad \Rightarrow \qquad Tx \in \mathcal{E}.$$

Hence, from Lemma 21.1 we infer that $T$ possesses a hyperinvariant subspace, against our assumption. □

Lomonosov Theorem 21.1 has appeared in [49]. For some time it was not clear whether there could exist nonscalar operators to which the theorem does not apply. An answer in that direction has been given by Hadwin, Nordgren, Radjavi and Rosenthal in the paper [32], where the authors provide an example of an operator $T$ with the following property: if $K$ is any compact operator which commutes with a nonzero nonscalar operator commuting with $T$, then $K$ is the null operator.

However, the problem of invariant subspaces in Banach spaces has been solved (negatively) only some years later by Enflo [24], who presented the first construction of an operator on a complex Banach space having no invariant subspace. A good reference for the subject is the book of Beauzamy [5].

# Chapter 22
# Measure Preserving Maps on Compact Hausdorff Spaces




Let $X$ be a compact Hausdorff space, and let $P(X)$ be the set of all Borel probability measures on $X$. By means of the Riesz representation theorem, the dual space of $C(X)$ can be identified with the space $M(X)$ of complex regular Borel measures on $X$. Recall that the norm $\|\mu\|$ of an element $\mu \in M(X)$ is given by the total variation of $\mu$. It is straightforward to check that $P(X)$ is convex and closed in the weak$^*$ topology of $M(X)$. Moreover, $P(X)$ is weakly$^*$ compact. Indeed, it is a weakly$^*$ closed subset of the unit ball of $M(X)$, which is weakly$^*$ compact by the Banach-Alaoglu theorem.

**Definition 22.1** Let $\mu \in P(X)$. A $\mu$-measurable map $f : X \to X$ is said to be *measure preserving* with respect to $\mu$ if

$$\mu(B) = \mu(f^{-1}(B)), \qquad \forall B \in \mathcal{B}(X),$$

where $\mathcal{B}(X)$ denotes the collection of Borel subsets of $X$. Such a $\mu$ is said to be an *invariant measure* for $f$.

Note that if $f$ is a $\mu$-measurable map, then the measure $\mu_f$ defined by

$$\mu_f(B) = \mu(f^{-1}(B)), \qquad \forall B \in \mathcal{B}(X),$$

belongs to $P(X)$. In particular, if $f$ is continuous, and therefore measurable with respect to every $\mu \in P(X)$, we have a map $\tilde{f} : P(X) \to P(X)$ defined by

$$\tilde{f}(\mu) = \mu_f.$$

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In addition, by applying the monotone convergence theorem to an increasing sequence of simple functions, it is easy to see that

$$\int_X g \, d\mu_f = \int_X g \circ f \, d\mu, \qquad \forall g \in C(X).$$

**Lemma 22.1** *Let $f : X \to X$ be continuous. Then the map $\tilde{f} : P(X) \to P(X)$ is continuous in the weak$^*$ topology.*

*Proof* Let $\{\mu_\iota\}_{\iota \in \mathcal{J}} \subset P(X)$ be a net converging to some $\mu \in P(X)$. Then, for every $g \in C(X)$,

$$\begin{aligned} \lim_{\iota \in \mathcal{J}} \int_X g \, d\mu_{\iota f} &= \lim_{\iota \in \mathcal{J}} \int_X g \circ f \, d\mu_\iota \\ &= \int_X g \circ f \, d\mu \\ &= \int_X g \, d\mu_f, \end{aligned}$$

which entails the claimed continuity. □

*Remark* We briefly recall the notion of net, used in the proof above. A *directed set* $\mathcal{J}$ is a partially order set (with respect to some order $\geq$) with the property that for any two elements $\upsilon, \tau \in \mathcal{J}$ there is $\iota \in \mathcal{J}$ such that $\iota \geq \upsilon$ and $\iota \geq \tau$. A *net* is a function $\iota \mapsto x_\iota$ mapping $\mathcal{J}$ into some topological space $X$. Given a net $x_\iota$ in $X$, we say that $x_\iota$ converges to $x$, and we write

$$\lim_{\iota \in \mathcal{J}} x_\iota = x,$$

if for every neighborhood $U$ of $x$ there exists $\upsilon = \upsilon(U) \in \mathcal{J}$ such that

$$x_\iota \in U, \qquad \forall \iota \geq \upsilon.$$

The topology of $X$ is fully determined by the knowledge of the convergent nets therein. This concept of net generalizes the one of sequence, which is enough to handle metric spaces (or, more generally, first-countable topological spaces), whose topology is instead fully determined by the convergent sequences.

We are now interested to find elements of $P(X)$ that are invariant measures for $f$. This is the same as finding a fixed point for the map $\tilde{f}$.

**Theorem 22.1** *Let $f : X \to X$ be a continuous map. Then there exists an invariant measure $\mu$ for $f$.*

*Proof* On account of the above discussion, $\tilde{f}$ is a continuous map of a compact convex subset of $M(X)$ into itself, and the existence of a fixed point is then guaranteed by the Schauder-Tychonoff Theorem 10.1. □

Clearly, an invariant measure for $f$ might not be unique. In particular, observe that if $x$ is a fixed point for $f$, then the Dirac measure $\delta_x$ is invariant for $f$.

More details on measure preserving maps and their link with ergodic theory can be found in the book [74].

# Chapter 23
# Invariant Means on Semigroups




Let $S$ be a semigroup, that is, a set endowed with an associative product

$$(s, t) \mapsto st.$$

We consider the (real) Banach space of all real-valued bounded functions on $S$, namely,

$$\ell^\infty(S) = \left\{ f : S \to \mathbb{R} \quad \text{such that} \quad \|f\| \doteq \sup_{s \in S} |f(s)| < \infty \right\}.$$

An element $f \in \ell^\infty(S)$ is called *positive* if $f(s) \geq 0$ for every $s \in S$. A linear functional $\Lambda : \ell^\infty(S) \to \mathbb{R}$ is called *positive* if $\Lambda f \geq 0$ for every positive element $f \in \ell^\infty(S)$. We agree to denote a constant function on $S$ by the value of the constant.

We begin by recalling a result that actually holds for more general situations. In what follows, $\|\cdot\|_*$ is the norm in the dual space $\ell^\infty(S)^*$.

**Lemma 23.1** *Let $\Lambda \in \ell^\infty(S)^*$, with $\|\Lambda\|_* = \Lambda 1 = 1$. Then $\Lambda$ is positive.*

*Proof* Assume not. Then there is $f \in \ell^\infty(S)$, such that $f \geq 0$ and that $\Lambda f = \beta < 0$. For $\varepsilon > 0$ small, we have

$$\|1 - \varepsilon f\| = \sup_{s \in S} |1 - \varepsilon f(s)| \leq 1.$$

Hence

$$1 < 1 - \varepsilon\beta = |1 - \varepsilon\beta| = |\Lambda(1 - \varepsilon f)| \leq \|1 - \varepsilon f\| \leq 1,$$

leading to a contradiction. □

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**Definition 23.1** If $s \in S$, we define the *left s-translation operator*

$$L_s : \ell^\infty(S) \to \ell^\infty(S)$$

to be

$$(L_s f)(\sigma) = f(s\sigma), \qquad \forall \sigma \in S.$$

In an analogous manner, we can define the *right s-translation operator* $R_s$ by

$$(R_s f)(\sigma) = f(\sigma s), \qquad \forall \sigma \in S.$$

.

**Definition 23.2** A (left) *invariant mean* on $S$ is a positive linear functional $\Lambda$ on $\ell^\infty(S)$ satisfying the following conditions:

- $\Lambda 1 = 1$.
- $\Lambda(L_s f) = \Lambda f$ for every $s \in S$ and every $f \in \ell^\infty(S)$.

When such a functional exists, $S$ is said to be (left) *amenable*.

Clearly, we can give the above definition replacing left with right or two-sided. The distinction is relevant when $S$ is not abelian.

**Example** (Banach) Let $S = (\mathbb{N}, +)$. Then $\ell^\infty(\mathbb{N})$ is merely the space $\ell^\infty$ of bounded (real) sequences. An invariant mean in this case is called a *Banach generalized limit*. The reason is that if $\Lambda$ is an invariant mean on $\mathbb{N}$ and $x = \{x_n\}_{n\in\mathbb{N}} \in \ell^\infty$ is such that $\lim_{n\to\infty} x_n = \alpha \in \mathbb{R}$, then $\Lambda x = \alpha$. Indeed, for any $\varepsilon > 0$, we can choose $n_0$ such that $\alpha - \varepsilon \le x_n \le \alpha + \varepsilon$ for every $n \ge n_0$. Hence, if we define $y = \{y_n\}_{n\in\mathbb{N}} \in \ell^\infty$ by $y_n = x_{n+n_0}$, we have $\Lambda x = \Lambda y$, and

$$\alpha - \varepsilon = \Lambda(\alpha - \varepsilon) \le \Lambda y \le \Lambda(\alpha + \varepsilon) \le \alpha + \varepsilon,$$

which yields the equality $\Lambda x = \alpha$. To prove the existence of an invariant mean, one has to consider the subspace $\mathcal{M}$ of $\ell^\infty$ given by

$$\mathcal{M} = \left\{ x = \{x_n\}_{n\in\mathbb{N}} \in \ell^\infty : \lim_{n\to\infty} \frac{x_0 + \cdots + x_n}{n+1} = \alpha_x \in \mathbb{R} \right\},$$

and define the linear functional $\Lambda_0$ on $\mathcal{M}$ as $\Lambda_0 x = \alpha_x$. Setting

$$p(x) = \limsup_{n\to\infty} \frac{x_0 + \cdots + x_n}{n+1}, \qquad \forall x = \{x_n\}_{n\in\mathbb{N}} \in \ell^\infty,$$

the Hahn-Banach theorem (cf. Introduction) allows to extend $\Lambda_0$ to a functional $\Lambda$ defined on the whole space and satisfying

$$-p(-x) \le \Lambda x \le p(x), \qquad \forall x \in \ell^\infty,$$

that is,

$$\liminf_{n\to\infty} \frac{x_0 + \cdots + x_n}{n+1} \le \Lambda x \le \limsup_{n\to\infty} \frac{x_0 + \cdots + x_n}{n+1}, \qquad \forall x \in \ell^\infty.$$

In particular, $\Lambda$ is continuous (for it is clearly continuous at 0).

The next result, due to Day [17] and based on the Markov-Kakutani Theorem 12.1, provides an elegant generalization of the example above.

**Theorem 23.1** (Day) *Let $S$ be an abelian semigroup. Then $S$ is amenable.*

*Proof* Denote

$$K = \big\{\Lambda \in \ell^\infty(S)^* : \|\Lambda\|_* = \Lambda 1 = 1\big\}.$$

In particular, if $\Lambda \in K$ then $\Lambda$ is positive by Lemma 23.1. The set $K$ is clearly convex. Besides, it is compact in the weak* topology of $\ell^\infty(S)^*$. Indeed, it is immediate to see that it is a weakly* closed subset of the unit ball of $\ell^\infty(S)^*$, which is weakly* compact by the Banach-Alaoglu theorem. We now define the family of linear operators $T_s : \ell^\infty(S)^* \longrightarrow \ell^\infty(S)^*$, depending on $s \in S$, as

$$(T_s\Lambda)(f) = \Lambda(L_s f), \qquad \forall f \in \ell^\infty(S).$$

First we show that $T_s$ is continuous in the weak* topology for every $s \in S$. Of course, it is enough to show the continuity at zero. Thus, let $V$ be a neighborhood of zero of the local base for the weak* topology, that is,

$$V = \big\{\Lambda \in \ell^\infty(S)^* \,:\, |\Lambda f_j| < \varepsilon_j,\ j = 1, \ldots, n\big\},$$

for some $\varepsilon_1, \ldots, \varepsilon_n > 0$ and $f_1, \ldots, f_n \in \ell^\infty(S)$. Then

$$\begin{aligned} T_s^{-1}(V) &= \big\{\Lambda \in \ell^\infty(S)^* \,:\, |(T_s\Lambda)(f_j)| < \varepsilon_j,\ j = 1, \ldots, n\big\} \\ &= \big\{\Lambda \in \ell^\infty(S)^* \,:\, |\Lambda(L_s f_j)| < \varepsilon_j,\ j = 1, \ldots, n\big\} \end{aligned}$$

is an open neighborhood of zero as well.

The second step is to prove that $T_s K \subset K$. Indeed,

$$(T_s\Lambda)(1) = \Lambda(L_s 1) = \Lambda 1 = 1,$$

and

$$\|T_s\Lambda\|_* = \sup_{\|f\|\le 1} |(T_s\Lambda)(f)| = \sup_{\|f\|\le 1} |\Lambda(L_s f)| \le \sup_{\|f\|\le 1} |\Lambda f| = \|\Lambda\|_* = 1,$$

since $\|L_s f\| \leq \|f\|$.

Finally, for every $s, t \in S$,

$$T_s T_t \Lambda = T_s(\Lambda \circ L_t) = \Lambda \circ L_t \circ L_s = \Lambda \circ L_{st} = \Lambda \circ L_{ts} = T_t T_s \Lambda,$$

meaning that the family of linear operators $T_s$ is abelian.

In summary, the hypotheses of Theorem 12.1 are satisfied. Hence, there is $\Lambda \in K$ such that $T_s \Lambda = \Lambda$ for every $s \in S$, which is the same as saying that

$$\Lambda(L_s f) = \Lambda f,$$

for every $s \in S$ and every $f \in \ell^\infty(S)$.

# Chapter 24
# Haar Measures




**Definition 24.1** A *topological group* is a group $G$ endowed with a Hausdorff topology that makes the group operations continuous; namely, the map $(x, y) \mapsto xy$ is continuous for every $x, y \in G$, and the map $x \mapsto x^{-1}$ is continuous for every $x \in G$.

For any fixed $y \in G$ the maps $x \mapsto xy$ and $x \mapsto yx$ are homeomorphisms of $G$ onto $G$, and so is the map $x \mapsto x^{-1}$. Hence the topology of $G$ is uniquely determined by any local base at the identity element $e$. Indeed, if $U$ is a neighborhood of some $x \in G$, the sets

$$x^{-1}U = \left\{x^{-1}y : y \in U\right\} \qquad \text{and} \qquad Ux^{-1} = \left\{yx^{-1} : y \in U\right\}$$

are neighborhoods of $e$. In what follows, we will denote by $\mathcal{U}$ the family of open subsets of $G$ containing the identity $e$.

**Lemma 24.1** *Let $G$ be topological group, and let $U \in \mathcal{U}$. Then there exists $V \in \mathcal{U}$ such that $VV = \left\{xy : x, y \in V\right\} \subset U$ and $V^{-1} = \left\{x^{-1} : x \in V\right\} = V$.*

*Proof* Using the continuity of the product at $(e, e)$, there exist $W_1, W_2 \in \mathcal{U}$ such that $xy \in U$ whenever $x \in W_1$ and $y \in W_2$. Let then $W = W_1 \cap W_2$, so that $WW \subset U$. On the other hand, from the continuity of inverse at $e$, we have that $W^{-1} \in \mathcal{U}$. Then, just define $V = W \cap W^{-1}$. □

By the same token, for every $n \geq 3$ there exists $V \in \mathcal{U}$ such that $V \cdots V$ ($n$-times) is contained in $U$ and $V^{-1} = V$. The next topological lemma will be used in the proof of the main result of the chapter.

**Lemma 24.2** *Let $G$ be a compact topological group, and let $K_1, K_2 \subset G$ be disjoint compact sets. Then there exists $U \in \mathcal{U}$ such that no (left or right) translate of $U$ meets both $K_1$ and $K_2$.*

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*Proof* Let $O \subset G$ be an open set such that $O \supset K_1$ and $O \cap K_2 = \emptyset$. For every $x \in K_1$, we have that

$$W_x = x^{-1}O \cap Ox^{-1} \in \mathcal{U}.$$

Then, on account of Lemma 24.1, there is $U_x \in \mathcal{U}$ such that $U_x = U_x^{-1}$ and $U_x U_x U_x \subset W_x$. Consider the set

$$V_x = xU_x \cap U_x x.$$

Since $U_x \in \mathcal{U}$, it follows that $\bigcup_{x \in K_1} V_x$ is an open cover of $K_1$, and by compactness there are $x_1, \ldots, x_n \in K_1$ such that $K_1 \subset \bigcup_{i=1}^n V_{x_i}$. Finally, define

$$U = \bigcap_{i=1}^n U_{x_i} \in \mathcal{U}.$$

Assume now that some translate of $U$ meets both $K_1$ and $K_2$. Then there are $k_1 \in K_1$ and $k_2 \in K_2$ such that either

$$k_2 = k_1 y^{-1} z \qquad \text{or} \qquad k_2 = y z^{-1} k_1,$$

for some $y, z \in U$. Let us examine the first case (being the second analogous). We have that $k_1 \in V_{x_i} \subset x_i U_{x_i}$ for some $i \in \{1, \ldots, n\}$. Hence,

$$k_2 \in x_i U_{x_i} U^{-1} U \subset x_i U_{x_i} U_{x_i}^{-1} U_{x_i} = x_i U_{x_i} U_{x_i} U_{x_i} \subset x_i W_{x_i} \subset x_i x_i^{-1} O = O,$$

which is a contradiction. □

Note that the proof actually holds for locally compact topological groups.

**Definition 24.2** Let $G$ be a compact topological group. A *Haar measure* on $G$ is a regular Borel probability measure $\mu$ which is simultaneously *left invariant*, i.e.,

$$\int_G f(x)\, d\mu(x) = \int_G f(yx)\, d\mu(x), \qquad \forall y \in G,\ \forall f \in C(G), \tag{24.1}$$

and *right invariant*, i.e.,

$$\int_G f(x)\, d\mu(x) = \int_G f(xy)\, d\mu(x), \qquad \forall y \in G,\ \forall f \in C(G). \tag{24.2}$$

It is readily seen that a Haar measure on $G$, if it exists, is unique. Indeed, if $\mu$ and $\nu$ are two Borel probability measures on $G$, with $\mu$ left invariant and $\nu$ right invariant, for every $f \in C(G)$ the Fubini theorem yields

$$\begin{aligned}\int_G f(x)\,d\mu(x) &= \int_G f(yx)\,d\mu(x)\\ &= \int_G \left[\int_G f(yx)\,d\mu(x)\right] d\nu(y)\\ &= \int_G \left[\int_G f(yx)\,d\nu(y)\right] d\mu(x)\\ &= \int_G f(yx)\,d\nu(y)\\ &= \int_G f(y)\,d\nu(y),\end{aligned}$$

forcing the equality $\mu = \nu$.

**Theorem 24.1** *Let $G$ be a compact topological group. Then there exists a unique Haar measure $\mu$ on $G$. Moreover,*

$$\int_G f(x)\,d\mu(x) = \int_G f(x^{-1})\,d\mu(x), \qquad \forall f \in C(G). \tag{24.3}$$

*Proof* By means of the above uniqueness argument, it is enough to prove the existence of a regular Borel probability measure $\mu$ satisfying (24.1). In which case, let $\nu$ be the Borel probability measure given by $d\nu(x) = d\mu(x^{-1})$. For any $f \in C(G)$, define $g \in C(G)$ as $g(x) = f(x^{-1})$. Using (24.1), we get

$$\begin{aligned}\int_G f(xy)\,d\nu(x) &= \int_G g(y^{-1}x^{-1})\,d\mu(x^{-1})\\ &= \int_G g(y^{-1}x)\,d\mu(x)\\ &= \int_G g(x)\,d\mu(x)\\ &= \int_G f(x)\,d\nu(x).\end{aligned}$$

Hence $\nu$ fulfills (24.2), and consequently $\mu = \nu$. In turn, (24.3) follows.

We then proceed by proving the existence of $\mu$. Let $\mathcal{K}$ be the family of compact subsets of $G$. If $K \in \mathcal{K}$ and $U \in \mathcal{U}$, we can always cover $K$ by a finite number of translates of $U$. We define the *covering number* $[K : U]$ of $K$ by $U$ to be the smallest number of translates of $U$ required to cover $K$. For every $K \in \mathcal{K}$, we introduce the normalized covering ratio

$$\xi_K(U) = \frac{[K : U]}{[G : U]}, \qquad \forall U \in \mathcal{U}.$$

It is immediate to verify that $0 \le \xi_K(U) \le 1$, $\xi_{K_1}(U) \le \xi_{K_2}(U)$ for $K_1 \subset K_2$, and $\xi_{xK}(U) = \xi_K(U)$ for all $x \in G$. Moreover, if $K_1, K_2 \in \mathcal{K}$ are disjoint sets, exploiting Lemma 24.2, we find $U \in \mathcal{U}$ such that no translate of $U$ meets both $K_1$ and $K_2$. Hence, any cover of $K_1 \cup K_2$ by translates of $U$ is the disjoint union of covers of $K_1$ and $K_2$. The same holds replacing $U$ with any $V \in \mathcal{U}$, $V \subset U$. Thus

$$\xi_{K_1 \cup K_2}(V) = \xi_{K_1}(V) + \xi_{K_2}(V), \qquad \forall V \in \mathcal{U},\ V \subset U. \tag{24.4}$$

We now observe that $(\mathcal{U}, \cap)$ is an abelian semigroup, so Theorem 23.1 applies, yielding the existence of an invariant mean $\Lambda$ on $\mathcal{U}$. Since $\xi_K : \mathcal{U} \to \mathbb{R}$ belongs to $\ell^\infty(\mathcal{U})$, we can define

$$\psi(K) = \Lambda \xi_K, \qquad \forall K \in \mathcal{K}.$$

The following hold:

$$\begin{aligned}
&\psi(\emptyset) = 0.\\
&\psi(G) = 1.\\
&\psi(xK) = \psi(K), \qquad \forall x \in G.\\
&\psi(K_1) \le \psi(K_2), \qquad \text{if } K_1 \subset K_2.\\
&\psi(K_1 \cup K_2) = \psi(K_1) + \psi(K_2), \qquad \text{if } K_1 \cap K_2 = \emptyset.
\end{aligned}$$

In particular, the last two imply that $\psi$ is finitely subadditive on $\mathcal{K}$, that is,

$$\psi(K_1 \cup K_2) \le \psi(K_1) + \psi(K_2), \qquad \forall K_1, K_2 \in \mathcal{K}.$$

The first four properties are direct consequences of the definition of $\xi_K$. The fifth one amounts to showing that

$$\Lambda \xi = 0 \qquad \text{with} \qquad \xi = \xi_{K_1 \cup K_2} - \xi_{K_1} - \xi_{K_2}.$$

This follows from (24.4) and the fact that if $\xi \in \ell^\infty(\mathcal{U})$ and there is $U \in \mathcal{U}$ such that $\xi(V) = 0$ for every $V \in \mathcal{U}$, $V \subset U$, then $\Lambda \xi = 0$. Indeed, $\Lambda(L_U \xi) = \Lambda \xi$, and

$$(L_U \xi)(W) = \xi(U \cap W) = 0, \qquad \forall W \in \mathcal{U}.$$

Here, with reference to Definition 23.1, $L_U$ (which coincides with $R_U$) is the left $U$-translation operator on $\ell^\infty(\mathcal{U})$.

Next, for each open set $O \subset G$ we define

$$\mu^*(O) = \sup\big\{\psi(K) : K \subset O,\ K \text{ compact}\big\},$$

and for every $E \subset G$ we set

$$\mu^*(E) = \inf\big\{\mu^*(O) : O \supset E,\ O \text{ open}\big\}.$$

From the properties of $\psi$, we learn that $\mu^*(\emptyset) = 0$ and $\mu^*(G) = 1$, $\mu^*(E_1) \le \mu^*(E_2)$ if $E_1 \subset E_2 \subset G$, and $\mu^*$ is left invariant. Moreover $\mu^*$ is countably subadditive, i.e.,

$$\mu^*\Big(\bigcup_{j=1}^{\infty} E_j\Big) \le \sum_{j=1}^{\infty} \mu^*(E_j).$$

To see that, let $\varepsilon > 0$, and choose open sets $O_j \supset E_j$ with

$$\mu^*(O_j) \le \mu^*(E_j) + \frac{\varepsilon}{2^j},$$

and a compact set $K \subset \bigcup_{j=1}^{\infty} O_j$. Because of its compactness, $K \subset \bigcup_{j=1}^{n} O_j$ for some $n$. Consider a partition of the unity $\varphi_1, \ldots, \varphi_n$ for $K$ subordinate to the open cover $\{O_j\}$ (see Lemma 10.1), and define the compact sets

$$K_j = K \cap \Big\{x \in G \,:\, \varphi_j(x) \ge \frac{1}{n}\Big\}.$$

Note that

$$\bigcup_{j=1}^{n} K_j = K,$$

for $\varphi_1(x) + \ldots + \varphi_n(x) = 1$ whenever $x \in K$. Besides, as $\varphi_j$ is supported on $O_j$, we have that $K_j \subset O_j$. Thus, from the finite subadditivity of $\psi$, we get

$$\psi(K) \le \sum_{j=1}^{n} \psi(K_j) \le \sum_{j=1}^{n} \mu^*(O_j) \le \sum_{j=1}^{\infty} \mu^*(E_j) + \varepsilon.$$

Taking the supremum over all such $K$, and letting then $\varepsilon \to 0$, we end up with

$$\mu^*\Big(\bigcup_{j=1}^{n} E_j\Big) \le \sum_{j=1}^{\infty} \mu^*(E_j),$$

and the claimed countable subadditivity follows by letting $n \to \infty$.

We conclude that $\mu^*$ is an outer measure. Applying then the Carathéodory extension process, we build a measure $\mu$, which coincides with $\mu^*$ on the measurable sets, namely, those sets $E \subset G$ such that

$$\mu^*(T) = \mu^*(T \cap E) + \mu^*(T \cap E^C), \qquad \forall T \subset G.$$

The proof is finished if we show that the open sets are measurable, since $\mu$ is then a Borel outer regular measure (by construction) and hence regular (being finite). So let $O$ be an open set, and $T$ be any set. Let $\varepsilon > 0$, and take an open set $A \supset T$ such that

$$\mu^*(A) \leq \mu^*(T) + \varepsilon.$$

Choose a compact set $K \subset A \cap O$ with

$$\mu^*(A \cap O) \leq \psi(K) + \varepsilon,$$

and a compact set $K_0 \subset A \cap K^C$ with

$$\mu^*(A \cap K^C) \leq \psi(K_0) + \varepsilon.$$

Then, observing that $O^C \subset K^C$,

$$\begin{aligned} \mu^*(T \cap O) + \mu^*(T \cap O^C) &\leq \mu^*(A \cap O) + \mu^*(A \cap K^C) \\ &\leq \psi(K) + \psi(K_0) + 2\varepsilon \\ &= \psi(K \cup K_0) + 2\varepsilon \\ &\leq \mu^*(A) + 2\varepsilon \\ &\leq \mu^*(T) + 3\varepsilon. \end{aligned}$$

Since $\varepsilon$ is arbitrary, the desired conclusion immediately follows. □

**Example** The unit circle $\mathbb{S}^1$ is a topological group with the product

$$(x, y) \mapsto xy = e^{i(\vartheta+\varphi)} \qquad \text{where} \qquad x = e^{i\vartheta},\ y = e^{i\varphi}.$$

In this case, the Haar measure $\mu$ is given by

$$\mu(S) = \frac{1}{2\pi}\lambda(\zeta^{-1}(S)), \qquad \forall S \subset \mathbb{S}^1,$$

where $\zeta : [0, 2\pi) \to \mathbb{S}^1$ is the (bijective) function defined as $\zeta(t) = e^{it}$ and $\lambda$ is the Lebesgue measure. The normalization constant $(2\pi)^{-1}$ is chosen in order to meet the requirement $\mu(\mathbb{S}^1) = 1$.

A different proof of Theorem 24.1, that directly applies the Markov-Kakutani Theorem 12.1, can be found in [66]. The advantage of the approach presented here is that it can be easily modified to extend the result to the locally compact case. Indeed, when $G$ is a locally compact topological group it is still possible to talk of Haar measures (removing the requirement that the measure be finite). In this

case, however, we shall distinguish between left and right invariant ones. Indeed, the uniqueness argument does not apply, since the Fubini theorem may fail to hold.

**Definition 24.3** Let $G$ be a locally compact topological group. A *left Haar measure* on $G$ is a nonnull Radon measure $\mu$ on $G$ satisfying (24.1). Analogously, a *right Haar measure* on $G$ is a nonnull Radon measure $\nu$ on $G$ satisfying (24.2).

Recall that a *Radon measure* is a positive linear functional on $C_c(G)$ (the space of compactly supported continuous functions on $G$). From the Riesz representation theorem every such functional can be uniquely represented by a positive outer regular Borel measure, finite on compact sets, for which open sets are inner regular (see [67]).

**Theorem 24.2** *Let $G$ be a locally compact topological group. Then there exist left Haar measures and right Haar measures on $G$. Moreover, any two left (right) Haar measure differ by a multiplicative positive constant.*

The proof of Theorem 24.2 can be found in [65]. Some important properties of Haar measures on locally compact topological groups, generalizing (24.1)–(24.3), are listed in the following proposition (see, e.g., [23, 65]).

**Proposition 24.1** *Let $G$ be a locally compact topological group.*

1. *There exists a continuous strictly positive function $\Delta$ on $G$, with $\Delta(e) = 1$ and $\Delta(xy) = \Delta(x)\Delta(y)$ for all $x, y \in G$, such that if $\mu$ is any left Haar measure on $G$, then*

$$\int_G f(xy)\, d\mu(x) = \Delta(y) \int_G f(x)\, d\mu(x), \qquad \forall y \in G,\ \forall f \in C_c(G).$$

2. *For any left Haar measure $\mu$ on $G$ we have*

$$\int_G f(x^{-1})\, d\mu(x) = \int_G f(x)\, \Delta(x) d\mu(x), \qquad \forall f \in C_c(G).$$

3. *For any left Haar measure $\mu$ and any right Haar measure $\nu$ on $G$ there is $c > 0$ such that*

$$\int_G f(x)\, d\nu(x) = c \int_G f(x)\Delta(x)\, d\mu(x), \qquad \forall f \in C_c(G).$$

4. *Given any left or right Haar measure on $G$, every nonempty open set has nonnull measure, and $G$ has finite measure if and only if it is compact.*

In particular, from points 1 and 2 we deduce that if $\nu$ is any right Haar measure on $G$ then

$$\int_G f(yx)\, d\nu(x) = \frac{1}{\Delta(y)} \int_G f(x)\, d\nu(x), \qquad \forall y \in G,\ \forall f \in C_c(G).$$

Indeed, the measure $d\mu(x) = d\nu(x^{-1})$ is a left Haar measure, and the claim follows arguing as in the uniqueness part of the proof of Theorem 24.1.

*Remark* The function $\Delta$ is termed the *modular function*. A locally compact topological group is called *unimodular* if there exists a bi-invariant Radon measure. This happens if and only if $\Delta \equiv 1$. Besides compact groups, abelian groups are clearly unimodular.

A complete reference for Haar measures is Nachbin's book [53].

# Chapter 25
# Game Theory




We consider a game with $n \geq 2$ players, under the assumption that the players do not cooperate among themselves. Each player pursues a *strategy*, in dependence of the strategies of the other players. Denote the set of all possible strategies of the $k$th player by $K_k$, and set

$$K = K_1 \times \cdots \times K_n.$$

An element $x \in K$ is called a *strategy profile*. For each $k$, let $f_k : K \to \mathbb{R}$ be the *loss function* of the $k$th player. If

$$\sum_{k=1}^{n} f_k(x) = 0, \qquad \forall x \in K, \tag{25.1}$$

the game is said to be of *zero-sum*. The aim of each player is to minimize her/his loss, or, equivalently, to maximize her/his gain.

**Definition 25.1** A *Nash equilibrium* is a strategy profile with the property that no player can benefit by changing his strategy, while all other players keep their strategies unchanged. In formulas, it is an element $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_n) \in K$ such that

$$f_k(\bar{x}) \leq f_k(\bar{x}_1, \ldots, \bar{x}_{k-1}, x_k, \bar{x}_{k+1}, \ldots, \bar{x}_n), \qquad \forall x_k \in K_k, \tag{25.2}$$

for every $k = 1, \ldots, n$.

Strictly speaking, a Nash equilibrium suggests a convenient "cautious" strategy to be adopted by each player in the game. We said *a* strategy rather than *the* strategy, since a Nash equilibrium (if it exists) might not be unique.

We need of course further hypotheses on the sets $K_k$ and on the maps $f_k$. It is reasonable to assume that, with all the other strategies fixed, the loss function $f_k$ has a small variation in correspondence of a small variation of $x_k$. Also, loosely speaking, it is assumed that the average of losses corresponding to two different strategies of the

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$k$th player is grater than the loss corresponding to the "average" strategy. Convexity can suitably translate this issue.

The fundamental result of game theory is the celebrated theorem about non-cooperative games due to Nash [55], for which he was awarded the Nobel prize in economics.

**Theorem 25.1** (Nash) *For every $k = 1, \dots, n$, let $K_k$ be a nonempty, compact and convex subset of a locally convex space $X_k$. Assume that, for every $k$, the loss function $f_k$ is continuous on $K$. In addition, for every fixed $x_j \in K_j$ with $j \neq k$, the map*

$$f_k(x_1, \dots, x_{k-1}, \cdot, x_{k+1}, \dots, x_n) : K_k \to \mathbb{R}$$

*is convex. Then there exists $\bar{x} \in K$ satisfying (25.2), i.e., there is a Nash equilibrium.*

*Proof* Define $\Phi : K \times K \to \mathbb{R}$ as

$$\Phi(x, y) = \sum_{k=1}^{n} \big[ f_k(x) - f_k(x_1, \dots, x_{k-1}, y_k, x_{k+1}, \dots, x_n) \big].$$

Then $\Phi$ is continuous, and $\Phi(x, \cdot)$ is concave for every fixed $x \in K$. Recall that $K$ is compact by the Tychonoff theorem, being the product of compact spaces. From Lemma 13.1 there exists $\bar{x} \in K$ such that

$$\sup_{y \in K} \Phi(\bar{x}, y) \leq \sup_{y \in K} \Phi(y, y) = 0.$$

In particular, if we set $\bar{y} = (\bar{x}_1, \dots, \bar{x}_{k-1}, x_k, \bar{x}_{k+1}, \dots, \bar{x}_n)$, for an arbitrarily given $x_k \in K_k$, we get

$$\Phi(\bar{x}, \bar{y}) \leq 0.$$

The latter inequality holds for every $x_k \in K_k$, and every $k = 1, \dots, n$, which is nothing but (25.2). □

The hypotheses can be weakened if we consider a two-player zero-sum game (sometimes called a *duel*). In this case, on account of (25.1), we have

$$\Psi(x_1, x_2) \doteq f_1(x_1, x_2) = -f_2(x_1, x_2).$$

Therefore we can repeat the above proof taking $\Psi$ to be convex and lower semicontinuous in the first variable, and concave and upper semicontinuous in the second. Now $\Psi$ is the loss function of the first player or, equivalently, the gain function of the second one.

**Theorem 25.2** (von Neumann) *Let $K_1 \subset X_1$ and $K_2 \subset X_2$ be as in Theorem 25.1. Let $\Psi : K_1 \times K_2 \to \mathbb{R}$ be such that:*

- $\Psi(\cdot, x_2)$ *is lower semicontinuous and convex* $\forall x_2 \in K_2$.
- $\Psi(x_1, \cdot)$ *is upper semicontinuous and concave* $\forall x_1 \in K_1$.

*Then there exists a Nash equilibrium* $(\bar{x}_1, \bar{x}_2) \in K_1 \times K_2$, *that is,*

$$\Psi(\bar{x}_1, x_2) \leq \Psi(\bar{x}_1, \bar{x}_2) \leq \Psi(x_1, \bar{x}_2),$$

*for every* $x_1 \in K_1$ *and every* $x_2 \in K_2$.

*Proof* In this case $\Phi : (K_1 \times K_2) \times (K_1 \times K_2) \to \mathbb{R}$ has the form

$$\Phi((x_1, x_2), (y_1, y_2)) = -\Psi(y_1, x_2) + \Psi(x_1, y_2),$$

and the argument of the proof of Theorem 25.1 applies. Indeed, $\Phi(\cdot, (y_1, y_2))$ is lower semicontinuous, while $\Phi((x_1, x_2), \cdot)$ is concave and upper semicontinuous, hence bounded above, as it attains its maximum on the compact set $K_1 \times K_2$. □

Theorem 25.2 is known in the literature as the *minimax* theorem [73]. The reason is clear from the next corollary.

**Corollary 25.1** *In the hypotheses of Theorem 25.2, the equality*

$$\inf_{x_1 \in K_1} \sup_{x_2 \in K_2} \Psi(x_1, x_2) = \Psi(\bar{x}_1, \bar{x}_2) = \sup_{x_2 \in K_2} \inf_{x_1 \in K_1} \Psi(x_1, x_2)$$

*holds true.*

*Proof* Define

$$g(x_1) = \sup_{x_2 \in K_2} \Psi(x_1, x_2) \qquad \text{and} \qquad h(x_2) = \inf_{x_1 \in K_1} \Psi(x_1, x_2).$$

Then, for all $x_1 \in K_1$ and $x_2 \in K_2$ we have

$$h(x_2) \leq \Psi(x_1, x_2) \leq g(x_1),$$

which entails

$$\sup_{x_2 \in K_2} h(x_2) \leq \inf_{x_1 \in K_1} g(x_1).$$

On the other hand, by Theorem 25.2,

$$h(\bar{x}_2) = \inf_{x_1 \in K_1} \Psi(x_1, \bar{x}_2) = \Psi(\bar{x}_1, \bar{x}_2) = \sup_{x_2 \in K_2} \Psi(\bar{x}_1, x_2) = g(\bar{x}_1).$$

Hence

$$\sup_{x_2 \in K_2} h(x_2) \geq h(\bar{x}_2) = g(\bar{x}_1) \geq \inf_{x_1 \in K_1} g(x_1) \geq \sup_{x_2 \in K_2} h(x_2),$$

so that all the above inequalities are actually equalities. □

We conclude the chapter by considering a duel game where the sets $K_1$ and $K_2$ of all possible strategies of each player are finite. We also assume that, for $k = 1, 2$, player $k$ plays randomly the strategy $x_k \in K_k$ with probability $p_k(x_k)$. In this case the players are said to adopt a *mixed strategy*. Denoting the loss function (of the first player) by $\Psi$, the average loss function is given by

$$\hat{\Psi}(p_1, p_2) = \sum_{x_1 \in K_1} \sum_{x_2 \in K_2} p_1(x_1) p_2(x_2) \Psi(x_1, x_2),$$

defined on the set $\hat{K}_1 \times \hat{K}_2$, where

$$\hat{K}_k = \left\{ p_k : K_k \to [0, 1] \quad \text{such that} \sum_{x_k \in K_k} p_k(x_k) = 1 \right\}.$$

**Theorem 25.3** *Any duel with a finite number of strategy profiles admits a Nash equilibrium made of mixed strategies.*

*Proof* Just observe that $\hat{K}_k$ and $\hat{\Psi}$ fulfill the hypotheses of Theorem 25.2 (actually, $\hat{\Psi}$ is linear in both variables). □

# Part III
# Some Problems

# Chapter 26
# Problems




**1.** Let $X$ be a metric space (not necessarily complete). A map $f : X \to X$ is said to be *closed* if, whenever $x_n \to x$ and $f(x_n) \to y$, it follows that $y = f(x)$.

1. Show that $f$ closed implies $f$ continuous, but not the other way around.
2. Prove that the (possibly empty) set

$$\mathcal{F} = \{x \in X : x = f(x)\}$$

of the fixed points of a closed map $f$ is closed in $X$.

**2.** Let $(X, d)$ be a metric space. Assume that $\hat{d}$ is another distance on $X$ such that, for some $m > 1$,

$$\frac{1}{m} d(x, y) \le \hat{d}(x, y) \le m d(x, y), \qquad \forall x, y \in X.$$

Two such metrics are called *strongly equivalent*. Prove that if $f$ is a contraction with respect to $d$, then $f^n$ is a contraction with respect to $\hat{d}$ for $n$ large enough.

**3.** Let $(X, d)$ be the space $[0, \infty)$ with the usual distance. Define on $X$ another distance $\hat{d}$ by the rule

$$\hat{d}(x, y) = \left| \frac{x}{1+x} - \frac{y}{1+y} \right|.$$

Show that the two metrics are *topologically equivalent*, namely, they induce the same topology on $X$. Consider now the function $f : X \to X$ given by

$$f(x) = 2x + 1.$$

It is clear that $f$ is not a contraction on $(X, d)$, nor is a contraction any iterate $f^n$. This is an obvious consequence of $f$ being fixed point free. Nonetheless, verify that

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$f$ is a contraction on $(X, \hat{d})$, with Lipschitz constant $\lambda = 1/2$. How does this fact match with the conclusions of Problem 2 and with those of the BCP Theorem 1.1?

**4.** Define the sequence $x_{n+1} = \sqrt{2 + \sqrt{x_n}}$ with $x_0 = \sqrt{2}$. Prove that $x_n$ converges to a root $\bar{x}$ of the equation

$$x^4 - 4x^2 - x + 4 = 0,$$

with $\sqrt{3} < \bar{x} < 2$. Find $n$ such that

$$|x_n - \bar{x}| < 10^{-100}.$$

**5.** For $0 < \alpha \leq 1$ and $\beta \geq 1$, consider the map $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f(x, y) = (\alpha y, \beta x).$$

1. Show that $f$ is not a contraction.
2. Find a necessary and sufficient condition involving the parameters $\alpha, \beta$ in order for $f^n$ to be a contraction for some $n \geq 2$.

**6.** Let $(X, d)$ be a complete metric space, and let $f : X \to X$ be a contraction with Lipschitz constant $\lambda$. Prove the contraction inequality

$$d(x, y) \leq \frac{1}{1 - \lambda}\big[d(x, f(x)) + d(y, f(y))\big].$$

Use this inequality to provide an alternative proof of the BCP Theorem 1.1.
$\diamond$ This proof of the BCP is due to Palais [56].

**7.** Let $X$ be a Banach space.

1. Let $f : X \to X$ be a contraction. Prove that, for every given $y \in X$, the equation

$$x - f(x) = y$$

has a unique solution $x \in X$.

2. Let $T \in L(X)$. Find the most general condition on $T$ under which the equation

$$x - Tx = y$$

has as solution $x \in X$ for every given $y \in X$. Repeat the exercise with the further request that the solution be unique.

**8.** Let $X$ be a Banach space, and let $T \in L(X)$. If $\|T\|_{L(X)} < 1$, prove that $I - T$ is invertible in $L(X)$. Find an explicit bound for the norm $\|(I - T)^{-1}\|_{L(X)}$ in terms of the norm $\|T\|_{L(X)}$.
***Hint***: Show that $I - T$ is bounded below and onto, and invoke the inverse mapping theorem in $L(X)$.

**9.** Given a Banach space $X$, let $T \in L(X)$, and let $f : X \to X$ be a Lipschitz function with Lipschitz constant $\lambda$. If $I - T$ is invertible with $(I - T)^{-1} \in L(X)$ and

$$\lambda \|(I - T)^{-1}\|_{L(X)} < 1,$$

prove that the map

$$x \mapsto Tx + f(x)$$

has a unique fixed point. Show that the same conclusion holds if we require instead

$$\|T\|_{L(X)} < 1 \qquad \text{and} \qquad \lambda < 1 - \|T\|_{L(X)}.$$

***Hint***: For the second part, use Problem 8.

**10.** Let $X = C([0, 1])$, and let $F \in C^1(\mathbb{R})$ be such that $F' \geq \mu > 0$. Given $k \in L^1(0, 1)$ and $v \in X$, prove that the functional equation

$$F(u(t)) + \int_0^t k(t - s)u(s)\, ds = v(t)$$

has a unique solution $u \in X$.
***Hint***: Consider the map $w \mapsto \Phi(w) = u$, with $u$ solution to

$$F(u(t)) + \int_0^t k(t - s)w(s)\, ds = v(t).$$

Show that $\Phi$ is a contraction on $C([0, t_*])$ for some $t_* \leq 1$, possibly very small. Repeat the argument on $[t_*, 2t_*]$, where now $u$ is prescribed on $[0, t_*]$ to be the solution found for $t \in [0, t_*]$. Iterate the procedure $n$ times until $nt_*$ reaches 1.

**11.** Let $X$ be a Banach space. For a given $r > 0$, consider the complete metric space $B_r = \overline{B}_X(0, r)$, and let $f : B_r \to X$ be such that $f(\partial B_r) \subset B_r$. Prove that if $f$ is a contraction from $B_r$ to $X$, that is, there is $\lambda < 1$ such that

$$\|f(x) - f(y)\| \leq \lambda \|x - y\|, \qquad \forall x, y \in B_r,$$

then $f$ has a unique fixed point.
***Hint***: Consider the map $F(x) = \frac{1}{2}(x + f(x))$ and show that $F$ maps $B_r$ into $B_r$, by adding and subtracting a suitable $f(y)$.

**12.** Let $B_r$ as in Problem 11, and let $f$ be a contraction from $B_r$ to $X$. Prove that if $f(-x) = -f(x)$ for every $x \in \partial B_r$, then $f$ has a unique fixed point.
***Hint***: The equality $f(x) = \frac{1}{2}(f(x) - f(-x))$ holds on $\partial B_r$.

**13.** Show that the function $\varphi$ of the Boyd-Wong Theorem 2.1 can be replaced by an increasing (or even strictly increasing) function still satisfying the assumptions of the theorem.

**14.** Prove Theorem 2.1 by assuming $\varphi$ only right continuous (or even only right upper semicontinuous) instead of continuous.

**15.** Let the space $X$ in the statement of Theorem 2.1 be a closed ball of a Banach space, and replace $\varphi$ continuous with $\varphi$ only right continuous (or even only right upper semicontinuous). Prove that there is a continuous increasing function $\psi : [0, \infty) \to [0, \infty)$ such that $\psi(r) < r$ if $r > 0$, and

$$d(f(x), f(y)) \le \psi(d(x, y)), \qquad \forall x, y \in X.$$

***Hint***: Define

$$\hat{\varrho}(r) = \sup_{d(x,y)\le r} d(f(x), f(y)) \qquad \text{and} \qquad \varrho(r) = \lim_{s \downarrow r} \hat{\varrho}(s).$$

First verify that $\hat{\varrho}(r) < r$ for every $r > 0$ (this is the more difficult step). Then verify that $\varrho(r) < r$ for every $r > 0$. At this point, knowing that $\varrho$ is increasing, right continuous and $\varrho(r) < r$, construct an increasing continuous function $\psi$ with the desired properties (reason as in the proof of Lemma 4.1). The detailed solution of this problem can be found in [57].

**16.** Let $f : [0, \infty) \to [0, \infty)$ be a continuous function such that $f(x) < x$ for every $x > 0$. Assume in addition that, for some $a > 0$ and $p > 0$,

$$f(x) = x - ax^{1+p} + o(x^{1+p}),$$

where $o(x^{1+p})/x^{1+p} \to 0$ when $x \to 0$. Prove that the unique fixed point of $f$ is $\bar{x} = 0$. Moreover, choosing any $x_0 > 0$ and defining the sequence $x_n = f^n(x_0)$, prove the asymptotic relation (cf. Theorem 2.2)

$$x_n \sim \frac{1}{(pan)^{1/p}}.$$

Find (for an arbitrarily given $p$) an example of a function $f$ of this kind, but such that $f$ does not comply with the assumptions of the Boyd-Wong Theorem 2.1.

**17.** For $r > 0$, let $f : [0, r] \to [0, r]$ be a function satisfying the hypotheses of the Caristi Theorem 3.1 for some lower semicontinuous $\psi : [0, r] \to [0, \infty)$. Assume that $\bar{x} = 0$ is the unique fixed point of $f$. Assume also that $f(x) > x$ for any $x > 0$ in a neighborhood of 0. Prove that there is a point $x_0 \neq 0$ such that $f(x_0) = 0$.
***Hint***: Prove that the restriction of $\psi$ on $(0, r]$ attains its minimum. To this end, argue by contradiction, and consider a sequence $x_n \in [1/n, r]$ for which

$$\psi(x_n) = \min\left\{\psi(x) \,:\, x \in [1/n, r]\right\}.$$

Show that $x_n \to 0$ and $x_n \not\to 0$ at the same time.

**18.** Define the function $f : [0, 2] \to [0, 2]$ as

$$f(x) = \begin{cases} -x^2 + 2x & 0 \le x < 1, \\ 3/2 & x = 1, \\ -x^2 + 4x - 2 & 1 < x < 2, \\ 0 & x = 2. \end{cases}$$

Prove that the assumptions of the Caristi Theorem 3.1 are satisfied for some lower semicontinuous $\psi : [0, 2] \to [0, \infty)$. Verify that, except for $x_0 = 0$ and $x_0 = 2$, no sequence of the form $f^n(x_0)$ reaches the (unique in this case) fixed point $\bar{x}$. Note that we are exactly in the framework of Problem 17.
Repeat the exercise with

$$f(x) = \begin{cases} -x^2 + 2x & 0 \le x < 1, \\ 1 & 1 \le x \le 2. \end{cases}$$

What happens now to the sequence $f^n(x_0)$?
***Hint***: Choose the $\psi$ of the example at Chap. 3, possibly by suitably modifying it at the point $x = 1$.

**19.** Find an example of $f : [0, 1] \to [0, 1]$, satisfying the assumptions of the Caristi Theorem 3.1, with a unique fixed point $\bar{x}$ but such that

$$\liminf_{n\to\infty} |f^n(x_0) - \bar{x}| > 0, \qquad \forall x_0 \neq \bar{x}.$$

Conclude that such a function $f$ cannot be a weak contraction.
***Hint***: Take $f(0) = 0$ and $f(x) < x$ for $x > 0$, with $f(x)$ approaching $x$ infinitely many times in correspondence to a sequence $x_n \to 0$.

**20.** Define the function $f : [0, \infty) \to [0, \infty)$ as

$$f(x) = \begin{cases} \dfrac{2(1 - e^{-x})}{x} + 1 & x > 0, \\ 3 & x = 0. \end{cases}$$

Prove that $f$ is a weak contraction. Is it possible to apply Theorem 4.1?

**21.** Define the linear operator $f \mapsto Tf$ on $L^2(0, 1)$ as

$$[Tf](t) = tf(t).$$

1. Prove that $T$ is a weak contraction on $L^2(0, 1)$ but its operator norm equals 1 (hence it is not a contraction).
2. Show that the unique fixed point of $T$ is 0. Use the Riesz mean ergodic Theorem 8.1 to show that, for any $f \in L^2(0, 1)$, the sequence

$$g_n(t) = \frac{1}{n+1} \sum_{k=0}^{n} t^k f(t)$$

converges to 0 in $L^2(0, 1)$.

$\diamond$ In particular, point 1 shows that (contrary to what happens for linear functionals) not all bounded linear operators on an infinite-dimensional reflexive Banach space realize their norms.

**22.** For a given $\varphi \in L^2(0, 1)$ and a given $\psi \in L^\infty(0, 1)$, define the selfmap $f \mapsto \Phi(f)$ on $L^2(0, 1)$ as

$$[\Phi(f)](t) = \varphi(t) + \psi(t) f(t).$$

Find conditions on $\varphi$, $\psi$ in order for $\Phi$ to be a contraction on $L^2(0, 1)$. Find more general conditions on $\varphi$, $\psi$ in order for $\Phi$ to have a unique fixed point.

**23.** Let $X$ be a complete metric space, and let $f : X \to X$ be a weak contraction. Assume that there exists $x_0 \in X$ such that the sequence $x_n = f^n(x_0)$ admits a convergent subsequence. Prove that $f$ has a unique fixed point $\bar{x}$. Moreover, prove that $x_n \to \bar{x}$.
***Hint***: For $x_{n_j} \to \bar{x}$, one has $d(x_{n_j}, f(x_{n_j})) \to d(\bar{x}, f(\bar{x}))$.

**24.** Provide an alternative proof of the fact that a weak contraction $f$ on a compact metric space $X$ has a unique fixed point (cf. Theorem 4.1), by completing the following steps:

1. Show that $X_\infty = \bigcap_n f^n(X)$ is nonempty and compact.
2. Show that $f(X_\infty) = X_\infty$.
3. Show that there exist $x, y \in X_\infty$ such that $\operatorname{diam}(X_\infty) = d(x, y)$.
4. Use the fact that $f$ is a weak contraction from $X_\infty$ onto $X_\infty$ to show that $\operatorname{diam}(X_\infty) = 0$, meaning that $X_\infty$ reduces to a single point.

**25.** Find an example of a weak contraction on a bounded complete metric space without a fixed point.
***Hint***: Refine the example at Chap. 6, by considering the map

$$f(x) = (1, a_0 x_0, a_1 x_1, a_2 x_2, \ldots),$$

where $a_n$ is a suitable sequence.

**26.** With reference to the proof of Proposition 5.2, let $\zeta : [0, 1] \to \mathbb{R}$ be a continuous function such that $\zeta(r) > 0$ for every $r > 0$. Construct a strictly increasing continuous function $\mu : [0, 1] \to \mathbb{R}$ such that $\mu(0) = 0$ and $\mu(r) \leq \zeta(r)$ for every $r > 0$.

**27.** Work the details of the example at Chap. 5 for the function $f : [1, \infty) \to [1, \infty)$ defined as

$$f(x) = -2 + x - 2\sqrt{x} + 4\sqrt[4]{x}.$$

Prove that $f$ is a contraction of $\varepsilon$-type.

**28.** Let $X$ be a metric space, and let $f, f_n : X \to X$ with $f$ continuous. Assume that each $f_n$ has at least a fixed point $\bar{x}_n = f_n(\bar{x}_n)$, and denote by $\mathcal{F}$ the (possibly empty) set of all limit points of any such sequence $\bar{x}_n$. Prove that if $f_n$ converges uniformly to $f$, then any $\bar{x} \in \mathcal{F}$ is a fixed point of $f$.

**29.** Let $f : \mathbb{D}^n \to \mathbb{D}^n$ be continuous (recall that $\mathbb{D}^n$ is the unit disk of $\mathbb{R}^n$). For $(t, x) \in \mathbb{R} \times \mathbb{R}^n$ define the map

$$g(t, x) = \begin{cases} (1 - \|x\|) f(x) + \|x\| x & t > 0, \\ x & t \le 0. \end{cases}$$

1. Prove that $g$ defines a continuous function from $\mathbb{S}^n$ to $\mathbb{R}^n$ (where $\mathbb{S}^n$ is the unit sphere of $\mathbb{R}^{n+1}$).
2. Assume that $g(t, x) \neq g(-t, x)$ for every $(t, x) \in \mathbb{S}^n$ with $t \neq 0$. Prove that no fixed point of $f$ lies in the interior of $\mathbb{D}^n$, hence $f$ has a fixed point $\bar{x} \in \mathbb{S}^{n-1}$.

**30.** A map $f : \mathbb{S}^n \to \mathbb{S}^{n-1}$ is called *odd* if $f(x) = -f(-x)$ for all $x \in \mathbb{S}^n$. A famous result in algebraic topology, the *Borsuk-Ulam theorem* (see, e.g., [51]), states that for all $n \ge 1$ there is no continuous odd map from $\mathbb{S}^n$ to $\mathbb{S}^{n-1}$. Accordingly, prove the following facts:

1. Any continuous map $g : \mathbb{S}^n \to \mathbb{R}^n$ identifies a pair of antipodes, namely, there is $x \in \mathbb{S}^n$ such that $g(x) = g(-x)$.
2. Provide an elementary proof of point 1 for the case $n = 1$. This is equivalent to show that, given any continuous function $g : [0, 2\pi] \to \mathbb{R}$ satisfying $g(0) = g(2\pi)$, there exists $x \in [0, \pi]$ such that $g(x) = g(x + \pi)$.
3. There is no continuous map $h$ from $\mathbb{D}^n$ to $\mathbb{S}^{n-1}$ with the property that

$$h(x) = -h(-x), \qquad \forall x \in \mathbb{S}^{n-1}.$$

4. Conclude that $\mathbb{S}^{n-1}$ is not a retract of $\mathbb{D}^n$, and deduce the Brouwer fixed point Theorem 9.1.

***Hint***: To prove 3, argue by contradiction, and consider the northern hemisphere

$$\mathbb{S}^n_+ = \left\{ x = (x_1, \dots, x_n, x_{n+1}) \in \mathbb{S}^n \ : \ x_{n+1} \ge 0 \right\},$$

along with the projection $p : \mathbb{S}^n_+ \to \mathbb{D}^n$ given by $p(x) = (x_1, \dots, x_n)$. Then define the map $f : \mathbb{S}^n \to \mathbb{S}^{n-1}$ as

$$f(x) = h(p(x)) \qquad \text{and} \qquad f(-x) = -h(p(x))$$

for $x \in \mathbb{S}^n_+$. Show that such an $f$ is forbidden by the Borsuk-Ulam theorem.

$\diamond$ Point 1 in dimension $n = 2$ has an interesting interpretation in meteorology: at any given moment, there exists a pair of antipodal points on the surface of the earth sharing the same temperature and the same barometric pressure (assuming the temperature and the barometric pressure to vary with continuity).

**31.** Let $K$ be the subset of the Hilbert space $\ell^2$ defined as

$$K = \big\{x = (x_0, x_1, x_2, \ldots) \,:\, |x_n| \le 1/(n+1)\big\}.$$

Prove that any continuous map $f : K \to K$ admits a fixed point.

**32.** Let $B_r$ as in Problem 11, and let $f : B_r \to X$ be a continuous and compact map satisfying

$$\|f(x)\|^2 \le \|f(x) - x\|^2 + r^2, \qquad \forall x \in \partial B_r.$$

Prove that $f$ has a fixed point.
***Hint***: Argue by contradiction. Setting $g(x) = x$ if $x \in B_r$ and $g(x) = rx/\|x\|$ if $x \notin B_r$, study the map $x \mapsto g(f(x))$.

**33.** Prove the following more general version of the Gronwall Lemma 16.1, valid for $t_0 = a$:
Let $\varphi : [a, b] \to \mathbb{R}$ be a bounded measurable function, and let $k \in L^1(a, b)$ be nonnegative. Finally, let $q : [a, b] \to \mathbb{R}$ be an increasing function. Assume to have the inequality

$$\varphi(t) \le q(t) + \int_a^t k(s)\varphi(s)\, ds,$$

for every $t \in [a, b]$. Then the inequality

$$\varphi(t) \le q(t)\, \exp\left[\, \int_a^t k(s)\, ds \,\right]$$

holds for every $t \in [a, b]$.

**34.** Prove the following differential version of the Gronwall Lemma 16.1.
Let $\varphi : [a, b] \to \mathbb{R}$ be an absolutely continuous function and let $k, \psi \in L^1(a, b)$ be nonnegative. Assume that the differential inequality

$$\varphi'(t) \le k(t)\varphi(t) + \psi(t)$$

holds for almost every $t \in [a, b]$. Then the inequality

$$\varphi(t) \le \left[\varphi(a) + \int_a^t \psi(s)\, ds\right] \exp\left[\, \int_a^t k(s)\, ds \,\right]$$

holds for every $t \in [a, b]$.

***Hint***: Integrate the differential inequality and use Problem 33.

**35.** Let $f \in C^1(\mathbb{R})$, and denote by $Z_f = \{y \in \mathbb{R} : f(y) = 0\}$ the zero-set of $f$. Assume that $Z_f$ contains at most a finite number of elements (possibly none). Call these elements $\bar{y}_1 < \ldots < \bar{y}_n$. Consider the ordinary differential equation

$$y' = f(y),$$

and prove the following facts:

1. The only stationary (i.e., constant) solutions are $y(t) = \bar{y}_j$ with $j = 1, \ldots, n$. Any other solution is strictly monotonic.
2. Let $y(t)$ be a solution with $\bar{y}_j < y(0) < \bar{y}_{j+1}$ for some $j = 1, \ldots, n-1$. Then $y(t)$ is defined for all $t$, and either

$$\lim_{t \to -\infty} y(t) = \bar{y}_j \quad \text{and} \quad \lim_{t \to \infty} y(t) = \bar{y}_{j+1},$$

   or

$$\lim_{t \to -\infty} y(t) = \bar{y}_{j+1} \quad \text{and} \quad \lim_{t \to \infty} y(t) = \bar{y}_j.$$

3. Let $y(t)$ be a solution with $y(0) > \bar{y}_n$. Then either

$$\lim_{t \to -\infty} y(t) = \bar{y}_n \quad \text{and} \quad \lim_{t \to t_\infty^+} y(t) = \infty,$$

   or

$$\lim_{t \to t_\infty^-} y(t) = \infty \quad \text{and} \quad \lim_{t \to \infty} y(t) = \bar{y}_n,$$

   where $t_\infty^+ \in (0, \infty]$ and $t_\infty^- \in [-\infty, 0)$. Find the condition in order to have $t_\infty^+ < \infty$ (or $t_\infty^- > -\infty$).
4. Discuss the case of a solution $y(t)$ with $y(0) < \bar{y}_1$.
5. Discuss what happens when $Z_f$ is empty.

**36.** Let $f \in C^1(\mathbb{R})$, and assume that the zero-set $Z_f$ of $f$ has empty interior. For an arbitrarily given $\bar{y}_0 \in Z_f$, is it always true that we can find a solution $y(t)$ of the ordinary differential equation

$$y' = f(y)$$

such that $y(t) \neq \bar{y}_0$ and $y(t) \to \bar{y}_0$ when $t \to \infty$ or $t \to -\infty$?

**37.** Define the function $f : \mathbb{R}^2 \to \mathbb{R}$ as

$$f(t, y) = \begin{cases} e^{-y^2/\sqrt{|t|}} & t \neq 0, \\ 0 & t = 0. \end{cases}$$

Prove that the Cauchy problem

$$\begin{cases} y'(t) = f(t, y(t)), \\ y(0) = y_0, \end{cases}$$

has a unique solution in a neighborhood of $t = 0$ for all initial data $y_0 \neq 0$.

**38.** Show that the solution to Cauchy problem

$$\begin{cases} y' = \sqrt{|y|}, \\ y(0) = 0, \end{cases}$$

is not unique. Discuss the result in light of Theorems 16.1 and 16.3.

**39.** Exhibit a continuous function $f : \mathbb{R} \to \mathbb{R}$ with $f(0) = 0$ such that $f$ is not Lipschitz continuous in a neighborhood of 0, but nonetheless the Cauchy problem

$$\begin{cases} y' = f(y), \\ y(0) = 0, \end{cases}$$

has a unique solution (the trivial one).

**40.** Consider again Problem 39, making this time the following further assumption: $f$ is Lipschitz continuous in the neighborhood of every point $y \neq 0$, and $f(y) \neq 0$ if $y \neq 0$. Find the correct condition to impose on $f$ in order to have a unique solution $y$ such that $y(0) = 0$.

**41.** Let $f \in C^1(\mathbb{R})$ and $g \in C([0, 1])$. For an arbitrarily fixed $p \in \mathbb{R}$, consider the Cauchy problem in the unknown $u = u(x) : [0, 1] \to \mathbb{R}$

$$\begin{cases} u'(x) + f(u(x)) + g(x) = p \quad 0 < x \leq 1, \\ u(0) = 0. \end{cases}$$

Prove that there exists a unique $p \in \mathbb{R}$ such that the corresponding solution $u$ is defined on the whole interval $[0, 1]$ and $u(1) = 0$.
***Hint***: For the uniqueness part, use the differential version of the Gronwall lemma discussed in Problem 34. Concerning existence, given any $p \in \mathbb{R}$, consider the solution $u_{r,p}$ to the Cauchy problem where $f$ is replaced by a $C^1$ cut-off $f_r$ such that

$$f_r(u) = \begin{cases} f(r+1) & \text{if } u \geq r+1, \\ f(u) & \text{if } |u| \leq r, \\ f(-r-1) & \text{if } u \leq -r-1. \end{cases}$$

Then note that $u_{r,p}$ is defined on the whole interval $[0, 1]$, and $|u_{r,p}(1) - p|$ is bounded, with a bound independent of $p$. Conclude that there is a unique $p = p_r$

such that $u_{r,p}(1) = 0$. Finally, prove that the sequence $u_{r,p_r}$ converges (up to a subsequence) to the sought solution $u$ as $r \to \infty$. This step requires some a priori estimates on $u_{r,p_r}$, and passing to the limit within the integral equation satisfied by $u_{r,p_r}$.
$\diamond$ This problem is a quite difficult one, and is discussed in detail in [19].

**42.** For $g \in L^2(0,1)$, consider the nonlinear elliptic boundary value problem in the unknown $u : [0,1] \to \mathbb{R}$

$$\begin{cases} -u'' + uu' = g & \text{in } (0,1), \\ u(0) = u(1) = 0. \end{cases}$$

Using Problem 41, show that there is a unique weak solution $u \in H_0^1(0,1)$, which is actually more regular, as $u \in H^2(0,1)$. Recall that in the one-dimensional case the space $H_0^1(0,1)$ is the space of absolutely continuous functions $u : [0,1] \to \mathbb{R}$ with $u(0) = u(1) = 0$ and with derivative $u' \in L^2(0,1)$.
***Hint***: Write $-u'' + uu' = \big(-u' + \frac{u^2}{2}\big)'$, and let $g = G'$.

**43.** Let $\Omega \subset \mathbb{R}^n$ be a domain with smooth boundary $\partial\Omega$. Let $g \in H^{-1}(\Omega)$ (this is the way the dual space of $H_0^1(\Omega)$ is denoted), and define the operator

$$Lu = -\Delta u + \sum_{i=1}^{n} a_i \partial_{x_i} u + bu,$$

where $a_i, b \in L^\infty(\Omega)$. For $\mu \geq 0$, consider the linear elliptic boundary value problem in the unknown $u = u(x) : \overline{\Omega} \to \mathbb{R}$

$$\begin{cases} Lu + \mu u = g & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Prove that if $\mu$ is large enough, then the problem admits a unique weak solution $u \in H_0^1(\Omega)$.

**44.** Let $\Omega \subset \mathbb{R}^n$ be a domain with smooth boundary $\partial\Omega$, and let $g \in H^{-1}(\Omega)$. Consider the nonlinear elliptic boundary value problem for $u = u(x) : \overline{\Omega} \to \mathbb{R}$

$$\begin{cases} -\Delta u + f(\partial_{x_1} u, \dots, \partial_{x_n} u) = g & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

where $f$ is a Lipschitz function from $\mathbb{R}^n$ to $\mathbb{R}$, with Lipschitz constant $\lambda$. Prove that if $\lambda$ is small enough, then the problem admits a unique weak solution $u \in H_0^1(\Omega)$. Moreover, if $g \in L^2(\Omega)$ then the solution $u$ belongs to $H^2(\Omega)$ as well.

**45.** A strongly continuous semigroup $S(t)$ on a Banach space $X$ is said to be *exponentially stable* if there exist $M \geq 1$ and $\omega > 0$ such that

$$\|S(t)\|_{L(X)} \leq Me^{-\omega t}, \qquad \forall t \geq 0.$$

Prove the following facts:

1. If there exists $\tau > 0$ for which $\|S(\tau)\|_{L(X)} < 1$, then $S(t)$ is exponentially stable.
2. Assume there exists a positive function $h$ vanishing at infinity such that

$$\|S(t)x\|_X \leq C_x h(t), \qquad \forall x \in X,$$

where $C_x$ is a positive constant depending on $x$. Then $S(t)$ is exponentially stable.

***Hint***: For point 1, use the concatenation property of the semigroup together with Theorem 19.1. For point 2, use the uniform boundedness principle, and then apply point 1.

**46.** Let $S(t)$ be an exponentially stable semigroup on a Banach space $X$. In which case, it is well know that the infinitesimal generator $T$ is invertible in $L(X)$ (see, e.g., [59]). Let $f : [0, \infty) \to X$ be a continuous function such that

$$\lim_{t\to\infty} f(t) = f_\infty \in X.$$

Prove that for every $x_0 \in X$ we have the limit

$$\lim_{t\to\infty} \|x(t) + T^{-1} f_\infty\|_X = 0,$$

where $x(t)$ is the unique mild solution to the linear nonhomogeneous Cauchy problem

$$\begin{cases} x'(t) = Tx(t) + f(t) & t > 0, \\ x(0) = x_0. \end{cases}$$

***Hint***: The mild solution can be written explicitly by means of (19.3). Note also that (see Theorem 19.2)

$$\int_0^t S(s)Tw\,ds = S(t)w - w, \qquad \forall w \in \mathrm{dom}(T).$$

Use this equality with $w = T^{-1} f_\infty$.

**47.** Use the Banach generalized limit to show that not every continuous linear functional $\Lambda$ on the real Banach space $\ell^\infty$ can be given the representation

$$\Lambda x = \sum_{n=0}^{\infty} c_n x_n \qquad x = (x_0, x_1, x_2, \ldots),$$

for some numerical sequence $c_n$.

**48.** Let $X = L^2(0, 1)$ on the complex field. Given $g \in X$, consider the linear operator $T : X \to X$ defined as

$$[Tf](t) = \int_0^t g(t-s)\, f(s)\, ds.$$

Prove that $T$ has a hyperinvariant subspace.

**49.** Let $S = [0, 1] \times [0, 1]$. For $\varepsilon > 0$, define the function $\Psi : S \to \mathbb{R}$ as

$$\Psi(x, y) = \sqrt{y}\, e^{\varepsilon x^2 y}.$$

Prove that, for any $\varepsilon$ sufficiently small, there exists $(\bar{x}, \bar{y}) \in S$ such that

$$\Psi(\bar{x}, y) \le \Psi(\bar{x}, \bar{y}) \le \Psi(x, \bar{y}), \qquad \forall (x, y) \in S.$$

***Hint***: Apply Theorem 25.2.

**50.** Let $C$ be a closed bounded convex subset of a Banach space $X$. For every nonempty set $B \subset C$, we define the *Kuratowski measure of noncompactness* (see, e.g., [33])

$$\alpha(B) = \inf \big\{ d : B \text{ has a finite cover of balls of } X \text{ of diameter less than } d \big\}.$$

Some noteworthy properties of $\alpha$ are recalled here below:

1. $\alpha(B) = 0$ if and only if $B$ is relatively compact.
2. $\alpha(\overline{B}) = \alpha(B)$.
3. $B_1 \subset B_2$ implies $\alpha(B_1) \le \alpha(B_2)$.
4. $\alpha(B_1 \cup B_2) = \max \big\{ \alpha(B_1), \alpha(B_2) \big\}$.
5. $\alpha(\mathrm{co}(B)) = \alpha(B)$.
6. If $B_0 \supset B_1 \supset B_2 \supset \ldots$ are nonempty closed sets such that $\alpha(B_n) \to 0$, then the intersection $\bigcap_{n \ge 0} B_n$ is nonempty and compact.

A function $f : C \to C$ is called a *set contraction* if there is $\lambda < 1$ such that

$$\alpha(f(B)) \le \lambda \alpha(B), \qquad \forall B \subset C,\ B \neq \emptyset.$$

Prove that a continuous set contraction on $C$ has a fixed point.
***Hint***: Set $C_0 = C$ and define recursively

$$C_{n+1} = \overline{\mathrm{co}}(f(C_n)).$$

Show by induction that $C_0 \supset C_1 \supset C_2 \supset \ldots$. Define $K = \bigcap_{n \ge 0} C_n$, noting that $\alpha(C_n) \to 0$, and show that $f(K) \subset K$. Apply Theorem 10.1 to conclude.

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